BEFORE THE HONOURABLE
KERALA STATE ELECTRICITY REGULATORY COMMISSION
At its office at C V Raman Pillai Road, Vellayambalam, Thiruvananthapuram

FILING No	
CASE No	
IN THE MATTER OF:	Truing up of Accounts of Generation, Transmission and Distribution Strategic Business Units of the Kerala State Electricity Board Limited (KSEBL) for the year 2023-24.
IN THE MATTER OF :	Kerala State Electricity Board Limited, Vydyuthi Bhavanam, Pattom, Thiruvananthapuram – 4
	APPLICANT
	Deputy Chief Engineer (TRAC) In-charge of Chief Engineer (Commercial & Tariff) Kerala State Electricity Board Limited

AFFIDAVIT VERIFYING THE APPLICATION ACCOMPANYING THE PETITION FOR TRUING UP OF ACCOUNTS OF GENERATION, TRANSMISSION, AND DISTRIBUTION FUNCTIONS OF KSEBL FOR THE YEAR ~~2023~~ 2023 – 24.

I, **Rajan M. P.**, son of Sri. M.S. Podiyan, aged 51 years residing at JAYAS, Hilltop Haven, NAD(PO), Aluva, Ernakulam do solemnly affirm and say as follows:

I am working as Deputy Chief Engineer-in-charge of Chief Engineer (Commercial & Tariff), Kerala State Electricity Board Limited, Vydyuthi Bhavanam, Pattom, Thiruvananthapuram and the petitioner in the above matter and I am duly authorized by the Board to make this affidavit on its behalf. I solemnly affirm at Thiruvananthapuram on this the 28th day of November 2024 that the contents of the above petition are true to my information, knowledge and belief. I believe that no part of it is false and no material has been concealed there from.

Deponent

Deputy Chief Engineer (TRAC)
In-charge of Chief Engineer (Commercial & Tariff)
Kerala State Electricity Board Limited,
Vydyuthi Bhavanam, Pattom
Thiruvananthapuram – 695 004

VERIFICATION

I, the above-named deponent, solemnly affirm at Thiruvananthapuram on this, the 28th day of November 2024 that the contents of the affidavit are true to my information, knowledge and belief, that no part of it is false and that no material has been concealed there from.

Deponent

Deputy Chief Engineer (TRAC)
In-charge of Chief Engineer (Commercial & Tariff)
Kerala State Electricity Board Limited
Vydyuthi Bhavanam, Pattom
Thiruvananthapuram – 695 004.

Solemnly affirmed and signed before me.

28/11/2024

T. N. OMANA
Advocate & Notary
Thiruvananthapuram Revenue District
Kerala State South India

SI NO 1598/2024
dt 28/11/2024
on page no 102
Book no X

GOVT. OF KERALA NOTARY
SPECIAL ADHESIVE

BEFORE THE HONOROUBLE
KERALA STATE ELECTRICITY REGULATORY COMMISSION

In the matter of: Truing up of Accounts of Generation, Transmission and Distribution Strategic Business Units of the Kerala State Electricity Board Limited (KSEBL) for the year 2023-24.

Petitioner: Kerala State Electricity Board Limited,
Vydyuthi Bhavanam, Pattom, Thiruvananthapuram - 4

THE PETITIONER HUMBLY STATES THAT:

CHAPTER- 1: INTRODUCTION

1.1 The petitioner, Kerala State Electricity Board Limited (KSEBL), is a State Public Sector power utility company constituted by the Government of Kerala. The Assets, Rights and Liabilities of former State Electricity Board was re-vested into the company on October 31, 2013. The petitioner company had been submitting ARR, ERC & Tariff and Truing up petitions in accordance with the Tariff regulations notified by the Hon'ble Commission from time to time.

1.2 Hon'ble Commission notified the third MYT Regulation, 2021 for determination of Tariff on 16.11.2021, after repealing the earlier MYT Tariff Regulation, 2018, with a five-year control period starting from FY 2022-23 till 2026-27. The regulations contain a performance-based approval scheme along with a framework for sharing gains and losses and specifying the controllable and un-controllable factors.

1.3 KSEBL had submitted MYT petition for the above-mentioned control period containing ARR & ERC on 31.01.2022, capital investment plan for the five-year control period from 2022-23 to 2026-27 in respect of its three SBUs on 10.01.2022 along with tariff revision proposals for all five years of the control period on 10.02.2022, before the Hon'ble Commission, in strict compliance with the MYT Regulations, 2021. Hon'ble Commission has approved Aggregate Revenue Requirements (ARR) / Expected Revenue from Charges (ERC) of KSEBL for the control period from FY 2022-23 to 2026-27 vide order dated 25.06.2022 and also was pleased to revise the electricity tariff of all categories of consumers in the state along with bulk supply tariff applicable to other licensees for 2022-23. Further, Hon'ble Commission vide tariff order dated 31.10.2023 has approved the tariff revision for the period 01.11.2023 to 30.06.2024. The Commission has later extended the validity of the prevailing tariff for the period up to 30.11.2024. KSEBL has also filed a proposal for revision of tariff for the remaining control period from 1.7.2024 to 31.3.2027, on 02.08.2024. Hearings of the tariff petition have been conducted by the Commission and the Commission is yet to issue the tariff revision order.

1.4 Regulation 10 (1) (iii) of Tariff Regulation, 2021 stipulates filing of annual truing up petition. In compliance, the summary of the audited Annual statement of Accounts of KSEBL for the year 2023-24, the amount approved by Hon'ble Commission in the order dated 25.06.2022 and the amount sought for true up is furnished in the Table–1.1 given below. The Statutory Auditors of KSEBL has

conducted audit and issued certificate for the year 2023-24 and a copy of the audited accounts is submitted separately along with this petition.

Table-1.1: Comparison of ARR, ERC and Revenue Gap of KSEBL for 2023-24 (Rs. Cr.)				
No	**Particulars**	**Approved***	**Audited**	**Truing up**
1	ARR	20061.99	21583.97	22122.05
2	Non-Tariff Income	866.94	1360.73	1036.35
3	Net ARR	**19195.05**	**20223.24**	**21085.70**
4	ERC	16255.96	20441.75	19761.95
5	**Revenue surplus (gap) (3-4)**	**(2939.09)**	**218.51**	**(1323.75)**
**Approved vide Tariff Order dt.25.06.2022 in OP No. 11 / 2022*				

1.8 The difference between audited accounts and true-up values for 2023-24 are mainly due to the difference in the principles and methodologies adopted in the annual accounts and the same prescribed in the Tariff Regulation, 2021 notified by the Hon'ble Commission. The matter is explained in detail in the respective paragraphs in Chapters below. Function wise segregation of expenses, revenue and surplus sought for true up are summarised below.

Table-1.2: Business Unit wise ARR, ERC and Revenue Gap for 2023-24 (Rs Cr)				
No	**Particulars**	**SBU-G**	**SBU-T**	**SBU-D**
1	ARR (including RoE)	666.69	1629.10	22005.88
2	Non-Tariff Income	40.21	75.96	920.18
3	Net ARR	**626.48**	**1553.14**	**21085.70**
4	Transfer Value / ERC	**626.48**	**1553.14**	19761.95
5	Revenue surplus (gap) (3-4)			**(1323.75)**

1.9 Following chapters in this petition describe the Actual Expenses and Revenue of the petitioner in FY 2023-24 and the comparison with respective approved values.

CHAPTER – 2: TRUING-UP OF ARR OF SBU – GENERATION

2. SBU –Generation – Overview

SBU-G is entrusted with the construction, operation, and maintenance of various types of generating stations of KSEBL. The generation capacity of SBU G comprises a mix of Hydel, Thermal, solar and wind power stations.

2.1 The total installed capacity as on 31.03.2024 is **2308.196 MW** of which 90.83 % is Hydel, 6.93% is Thermal, 2.15 % from solar and rest from the wind. The source wise generation capacity details are given in the table- G 1 below:

Table- G 1: Source wise Generation Capacity

Source	Installed Capacity (MW) as on 31.03.2024	Percentage share
Hydel	2096.611	90.83%
Thermal	159.96	6.93%
Wind	2.025	0.09%
Solar	49.6	2.15%
Total (Generation)	**2308.196**	100.00%

2.2 Energy generated by the KSEBL's Generation Business Unit was 5721.17 **MU out** of which 5663.67 **MU** was from own Hydro-electric Stations and remaining 57.50 **MU** was generated from thermal, wind and solar stations. Source wise generation approved by the Hon'ble Commission and actually generated during the year is furnished below**.** Details of energy generated, and expenses are furnished in paragraphs hereunder.

Table – G2: Energy Generation from own stations for FY 2023-24 (MU)

Source	KSERC approval	Audited Accounts	Difference
Gross Generation			
Hydro	7163.06	5663.67	-1499.39
BDPP	0.00	0.00	0.00
KDPP	0.00	0.36	0.36
Wind	0.00	1.21	1.21
Solar	72.68	55.93	-16.75
Gross Subtotal	**7235.74**	**5721.17**	-1514.57
Aux consumption			
Hydro	85.85	63.41	-22.44
BDPP	0.00	0.00	0.00
KDPP	0.00	0.36	0.36
Wind	0.00	0.00	0.00
Solar	0.18	0.00	-0.18
Aux Sub total	**86.03**	**63.77**	-22.26
Net Generation			
Hydro	7077.22	5600.26	-1476.96
BDPP	0.00	0.00	0.00
KDPP	0.00	-0.36	-0.36
Wind	0.00	1.21	1.21
Solar	72.50	55.93	-16.57
Subtotal	**7149.72**	**5657.04**	**-1492.68**

2.3 Hydro Generation:

Approved net hydro generation for the FY 2023-24 as per the MYT Order dated 25.06.2022 was 7077.22 **MU** whereas the actual hydro generation was 5600.26 **MU,** resulting in reduced generation of **(**-)1476.96 **MU**. This decrease in generation was mainly due to lower inflow received during the financial year than anticipated.

The anticipated and actual inflow during the year 2023-24 are given in the table- G 3 below:

Table – G3: The anticipated and actual inflow during the year 2023-24					
Months	**Anticipated Inflow (MU)**	**Actual Gross Inflow (MU)**	**Actual Spill (MU)**	**Actual Net Inflow (MU)**	**Surplus (+) / Deficit (-) (MU)**
Apr-23	117.60	130.19	6.00	124.19	6.59
May-23	197.54	165.82	0.44	165.38	-32.16
Jun-23	770.41	224.38	0.00	224.38	-546.03
Jul-23	1452.24	1280.07	67.81	1212.26	-239.98
Aug-23	1281.67	307.25	0.00	307.25	-974.42
Sep-23	957.57	804.86	0.00	804.86	-152.71
Oct-23	774.78	815.45	2.70	812.75	37.97
Nov-23	526.88	575.04	88.59	486.45	-40.43
Dec-23	250.23	371.06	0.00	371.06	120.83
Jan-24	135.41	217.18	0.00	217.18	81.77
Feb-24	80.17	121.01	0.00	121.01	40.84
Mar-24	99.65	65.32	0.00	65.32	-34.33
Total	**6644.16**	**5077.64**	**165.53**	**4912.09**	**-1732.06**

2.3.1 From the table, it may kindly be seen that the inflow received was much lower than the level approved by the Hon'ble Commission.

2.3.2 This was the year in which rainfall was much lesser compared to earlier years. The expected rainfall was calculated based on the average of the previous 20 years of rainfall. Monsoon rain was much lesser and hence the major dams were not filled as expected.

2.3.3 In addition to this, it may kindly note that, the spillage of major dams is governed by the rule curve. In the backdrop of unprecedented extreme rainfall occurred in August 2018 in Kerala, it was decided by KSEBL to prepare Emergency Action Plans for the dams and to conduct a study on reservoir water level of selected dams and preparation of Rule Curves for five major dams. The rule levels for the reservoirs were developed in consideration of the suggestions of Central Water Commission (CWC). The Government of Kerala accorded sanction to KSEBL for adopting the alert levels, rules curves as suggested by CWC for Idukki, Idamalayar, Kakki, Banasurasagar and Sholayar dams owned by KSEBL. The operational strategy for the five major reservoirs of KSEBL were updated by reviewing the rule levels. The rule levels of dams were fixed to create dynamic flood cushion for moderating the floods, particularly in the early periods of monsoon.

2.3.4 The details of hydel generation (Including auxiliary consumption) and station wise generation details are given in Table- G5 and G6 respectively hereunder.

2.3 Projects commissioned during the FY 2023-24 with write-up, DPR cost, actual cost incurred, reason for time/cost overrun

2.3.1 PERUVANNAMUZHI SHEP

General Overview

Peruvannamuzhi Small Hydro Electric Project (2x3MW) is constructed in Chakkittappara and Chembanoda villages of Koyilandi Taluk in Kozhikode District. This SHEP envisages the utilization of the excess inflow over and above the demand for Irrigation and water supply at the existing Peruvannamuzhi Irrigation Dam by the construction of water conducting system of about 700 m length and a Power House with an installed capacity of 2x3MW, 2 Nos. of Horizontal double regulated Kaplan turbines, for generating an average annual energy of 24.70 MU.

The power generated from Peruvannamuzhi SHEP is evacuated to 110kV Substation Chakkittappara, by drawing 3.60km, 33kV line by installing 110/33kV 16MVA Transformer and 110/33kV bay at 110kV Substation Chakkittappara and constructing 3.60km, 33kV line from Peruvannamuzhi SHEP to 110kV Substation Chakkittappara. The total generation as on 01.11.2024 is 24.911MU.

Component Parts of the Project

(i) Civil Structures:

The main civil components of the Project are Intake, Leading Channel and Trash Rack structure, Head Race Tunnel, Surge shaft, Pressure shaft, Penstock pipe & Feeder pipes and Powerhouse, for installing 2 Generators with 3 MW capacity of each with horizontal Kaplan Turbine, Tail Race pool, Tail Race channel and Switch yard.

(ii) Land:

The uniqueness of this project is that, the forest is not destroyed in connection with its construction. No forest land is involved. Sanction was accorded from the Board to acquire 0.507 Ha of land for the project, vide BO dated 23.08.2016. The total extent of land used for the implementation of project is 4.877Ha, of which 4.423 Ha of land belongs to Irrigation Department. Permissive sanction and NOC for the construction of project in this land has been accorded from Govt. The private land used for the project is only an extent of 0.454 Ha and it has been acquired through negotiable purchase process.

(iii) Administrative & Technical Sanction

Administrative sanction was for Rs. 92.61Cr., vide BO dated 17.03.2020(Civil works: Rs.50.52Cr, E&M works: Rs.37.88Cr, Transmission works: Rs.4.21 Cr, TOTAL: Rs.92.61 Cr.), by revising the E&M cost alone.

(iv) Progress of Works

The Contract for the execution of E&M works of Peruvannamuzhy SHEP (2x3 MW 24.7 MU) was awarded to M/s Kirloskar Brothers Limited (KBL), Pune for the contract amount of Rs

29.39 Cr (Supply- Rs. 27.03 Cr., Service: Rs. 2.36 Cr.) and the work order was issued on 25.01.2021. The agreement was executed on 08.02.2021 with a completion period of 18 months up to 10.08.2022. The project was commissioned on 16.07.2023 and the DLP is up to 06-08-2026. The details are follows:

A. Date of completion of model test as per the approved schedule was 10.03.2023 for the first time for carrying out the model test, as the test bench for conducting the Model test of the turbine is only available during the request period.

B. M/s. KBL was granted time extension up to 30.06.2023 citing impact of Covid19 as the reason for delay the execution of E&M works of the project and conducting Model Test at IIT, Roorkee.

C. Erection work of Unit 1 was completed on 28.05.2023 and Unit II on 05.07.2023. Required water level was achieved only from 30.06.2023. Hence, the mechanical spinning of Unit 1 was done on 30.06.2023 and Unit II on 05.07.2023. The Unit 1 and Unit 2 were synchronized on 09/07/2023 and the commercial operation started on 16.07.2023 and 19.07.2023 respectively.

D. The project was officially inaugurated on 26.02.2024. Performance Guarantee Test was conducted by IIT Roorkee from 29-02-2024 to 02-03-2024 and the results were found satisfactory. The balance punch points with E&M contractor, M/s KBL was also completed on 26-07-2024.

(v) Final Project Cost

The actual project cost after COD is Rs.109.34 Cr of which Rs 104.34cr has been capitalized till this financial year.

2.5 Auxiliary Consumption:

2.5.1 As per Tariff regulation, 2021 the auxiliary consumption of the generating station includes auxiliary energy consumption of a generating station or a generating unit means the quantum of energy consumed by auxiliary equipment of the generating station or generating unit including switchyard of the generating station and the transformer losses within the generating station or generating unit. The metered energy for auxiliary consumption does not include the energy consumption of excitation system, generator transformer (transformer losses). Hence KSEBL has worked out a methodology to capture actual auxiliary consumption of power station and its switchyard.

Table – G4: Energy Generation from own stations for FY 2023-24 (MU)

Source	KSERC approved Gross Generation	KSERC approved Aux Generation	Aux % of Gross Generation	Truing up values for Gross Generation	Truing up values for AuX Generation	Truing up Aux % of Gross Generation	Difference in percentage
Hydro	7,163.06	85.85	1.20%	5663.67	63.41	1.12%	-0.08%
BDPP	0.00	0.00	0.00%	0.00	0.00	0.00%	0.00%
KDPP	0.00	0.00	0.00%	0.36	0.36	0.00%	0.00%
Wind	0.00	0.00	0.00%	1.21	0.00	0.09%	0.09%
Solar	72.68	0.18	0.25%	55.93	0.00	0.00%	-0.25%
Gross Subtotal	7,235.74	86.03	1.19%	5721.17	63.77	1.11%	-0.07%

2.5.2 The auxiliary consumption approved by the Hon'ble commission during the FY 2023-24 is 86.03 MU consisting of 85.85 MU for hydel stations and 0.18 MU for solar stations. But as per the audited account, the actual auxiliary consumption is 63.77 MU, out of which 63.41 MU is auxiliary consumption of hydel stations.

2.5.3 From the table above, it can be seen that the actual generation is less than the approved generation. Consequently, there is a decline in the actual auxiliary consumption. The details of gross generation, percentage of auxiliary consumption along with trued up gross generation and auxiliary consumption are shown in the table G4 above. On comparing approved percentage of auxiliary generation with that of trued up, it is seen that percentage auxiliary consumption is less than that of the approved value and is reasonable.

2.5.4 It is further submitted that the auxiliary energy consumption inside any generating station can be classified into three categories, viz.

(a) General auxiliaries
(b) Operational auxiliaries
(c) Maintenance auxiliaries.

2.5.5 The general auxiliaries include, the auxiliary consumption of lighting loads, dewatering pumps, firefighting system etc. These auxiliaries will be in service for all the time irrespective of actual generation. The power consumption of such auxiliaries is designed during the initial stages of powerhouse design hence further optimisation on such power consumption is very difficult.

2.5.6 The general auxiliaries include, the auxiliary consumption of lighting loads, dewatering pumps, firefighting system etc. These auxiliaries will be in service for all the time irrespective of actual generation. The power consumption of such auxiliaries is designed during the initial stages of powerhouse design hence further optimisation on such power consumption is very difficult.

2.5.7 The operational auxiliaries mainly contain lubricating oil system, high pressure oil pumps, cooling water system, compressed air oil system etc. The energy consumption of such equipment's will be dominant during the period of generation.

2.5.8 The maintenance auxiliaries including oil purification equipments, turbine dewatering system, lifting devices etc. Such auxiliaries are essential for daily/monthly/annual maintenance service and the consumption of such auxiliaries depends on its rating, which are finalised in the design manufacture.

2.5.9 Hence during generation period, the general auxiliaries, operational auxiliaries, and maintenance (daily/monthly) will be in service and during weak generation period, the general auxiliaries and maintenance (daily/monthly/yearly) will be dominant. As the auxiliary consumption is measured in terms of gross generation during the year, if the gross generation is lower, then the percentage of auxiliary consumption in relation to such reduced generation would be on a higher side and vice versa. This pattern is more evident in the case of small hydro stations, where the generation is possible only during monsoon period.

2.5.10 In the case of thermal stations, it may kindly be noted that the auxiliary equipments such as Lube oil circulating pump, Compressors, Air Conditioning systems, Battery chargers, Fire system pumps, general lighting etc consumes power even without generation on up keeping of machineries as recommended by the Original Equipment Manufacturer (OEM). The auxiliary energy consumption of such devices is unavoidable for the upkeeping the generator.

2.5.11 The following auxiliary equipment consumes power even without generation as the part of up keeping of machineries as recommended by the OEM.

(a) Lube oil circulating pump - Needs to operate once weekly
(b) Jacket water Circulating Pumps - Needs to operate once weekly
(c) Auxiliary boiler and all its accessories once monthly
(d) Compressors once weekly
(e) Air Conditioning systems
(f) Battery chargers 175A/110V will be in continuous operation
(g) Station transformers in no load
(h) Fire system pumps once weekly
(i) Water Pumps, lighting system
(j) Slow barring of engines once monthly

2.5.12 Hon'ble Commission may kindly note that the auxiliary energy consumption of such devices is unavoidable for the upkeeping of the generator.

Table-G7: Station Wise Hydel Generation for 2023-24

No	Hydro Electric Stations	Generation (MU)	Aux Generation	Net Generation	% Aux with Gross Gen
1	Idamalayar	198.20	2.29	195.90	1.16%
2	Idukki	1644.53	19.77	1624.76	1.20%
3	Kakkad	189.15	2.00	187.15	1.06%
4	Kallada	60.56	0.46	60.10	0.77%
5	Kuttiyadi	98.34	0.67	97.67	0.68%
6	KES	152.32	1.74	150.58	1.14%
7	KAES	271.94	2.88	269.05	1.06%
8	KuttiyadiTail Race	5.45	0.06	5.39	1.08%
9	Lower Meenmutty	4.64	0.09	4.55	1.93%
10	Lower Periyar	489.67	5.73	483.95	1.17%
11	Maduppatty	2.34	0.03	2.31	1.26%
12	Malampuzha	1.00	0.02	0.98	1.57%
13	Malankara	24.94	0.26	24.69	1.03%
14	Neriamangalam	182.62	2.22	180.41	1.21%
15	NES	110.52	1.29	109.24	1.16%
16	Pallivasal	157.32	1.11	156.21	0.71%
17	Panniyar	164.71	1.93	162.78	1.17%
18	Peppara	5.61	0.01	5.60	0.16%
19	Poringalkuthu (PLB)	96.12	1.12	95.00	1.17%
20	PLBE	65.38	0.79	64.60	1.20%
21	Poringalkuthu new	88.72	1.03	87.68	1.16%
22	Sabarigiri	1130.92	12.21	1118.71	1.08%
23	Sengualm	132.10	1.62	130.49	1.22%
24	Vellathhoval SHEP	5.50	0.10	5.40	1.86%

No	Hydro Electric Stations	Generation (MU)	Aux Generation	Net Generation	% Aux with Gross Gen
25	Sholayar	250.07	2.99	247.08	1.20%
26	Chembukadvu - I	3.26	0.02	3.25	0.56%
27	Chembukadavu- II	4.99	0.03	4.96	0.53%
28	Urumi - I	5.27	0.03	5.24	0.57%
29	Urumi - II	0.60	0.04	0.57	5.98%
30	Poozhithode	9.77	0.04	9.72	0.45%
31	Ranni-Perinadu	8.02	0.15	7.88	1.82%
32	Peechi	1.00	0.04	0.97	3.53%
33	Vilangadu	14.30	0.08	14.23	0.53%
34	Chimmony	2.33	0.02	2.31	0.85%
35	Addyanpara	7.70	0.03	7.66	0.45%
36	Barapole	35.07	0.15	34.92	0.42%
37	Perunthenaruvi	8.10	0.09	8.01	1.08%
38	Kakkayam	6.86	0.11	6.75	1.59%
39	Chathankottunada	8.77	0.08	8.69	0.91%
40	Upper Kallar	4.70	0.01	4.69	0.27%
41	Poringal kuthu screw generation	0.05	0.00	0.05	0.00%
42	Peruvannamuzhy SHEP	10.19	0.10	10.09	0.99%
	Sub total	**5663.67**	**63.41**	**5600.26**	**1.12%**

2.6 Thermal Generation:

2.6.1 No thermal generation is reported for the year 2023-24. Due to high variable cost, thermal generation has been kept to the bare minimum. KSEBL managed the grid without resorting to thermal Generation. But upkeeping of the machines were carried out properly as per the OEM recommendation for meeting unexpected contingencies. As the thermal generating station was operated only for meeting the contingencies and KSEBL has strictly adhere to the merit order despatch strictly, as directed by the Hon'ble Commission in the tariff order dated 25-06-2022, it is submitted that the above thermal generation may be approved.

2.7 Solar & Wind projects:

2.7.1 Hon'ble Commission has approved energy generation from solar to the tune of 72.68 MU. No generation was approved from Wind for the FY 2023-24. But the actual generation from Solar is only 55.93 MU. Due to technical constraints, maintenance activities, weather conditions etc, the anticipated quantum of power could not be generated, in the case of Solar.

2.8 ARR ERC of SBU- G:

2.8.1 The actual ARR, ERC for the Generation Business Unit is furnished in **Form G-P&L.** A statement showing comparison of ARR & ERC approved by the Hon'ble Commission and actual for the

year as well as true up sought along with deviations are summarized below. Item-wise explanation is furnished thereafter. A gist of reasoning is provided hereunder.

TABLE – G8: ARR OF GENERATION BUSINESS UNIT (SBU-G) for 2023-24 (Rs Cr)					
No	**Particulars**	**ARR Approval**	**Actuals**	**TU Sought**	**Difference over approval**
1	Cost of Generation of Power	0.00	0.34	0.34	0.34
2	O&M Expenses for existing stations	170.38	291.33	175.66	-16.62
	O&M Expenses for new stations	35.70		13.80	
	O&M Expenses - Total	206.08		189.46	
3	Interest & Finance Charges	168.90	120.03	155.96	-12.94
4	Depreciation	166.31	773.85	159.67	-6.64
5	Repayment of master Trust Bond	21.99	0.00	21.99	0.00
6	Additional contribution to Master Trust	21.60	0.00	21.60	0.00
7	Amortisation of intangible assets	0.00	1.20	1.20	1.20
8	RoE	116.38	116.38	116.38	0.00
9	Exceptional Items		0.18	0.18	0.18
10	Others		0.07	0.07	0.07
11	ARR	701.26	1303.22	666.69	-34.57
12	Less Non-Tariff Income	10.81	33.21	40.21	29.40
13	**Net ARR** (Transferred to SBU-D)	690.45	1270.01	626.48	-63.97

2.8.2 It may kindly be seen that even though the actual net ARR is Rs 1270.01, which is more than the approved net ARR of Rs. **690.45** Cr, the amount sought for true up is only Rs. **626.64 Cr** which is lesser than the approved figure by Rs **63.97 Cr.**

2.9 Cost of generation of power

2.9.1 Rs 0.34 Cr was actually incurred during the year towards the cost of fuel and lubricants for generation of stations owned by KSEBL, which may please be approved. Lubricating systems is an essential part of powerplant maintenance. Any piece of equipment with moving parts depends on some type of lubricant to reduce friction and wear and to extend its life. In most cases, a lubricant also dissipates heat, prevents rust or corrosion, acts as a seal to outside contaminants, and flushes contaminants away from bearing surfaces. The hydraulic oil is used for the purpose of governor action for speed /load control.

2.9.2 Further, the diesel oil is used for boiler operation and lube oils are used for bleeding purpose in connection with regular routine maintenance to upkeep the machines of KDPP

in working condition. Hence governor oil and lubricating oils are very much essential and hence such cost may please be approved. The split-up details are given below:

Table–G9(A) : Cost of generation (Rs Cr)				
Particulars	KDPP	BDPP	Hydro stations	Total
Oil	0.01			0.01
HSD oil	0			0
Lub oil	0.02		0.05	0.07
Lubricants and consumable stores	0		0.26	0.26
Total	**0.03**	**0**	**0.31**	**0.34**

Table–G9(B) : Cost of generation (Rs Cr)						
Station wise Cost (Rs Cr)						
Station	**Heavy Fuel Oil (Furnace Oil, LSHS, Etc.)**	**H.S.D. Oil**	**Lub. Oil**	**For Hydel Power Generation**	**For Internal Combustion Power Gen.**	**TOTAL**
KDPP	0.01		0.02			0.03
Generation Circle, Moozhiyar				0.17		0.17
Generation Circle, Moolamattom			0.05	0.07		0.12
Generation Circle, Meencut						0
Generation Circle, Kothamangalam				0.02		0.02
Generation Circle, Thrissur						0
Total	**0.01**		**0.07**	**0.26**		0.34

2.10 Interest and Finance charges:

2.10.1 Actual Interest and finance charges for the year 2023-24 is Rs. 120.03 Cr as against the approved figure of Rs 168.90 Cr. An amount of Rs.155.96 Cr is claimed under the head for true up. The details are given below.

Table G 10: Interest and Finance charges (Rs Cr)					
No	**Particulars**	**Approved**	**Actual**	**TU**	**Difference**
1	Interest on Capital Liabilities	118.52	109.66	104.46	14.06
2	Interest on GPF	12.73	9.94	9.94	2.79
3	Interest on Working capital	6.87	5.25	9.21	-2.34
4	Interest on Master Trust Bonds	30.78	31.88	31.88	-1.10
5	Other Interests	0.00	0.59	0.47	-0.47
6	Sub Total	168.90	157.32	155.96	12.94
7	Less: Capitalized	0.00	37.29	0.00	0.00
8	**Balance**	**168.90**	**120.03**	**155.96**	**12.94**

2.10.2 **Interest on normative loan**: Regulation 26 and 29 of Tariff Regulation 2021 specifies the approval of loan Interest on normative basis. Based on this, while issuing orders on MYT petition, Hon'ble Commission approved Rs.1037.24 Cr as opening normative loan as at 01.04.2023. After considering provisional asset addition, contribution and grants etc., during the year 2023-24 and considering interest rate at 9.87%, the Hon'ble Commission had approved a normative interest of Rs. 118.52 Cr for 2023-24. Actual interest amount is Rs. 109.66 Cr as explained in chapter 5. Hon'ble Commission may kindly approve an amount of Rs. 104.46 Cr towards interest on normative loan for the year 2023-24.

2.10.3 **Interest on PF:** Hon'ble Commission approved Rs 12.73 Cr towards interest on PF. But as per the audited accounts, the actual interest paid on PF was Rs.9.94 Cr, which is less than the approval by Rs 2.79 Cr. The actual interest for SBU- G amounting to Rs.9.94 Cr may kindly be approved.

2.10.4 **Interest on bond issued to Master Trust:** State Government, as per notifications dated 31.10.2013 and 28.01.2015, ordered creation of a Master Trust for meeting the unfunded liability of pension, gratuity and leave surrender as on 31.10.2013, in respect of the personnel transferred from erstwhile KSEB to KSEBL. The total liability as on 31.10.2013 was estimated at Rs.12418.72 Cr and necessary funding arrangements put in place through issue of 2 series of Bonds. Hon'ble Commission recognized the unfunded pension liabilities as above and approved recovery of interest on KSEBL share of Bonds as per Tariff Regulations, 2021. The Hon'ble Commission approved Rs.30.78 Cr towards interest on Master Trust bonds. As detailed in chapter 5, the share of expenses for SBU G has been Rs. 31.88 Cr, which may kindly be approved.

2.10.5 **Interest on Additional Bond to Trust:** In addition to the interest on bonds actually issued, an additional amount of Rs.400 Cr (Rs. 30.78 Cr for SBU G) has been provisionally approved towards unfunded liability. Hon'ble Commission may kindly true up the amount of Rs.31.88 Cr as detailed in chapter 5.

2.10.6 **Interest on working capital**: Regulation 32(2) of tariff regulation 2021 allows interest on normative level of working capital at 2% higher than the base rate on the first rate of April of the year. Further as per regulation 3(12) of the tariff regulation 2021 base rate means the external bench mark linked lending rate(EBLR) declared by the State Bank of India as applicable on the first day of the respective FY. Base rate of SBI on 01.04.2023 has been 9.15% and therefore the normative interest rate for the computation of the working capital interest for the FY 2023-24 would be 11.15% (EBLR +2%)

As per Tariff regulation, 2021, KSEBL is eligible for a working capital requirement of Rs. 82.60 Cr and interest on working capital of Rs 9.21 Cr, as detailed below, which may kindly be approved as against Rs .84.49 Cr approved in the MYT order dated 25.06.2022.

Table G11-Interest on working capital – SBU G (Rs Cr)

Sl. No	Particulars	Amount
1	Cost of fuel for FY 2022-23	0.34
2	**Cost of fuel for one month**	**0.03**
	O&M expenses	
i	As per norms	175.66
ii	O&M Expenses for new stations	13.80
iii	Pay revision expenses	0.00
iv	Repayment of master Trust Bond	21.99
v	Additional contribution to Master Trust	21.60
3	Total (I to v)	233.05
4	**O&M Expenses for one month**	**19.42**
5	Approved GFA - Plants and Equipment for FY 2023-24	6315.58
6	**Cost of maintenance of spares at 1% of historical cost**	**63.16**
7	**Total requirement of working Capital=(2)+(4)+(6)**	**82.60**
8	Base rate as on 01-04-2023 (%)	9.15
9	Interest rate on working capital (%)=Base rate+2	11.15
10	**Interest on working capital = (7)*(9)**	**9.21**

2.10.7 In view of the above, the Hon'ble Commission may kindly true up interest and finance charges of Rs. **155.96** Cr as detailed in Table G 10.

2.11 O&M of SBU-G:

2.11.1 The Hon'ble commission had approved O&M cost of 170.38 Cr for existing stations and an amount of Rs 35.70 Cr. For new stations. It was arrived using the average yearly escalation specified in the Tariff regulation, 2021. But it has been expressly provided that

actual inflation for the year can be considered while true up. Thus, total approved O&M expenses for 2023-24 has been Rs. 206.08 Cr. Actual escalation rate for the year 2023-24 is 3.41%. Thus, eligible O&M expenses for the existing stations for 2023-24 is Rs.175.66 Cr. and an amount of Rs 13.80 Cr is claimed for new stations in True-up

2.11.2 Only one small hydel project was commissioned during the FY 2023-24, viz. **Peruvannamoozhi** SHEP of 6 MW which was commissioned on 07.06.2023. The O&M expense for small hydel projects less than 5 MW for the financial year 2023-24 as per the RE regulation 2020 is Rs 0.8 Cr.

2.11.3 Solar projects were also commissioned during the FY 2023-24 by adding a capacity of 2.06 MW. The O&M cost for the solar projects under SBUG for the FY 2023-24 amounts to Rs.2.55 Cr as shown in Annexure C.

2.11.4 In addition to these, projects completed after the base year of 2018-19 was also taken as it was not included for calculating the O&M cost for the existing stations. The table is shown in the Annexure B. The O&M expense for small hydel projects above 5MW is Rs 10.45 Cr.

2.11.5 The eligible O&M expense for the existing stations and new stations for the year 2023-24 is Rs.189.46Cr (Rs. 175.66 Cr + Rs. 13.80 Cr) as shown in Annexure C.

2.11.6 Actual O&M expenses of the Generation Business Unit was **Rs. 291.33 Cr.** Hon'ble Commission had approved **Rs 206.08 Cr** towards O&M expenses based on the norms specified in the Tariff Regulation, 2021. However, based on the commissioning of the new generating stations, a claim of Rs. 189.46 Cr is sought for true up for the year 2023-24. It is humbly submitted that the claim towards additional contribution to Master trust are claimed separately as detailed in chapter 5.

Table G 13 Details of O&M expenses for 2023-24 (Rs Cr)

No	Particulars	Approved	Actual	TU requirement	Difference
1	Employee Cost		220.12		
2	A&G Expenses		48.32		
3	R&M Expenses		22.89		
4	**Total**	**206.08**	**291.33**	**189.46**	**(-) 16.62**

2.11.7 Detailed explanation on the O&M expenses is furnished in Chapter 5 of this petition. It is humbly prayed that the Hon'ble Commission may be pleased to approve the O&M expense as claimed above.

2.12 Depreciation:

2.12.1 The approved depreciation for the Generation assets for the year 2023-24 was Rs 166.31 Cr whereas the actual amount was Rs. 773.85 Cr. Higher amount of actual depreciation is on account of provision of depreciation for assets revaluated at the time of revesting of assets and liability of KSEB to KSEBL. As per MYT Regulation for the control period

2022-27, depreciation is allowable as per the provisions contained in the Tariff Regulations, in which applicable rates as well as methodology to be followed are specified therein. Accordingly, allowable depreciation for SBU G has been worked out at **Rs. 159.67 Cr** as detailed in Chapter 5. Hon'ble Commission may kindly approve the same. Detailed explanation is furnished in chapter 5.

2.13 Amortisation of intangible assets towards software development cost

2.13.1 KSEBL has developed its own software for major functional areas such as human resource, accounting, billing viz ERP, HRIS, ORUMANET etc. The employee cost attributable to the software developments were amortised to the tune of Rs 1.20 Crs during 2023-24

2.14 Return on Equity:

2.14.1 Hon'ble Commission has approved Return on equity at Rs.116.38 Cr. Actual RoE for the Generation SBU as per audited accounts amounts to Rs. 116.38 Cr. Hon'ble Commission may kindly approve Rs.116.38 Cr as detailed in Chapter 5.

2.15 Non-Tariff Income:

2.15.1 Hon'ble Commission approved Rs. 10.81 Cr as non-tariff income in MYT order. Actual income earned as per accounts has been Rs.33.21 Cr. SBU G seeks to true up Rs.40.21 Cr under the head non-tariff income as detailed in chapter 5 after excluding income booked under claw back depreciation in accounts. The non-Tariff income includes income from sale of scrap, interest on advances to contractors, interest on staff loans and advances, rent from buildings etc. Hon'ble Commission may kindly true up Rs.40.21 Cr under this head.

2.16 Other items and Prior period expenditure:

2.16.1 Other expenses amounted of Rs.0.07Cr for the year 2023-24 as detailed in chapter 5, which may kindly be approved.

2.17 Exceptional items:

2.17.1 Share of SBU G as detailed in chapter 5 amounts to Rs.0.02 Cr. Hon'ble Commission may kindly true up the claim.

2.18 Transfer of Generation Costs:

2.18.1 ARR for SBU G sought for true up is **Rs. 666.69 Cr** and Non-tariff income is trued up for an amount of **Rs. 40.21 Cr**. The net ARR of SBU G transferred to SBU-D is **Rs. 626.48 Cr.** This cost is considered as the Cost of own generation in the ARR of SBU-D, which may kindly be approved.

2.19 ARR & Tariff Formats:

2.19.1 Formats relating to Generation Business Units as per the Tariff regulation 2021 are furnished hereafter. However, Forms G-3.1, G-6.5 to G-6.8, G-6.10 to G-7.1 which are not directly relevant or applicable or on which data are not readily available are not included in this submission.

2.20 The Next Chapter thereafter provides the truing up of Transmission Business Unit (SBU – T).

Annexure - A

2023-24

POWER PROJECTS IN KERALA as on March 2024 (31-03-2024)

Major Hydro Electric Projects						
Sl. No	Name of Station	Installed Capacity of the plant		Annual Generation Designed MU	Date of Commissioning	
		No. of Generators. X Capacity in (MW)	Plant Capacity in (MW)		Original date	After Renovation
I	**HEPs**					
1	Idukki	6 x 130	780.00	2398	12.02.1976/07.06.1976/24.12.1976/04.04.1985/22.03.1986/09.09.1986	(First stage three machines renovated)14.07.2020/18.01.2020/ 16.03.2019
2	Sabarigiri	4 x 55 + 2x 60	340.00	1338	18.04.1966/14.06.1966/29.12.1966/22.06.1967/09.09.1967/26.11.1967	03.12.2009/07.02.2009/17.03.2008/06.05.2014/05.05.2006/01.07.2005
3	Idamalayar	2 x 37.5	75.00	380	03.02.1987/28.02.1987	
4	Sholayar	3x18	54.00	233	09.05.1966/26.01.1968/14.05.1968	05.10.2020 /19.01.2020/18.09.2019
5	Pallivasal	3 x 5+ 3x7.5	37.50	284	19.03.1940/02.02.1941/10.02.1942/01.05.1948/01.10.1949/07.03.1951	19.11.2001/17.11.2001/20.11.2001/26.08.2002/21.08.2002/19.08.2002
6	Kuttiyadi	3x25	75.00	268	11.09.1972/01.11.1972/28.11.1972	
7	Kuttiyadi Extension	1x50	50.00	75	27.01.2001	
8	Kuttiyadi Additional Exten.Scheme	2x50	100.00	223	11.10.2010/30.10.2010	
9	Neriamangalam	3 x17.5	52.50	237	27.01.1961/09.04.1961/11.05.1963	20.10.2004/31.11.2005/29.09.2006
10	Lower Periyar	3 x 60	180.00	493	27.09.0997/22.10.1997/28.11.1997	
11	Poringalkuthu	4x9	36.00	191	06.03.1957/13.01.1958/24.04.1959/06.02.1960	14.04.2014/15.08.2014/08.02.2015/29.05.2015
12	Sengulam	4 x 12.8	51.20	182	01.05.1954/25.071954/ 11.11.1954/05.01.1955	30.11.2002/04.10.2002/05.12.2001/01.12.2001
13	Kakkad	2x25	50.00	262	13.10.1999	
14	Panniar	2 x 16.2	32.40	158	29.12.1963/26.01.1964	01.02.2003/23.11.2001
15	Neriamangalam Extn Scheme (NES)	1x25	25.00	58.27	27.05.2008	
	Sub Total (HEP)		**1938.60**	**6780.27**		

Small HEPs						
Sl.No	Name of Station	Installed Capacity of the plant		Firm annual Gen. Capability	Annual Generation Designed MU	Date of Commissioning
		No. of Generators. X Capacity in (MW)	Plant Capcity in (MW)	MU		
1	Poringalkuthu Micro (Screw Turbine)	1x 0.011	0.01	0.082	0.082	01.02.2016
2	Peechi	1x1.25	1.25	3.21	3.21	07.01.2013
3	Madupatty	1x2	2.00	6.40	6.4	14.01.1998
4	Upper Kallar	2x1	2.00		5.14	**30.09.2021**
5	Urumi -2	3x0.8	2.40	6.28	6.28	25.01.2004
6	Malampuzha	1x2.5	2.50	5.60	5.6	26.11.2001
7	Chimmony	1x2.5	2.50	6.70	6.7	22.05.2015
8	Chembukadavu - 1	3x0.9	2.70	6.59	6.59	19.08.2003
9	Peppara	1x3	3.00	11.50	11.5	15.06.1996
10	Kakkayam SHEP	2x1.5	3.00		10.39	16.07.2018
11	Lower Meenmutty	(1x0.5 + 2x1.5)	3.50	7.63	7.63	25.03.2006
12	Adyanpara	2x1.5+0.5	3.50	9.01	9.01	09.03.2015
13	Vellathooval	2x1.8	3.60	12.170	12.7	08.09.2016
14	Chembukadavu - 2	3x1.25	3.75	9.03	9.03	04.09.2003
15	Urumi -1	3x1.25	3.75	9.72	9.72	25.01.2004
16	KTR	3x1.25	3.75	17.01	15	19.06.2008/ 22.06.2008/ 23.10.2009
17	Ranni-Perinadu	2x2	4.00	16.73	16.73	16.02.2012
18	Poozhithode	3 x 1.6	4.80	10.97	10.97	25.06.2011
19	Perunthenaruvi	2x3	6.00	25.770	25.77	24.10.2017
20	Chathankottunada Stage-II	3x2	6.00		14.76	**25-06-2021**
21	Peruvannamoozhy	2x3	6.00		24.7	**19-07-2023**
22	Vilangad	3x2.5	7.50	22.63	22.63	09.01.2014
23	Malankara	3x3.5	10.50	65.00	65	23.10.2005
24	Kallada	2x7.5	15.00	65.00	65	05.09.1994/ 05.01.1994
25	Barapole	3x5	15.00	36.00	36	29.02.2016
26	PLBE	1x16	16.00	74.00	74	20.03.1999
27	Poringalkuthu 24MW	1x24	24.00		45.02	**04.05.2022**
	Sub Total (SHEP)		**158.01**	**427.03**	**525.562**	

Small HEPs greater than 5 MW						
Sl.No	Name of Station	Installed Capacity of the plant		Firm annual Gen. Capability	Annual Generation Designed MU	Date of Commissioning
		No. of Generators. X Capacity in (MW)	Plant Capcity in (MW)	MU		
1	Perunthenaruvi	2x3	6.00	25.770	25.77	24.10.2017
2	Chathankottunada Stage-II	3x2	6.00		14.76	**25-06-2021**
3	Peruvannamoozhy	2x3	6.00		24.7	**19-07-2023**
4	Vilangad	3x2.5	7.50	22.63	22.63	09.01.2014
5	Malankara	3x3.5	10.50	65.00	65	23.10.2005
6	Kallada	2x7.5	15.00	65.00	65	05.09.1994/ 05.01.1994
7	Barapole	3x5	15.00	36.00	36	29.02.2016
8	PLBE	1x16	16.00	74.00	74	20.03.1999
9	Poringalkuthu 24MW	1x24	24.00		45.02	**04.05.2022**
	Sub Total (SHEP)		**106.00**	**288.40**	**372.88**	

Small HEPs less than 5MW

Sl. No	**Name of Station**	**Installed Capacity of the plant**		**Firm annual Gen. Capability MU**	**Annual Generation Designed MU**	**Date of Commissioning**
		No. of Generator X Capacity in (MW)	**Plant Capcity in (MW)**			
1	Poringalkuthu Micro (Screw Turbine)	1x 0.011	0.01	0.082	0.082	01.02.2016
2	Peechi	1x1.25	1.25	3.21	3.21	07.01.2013
3	Madupatty	1x2	2.00	6.40	6.4	14.01.1998
4	Upper Kallar	2x1	2.00		5.14	**30.09.2021**
5	Urumi -2	3x0.8	2.40	6.28	6.28	25.01.2004
6	Malampuzha	1x2.5	2.50	5.60	5.6	26.11.2001
7	Chimmony	1x2.5	2.50	6.70	6.7	22.05.2015
8	Chembukadavu - 1	3x0.9	2.70	6.59	6.59	19.08.2003
9	Peppara	1x3	3.00	11.50	11.5	15.06.1996

Sl. No	Name of Station	Installed Capacity of the plant		Firm annual Gen. Capability MU	Annual Generation Designed MU	Date of Commissioning
		No. of Generator X Capacity in (MW)	Plant Capcity in (MW)			
10	Kakkayam SHEP	2x1.5	3.00		10.39	16.07.2018
11	Lower Meenmutty	(1x0.5 + 2x1.5)	3.50	7.63	7.63	25.03.2006
12	Adyanpara	2x1.5+0.5	3.50	9.01	9.01	09.03.2015
13	Vellathooval	2x1.8	3.60	12.170	12.7	08.09.2016
14	Chembukadavu - 2	3x1.25	3.75	9.03	9.03	04.09.2003
15	Urumi -1	3x1.25	3.75	9.72	9.72	25.01.2004
16	KTR	3x1.25	3.75	17.01	15	19.06.2008/22.06.2008/23.10.2009
17	Ranni-Perinadu	2x2	4.00	16.73	16.73	16.02.2012
18	Poozhithode	3 x 1.6	4.80	10.97	10.97	25.06.2011
	Sub Total (SHEP)		**52.01**	138.63	**152.682**	

Thermal Plants						
Sl. No	Name of Station	Installed Capacity of the plant		Firm annual Gen. Capability	Annual Generation Designed MU	Date of Commissioning
		No. of Generators. X Capacity in (MW)	Plant Capacity in (MW)	MU		
2	KDPP	6x16	96.00	672	672	~~01.09.1999~~/11.09.1999/18.09.1999/~~23.09.1999~~/30.09.1999/11.09.1999/25.10.1999/06.11.1999
	Sub Total (Thermal)		**96.00**	**672.00**	**672.00**	
Wind farm						
1	Kanjikode Wind Farm	9x0.225	**2.03**	4	**4**	18.05.1995

Annexure – B

Consolidated report of KSEB own Soura Solar Projects						
Sl No	**FY**	**Capacity addition during that FY(from all solar projects of KSEB MW**	**Total Capacity at the end of the FY MW**	**KSEB investment in Lakhs**	**Subsidy/ Grant portion in Lakhs**	**Total Investment in Lakhs**
1	2018-19	1.37	15.25	826.43	161.16	987.59
2	2019-20	1.17	16.42	403.30	115.66	518.96
3	2020-21	0.00	16.42	529.63	0.00	529.63
4	2021-22	13.86	30.28	5817.54	28.30	5856.26
5	2022-23	17.26	47.54	4432.35	202.64	4636.83
6	2023-24	2.06	49.60	3566.68	136.50	3703.18

Normative O&M cost allowable as per RE Regulations, 2020 for RE projects									
Particulars		**Base level O&M cost for 2019-20 (Rs. Lakh/ MW)**	**2020-21**	**2021-22**	**2022-23**	**2023-24**	**2024-25**	**2025-26**	**2026-27**
					(Rs. Lakh/MW) with Escalation factor of 5.72%				
ARR Approved values	**(i) Capacity below 5 MW**	32.41			38.3	40.49	42.8	45.25	47.84
	(ii) Capacity above 5 MW to 25 MW	23.47			27.73	29.32	31	32.77	34.64
	Solar projects	6			7.09	7.5	7.92	8.38	8.86
Actual Escalation factor			**3.91%**	**7.48%**	**7.06%**	**3.41%**			
Actual values	**(i) Capacity below 5 MW**	32.41	33.68	36.20	38.75	40.07			
	(ii) Capacity above 5 MW to 25 MW	23.47	24.39	26.21	28.06	29.02			
	Solar projects	6.00	6.23	6.70	7.17	7.42			

Annexure - C

Normative O&M cost allowable as per RE regulations 2020 for RE projects for new projects after the base year 2018-19						
Sl No	**Particulars**	**Total Installed Capacity in the FY 2023-24 in MW**	**Capacity in the FY 2018-19 MW**	**Capacity addition after 2018-19 base year MW**	**Normative O&M cost for the FY 2023-24 in Rs Lakhs/MW**	**O&M cost applicable for the FY 2023-24 in Cr**
1	SHEPs Below 5 MW	52.011	50.011	2	40.07	0.80
2	SHEPs above 5 MW	106	70	36	29.02	10.45
3	Solar Projects	49.60	15.25	34.35	7.42	2.55
	Total RE O&M Cost for new SHEP and Solar					**13.80**

SBU-G: O & M for Existing stations									
Year	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27
ARR escalation factor	4.84%				4.45%				
SBU-G O&M	135.51	142.07	148.95	156.16	163.12	170.38	177.97	185.90	194.18
Actual escalation factor			3.91%	7.48%	7.06%	3.41%			
TU O&M	135.51	142.07	147.62	158.67	169.87	**175.66**			

CHAPTER 3: TRUING UP OF ARR OF SBU – T & SLDC FOR 2023-24

3.1 The functions of the STU as envisaged in the Electricity Act is entrusted to the Strategic Business Unit – Transmission (SBU-T). As per the second Transfer Scheme, SBU-T is entrusted with the development and execution of long-term plans for evacuation and supply of power at voltages of and above 33KV, in consultation and co-ordination with the other SBUs. At present SBUT is also carrying out the functions of State Load Despatch Centre, as provided in Part-II of Schedule-A1 of the Second Transfer Scheme. As per the Second Transfer Scheme, all assets of voltage level of 66KV and above is assigned to SBU-Transmission. At the same time, SBU-T is entrusted with the construction of 33kV and above system. The voltage level wise SBU T capacity of Transmission system and number of substations and transmission lines at the beginning of the financial year, ie as on 1st April 2023 is summarized below:

Table-T 1: Transmission System statistics			
Sl. No	**Item**	**Unit**	**Quantity as on 01.04.2023**
1	400 kV Lines	Ckt-km	177.03
2	220 kV Lines	Ckt-km	3454.62
3	110 kV Lines	Ckt-km	5460.85
4	66 kV Lines	Ckt-km	1542.232
5	400 kV Substations	Nos	1
6	220 kV Substations	Nos	33
7	110 kV Substations	Nos	176
l8	66 kV Substations	Nos	54
9	Total Transmission Capacity	MVA	25344.5
10	Bays	Nos	2905

Projects commissioned during 2023-24

During the period 2023-24, following substations and lines were commissioned.

Table-T 2: Substation Commissioned during 2023-24				
Sl No	**Name of Substation**	**Voltage Level**	**New/ Upgradation**	**Date of Commissioning**
1	KOTTAYAM	400kV	New	13.05.2023
2	PALAKKAL	110kV	Upgradation	03.04.2023
3	KANNAMPULLY	110kV	Upgradation	26.10.2023
4	ADOOR	110kV	Upgradation	18.12.2023
5	EANATH	110kV	Upgradation	19.01.2024
6	THAMARASSERY	110kV	Upgradation	12.01.2024
7	KARUNAGAPPALLY	110kV	Upgradation	21.02.2024

Table-T 3: 400kV line Commissioned During 2023-24			
Sl No	**Name of line**	**Voltage Level**	**Date of Commissioning**
1	LILO of Tirunelveli-Kochi line	400kV	13.05.2023
Table-T 4: 220kV lines Commissioned During 2023-24			

Sl No	Name of line	Voltage Level	Date of Commissioning
1	From Kanjirod to Mundayad	220kV	01.11.2023

Table-T 5: 110 kV lines Commissioned During 2023-24			
Sl No	**Name of line**	**Voltage Level**	**Date of Commissioning**
1	Interlinking 110kV Substation, Kollam with 110kV Substation Kottiyam	110kV	13.05.2023
2	Mundayad-Azhikode	110kV	08.12.2022
3	Payyannur-Cheruvathur	110kV	14.07.2023
4	Cheruvathur-Ambalathara tap	110kV	17.06.2023 & 14.07.2023
5	Mangad-Thalipparamba	110kV	13.12.2023
6	Koodal-Pathanamthitta	110kV	05.10.2023.
7	Adoor-Pathanamthitta	110kV	03.12.2023.
8	Upgradation of Sasthamcotta -Karunagapally 66kV SC feeder to 110kV DC	110kV	21.02.2024
9	Upgradation of Kannampully-Nenmara 66kV line to 110kV	110kV	26.10.2023
10	Upgradation of 66 kV Thamarassery-Kunnamangalam SC line to 110 kV DC	110kV	04.11.2023

The summary of the asset created during 2023-24 is given below. Details of the capital works executed in 2023-24 are enclosed as Annexure- 1.

Table-T 6: Transmission System statistics - Asset created during 2023-24			
Sl. No	**Item**	**Unit**	**Asset Created during 2023-24**
1	400 kV Lines	Ckt-km	1.23
2	220 kV Lines	Ckt-km	18.12
3	110 kV Lines	Ckt-km	202.51
4	66 kV Lines	Ckt-km	-55.63*
5	400 kV Substations	Nos	1
6	220 kV Substations	Nos	0
7	110 kV Substations	Nos	6
8	66 kV Substations	Nos	-5
9	Total Transmission Capacity	MVA	785
10	Bays	Nos	24

* Upgradation of 66 kV line to 110 kV

The voltage level wise SBU T capacity of Transmission system and number of substations and transmission lines at the beginning of the financial year, ie as on 31st March 2024 is summarized below;

Table-T 7: Transmission System statistics			
Sl. No	**Item**	**Unit**	**Quantity as on 31.03.2024**
1	400 kV Lines	Ckt-km	178.256
2	220 kV Lines	Ckt-km	3472.74
3	110 kV Lines	Ckt-km	5663.36
4	66 kV Lines	Ckt-km	1486.603*
5	400 kV Substations	Nos	2
6	220 kV Substations	Nos	33
7	110 kV Substations	Nos	182
8	66 kV Substations	Nos	49
9	Total Transmission Capacity	MVA	26129.5
10	Bays	Nos	2929

* Reduction due to upgradation to 110 kV

3.2 The SBU-T handle total energy sold to consumers and licensees within the state, open access power drawn by the consumers, the quantum energy sold outside the state and energy transmitted by Captive Consumers. The ARR, ERC and Transfer cost to SBU-T are furnished as per Form T P&L and Form T1. Transfer price for SBU T as per accounts has been Rs. 1284.17 Cr against approval of Rs. 1533.35 Cr. Net ARR in line with the Tariff Regulation sought for true up in this petition amounts to Rs. 1553.14 Cr. Item wise explanation is furnished thereafter.

TABLE – T 8: ARR OF TRANSMISSION BUSINESS UNIT (SBU-T) & SLDC for 2023-24 (Rs Cr)				
No	**Particulars**	**Approved**	**Actual**	**Truing up requirement**
1	Interest on loan	316.61	144.78	350.82
2	Interest on GPF	26.50	18.85	18.85
3	Interest on Bonds issued to Master trust	64.13	61.17	61.17
4	Interest on Working Capital	23.99	11.85	30.86
5	Other interest	0.00	0.81	0.52
6	Interest & Finance Charge	**431.23**	**237.46**	**462.22**
7	Return on Equity (14%)	119.99	119.99	119.99
8	Depreciation	286.45	416.87	309.17
9	O&M expenses	644.81	588.95	625.20
10	Repayment of bond to master trust	45.81	0.00	45.81
11	Additional contribution to Master Trust	44.98	0.00	44.98
12	Edamon - Kochi line compensation	14.94	0.00	8.06
13	Pugalur – Thrissur line compensation	0.00	0.00	1.24
14	Amortising of intangible assets	0.00	1.20	1.20
15	Other expenses	0.00	0.56	0.56
16	Exceptional items	0.00	0.18	0.18
17	Incentive on Transmission Availability	0.00	0.00	10.49
19	**ARR**	1588.21	1365.21	1629.10
20	Less: Non-tariff Income	54.86	86.62	75.96
21	**Net ARR (Cost Transferred to SBU-D)**	**1533.35**	**1278.59**	**1553.14**

Interest and Finance charges:

3.3 An amount of Rs **431.23** Cr had been approved by the Hon Commission under this head. The actual expense incurred was **Rs. 237.46 Cr**. Item wise explanation is furnished below.

3.4 Interest on Normative loan: While issuing orders on MYT petition, Hon'ble Commission approved Rs. **2737.55** Cr as opening normative loan as on 01.04.2023. The interest was approved at 9.83%. Further, provisional approval of Rs. 1593.33 Cr was granted for the asset addition during 2023-24 with contribution and grant as Rs.340.23 Cr. Actual entitlement of normative loan as on 01.04.2023, GFA addition during 2023-24, repayment during the year and closing normative loan as on 31.03.2024, interest on normative loan along with SBU wise segregation are computed and explained in chapter 5. Based on the computation, an amount of Rs. 350.82 Cr is claimed towards interest on normative loan for the year 2023-24.

3.5 During the year, capital works executed by SBU-T was Rs.1305.84 Cr and Rs.625.41 Cr has been capitalized as per accounts. GFA addition includes decommissioning liability of Rs.0.12Cr. Out of the GFA addition during the year, assets worth Rs.18.98 Cr represent part capitalization within the meaning of Tariff Regulations. Further, Rs. 55.91Cr worth of part capitalized assets in earlier years were commissioned during the year. Thus, in order to arrive at the GFA addition allowable as per Tariff Regulations, 2021 part capitalized assets and decommissioning liability during the year were deducted from the GFA addition as per accounts. Thereafter, part commissioned assets in earlier years but commissioned in 2023-24 were considered to arrive at the asset addition eligible as per Tariff Regulation. This amounted to Rs.662.22 Cr as detailed in chapter 5.

3.6 Hon'ble Commission may kindly approve GFA addition of Rs.662.22 Cr for 2023-24. Consumer contribution, grants and subsidies received during the year amounted to Rs.46.10 Cr are as detailed **Appendix 5**. GFA addition to the tune of Rs. 616.12 Cr (Rs.662.22 Cr - Rs. 46.10 Cr) may kindly be approved as GFA addition eligible for normative loan and Rs. 614.36 Cr (excluding land value Rs. 1.76 Cr) as GFA eligible for depreciation.

3.7 Normative loan as on 31.03.2024 and normative interest for the year works out to Rs. 4176.72 Cr and Rs. 350.82 Cr respectively as detailed in chapter 5 of this petition may kindly be approved.

3.8 Interest on GPF: Interest paid on GPF during the year is as detailed in chapter 5. The share of interest on GPF for SBU-T along with approved figure is furnished below.

Table-T9: Interest on GPF for SBU-T for 2023-24

	Approved	Actual	True-up requirement
	Rs. Cr	Rs. Cr	Rs. Cr
SBU-T	26.50	18.85	18.85

3.9 Interest on Master Trust Bonds: Hon'ble Commission, by virtue of provision contained in Tariff Regulation, 2021 had approved Rs. 64.13 Cr towards interest on Master Trust bonds. Actual

expenses for SBU T have been Rs. 61.17 Cr as explained in Chapter 5. Thus, Hon'ble Commission may kindly approve Rs.61.17 Cr for the year 2023-24.

3.10 Additional contribution to Master Trust: Hon'ble Commission, by virtue of provision contained in Tariff Regulation, 2021 had approved Rs.44.98 Cr towards additional bond to Master Trust bonds. Hence Hon'ble Commission may kindly true up the Rs.44.98 Cr as detailed in chapter 5.

3.11 Interest on Working Capital: As per the Regulation 32 of Tariff Regulation 2021, interest on working capital is allowed normatively at a rate equal to two per cent higher than the base rate as on the first day of the financial year for which petition is filed. Hon'ble Commission approved the Interest on Working capital at a rate of 9.55% on the working capital of Rs.251.17 Cr and the interest on working capital approved for SBU-T for 2023-24 is Rs.23.99 Cr. Regulation 32(2) of Tariff Regulations 2021, allows Interest on normative level of working capital at two percent higher rate than the base rate as on first day of April of the year of ARR filing. Further, as per the Regulation 3(12) of the Tariff Regulations, 2021, Base rate" means the External Benchmark-Linked Lending Rate (EBLR) declared by the State Bank of India as applicable on first day of April of respective financial year. Base rate of SBI on 01.04.2023 has been 9.15% and therefore the normative interest rate for the computation of working capital interest for 2023-24 would be 11.15%. Based on the working capital requirement computed as per Tariff Regulation, a sum of Rs. 30.86 Cr is claimed as interest on working capital as detailed below, which may kindly be approved.

Table-T 10: Interest on working capital (Rs Cr)

Particulars	Approved	True up requirement
O&M Expenses As per norms	644.81	625.20
Additional contribution to Master Trust		44.98
Total O&M expenses		**670.18**
GFA Plants and Equipment	7059.84	**9241.16**
Transmission charges	1522.00	1542.64
O&M Expenses for one month	53.73	55.85
1% of Historical cost of plants & Equipment	70.60	92.41
Receivables (Transmission charges for one month)	126.83	128.55
Total requirement of working capital	251.17	**276.81**
Rate of interest	9.55%	11.15%
Interest on Working Capital	**23.99**	**30.86**

3.12 Other charges: Hon'ble Commission has not approved any amount towards other interest. Actual expenses incurred under this head for SBU-T amounts to Rs.0.52 Cr. Also, as per account Rs. 0.52 Cr incurred as other bank charges as detailed in chapter 5 and furnished below, may kindly be approved.

Table T11: Details of other charges under Interest and Finance Charges (Rs. Cr)		
No	**Particulars**	**SBU T**
1	Other Bank Charges	0.52
	Total	**0.52**

3.13 Summary of Interest & Finance charges: A comparison of the approved and actual values of various items of Interest & Finance charges are given below:

Table- T12: Summary of Interest and Finance charges (Rs Cr)				
No	**Particulars**	**Approved**	**Accounts**	**True-up requirement**
1	Interest on Outstanding Capital Liabilities	316.61	144.78	350.82
2	Interest on GPF	26.5	18.85	18.85
3	Interest on Master Trust Bonds	64.13	61.17	61.17
4	Interest on Working capital	23.99	11.85	30.86
5	Other charges	0	0.81	0.52
6	**Total**	**431.23**	**237.46**	**462.22**

3.14 O&M Expenses: As per the provisions of Regulation 58, the O&M expenses applicable for the SBU-T is based on the number of bays, MVA Capacity of the Power Transformer and transmission line length in ckt km. Further as per the regulation, The O&M expenses for any year of the Control Period shall be allowed by multiplying the O&M norms for that year with the actual number of bays existing at the beginning of the financial year, total MVA capacity of the transformers and transmission line length in circuit km at the beginning of the financial year and at 50 per cent of the circuit kilometre of transmission lines, MVA capacity of the transformers and number of Bays in the substation of the Transmission Licensee added during the year and put to its intended use. The no. bays, MVA capacity and circuit kilometres and the O&M cost based on the norms approved for 2023-24 and actual are as shown below:

Table - T 13: Approved and actual Cost Drivers of Transmission Business for 2022-23		
Item (66 kV & above)	**KSERC provisionally considered for 2022-23 end**	**Actual for 2022-23 end**
Substation Bays*	3065	2905
Tran Lines***	11773	10633.90
MVA Capapcity	24238	25344.5
* Excluding 33 kV bays	*** (in Ckt kms)	

Table - T 14: Approved and actual Cost Drivers of Transmission Business for 2023-24		
Item	**KSERC provisionally considered for 2023-24 addition**	**Actual for 2023-24 addition**
Substation Bays*	148	24
Tran Lines***	248	166.227
MVA Capapcity	1357	785
* Excluding 33 kV bays	*** (in Ckt kms)	

3.15 Even though the Hon'ble Commission in the Tariff Regulation considered an escalation rate of 4.454% for determining the norms for the control period 2022-27, Annexure 7 of Tariff

Regulation allows actual escalation rates based on actual inflation existing for each year of the control period at the time of truing up of the considered financial year. The O&M cost norm for SBU-T for 2023-24 as per the Regulation and the O&M cost based on the norms considering the actual inflation [7.06% for 2022-23 over to 2021-22 & 3.41% for 2023-24 over 2022-23] norms are submitted below.

Table T 15: O&M Norms		
Item	**Norms approved for 2023-24**	**Norms for 2023-24 (Arrived by applying actual inflation)**
Rs. lakh per Bay	7.770	7.885
Rs. lakh per MVA	0.860	0.873
Rs. lakh per circuit km	1.569	1.592

3.16 The O&M expense calculated based on norms for the period 2023-24 is Rs. 625.20 Cr. Against this, Rs. 619.51 Cr is for the asset at the beginning of the FY 2023-24 and Rs. 6.136 Cr is for the new asset created during the period 2023-24. The O&M details are tabulated below.

Table T 16: SBU T O&M Charges	
Item	**2023-24**
O&M cost for Bay (Rs lakh/Bay)	7.885
O&M cost (Rs lakh/circuit km)	1.59
O&M cost (Rs lakh/MVA)	0.873
Opening bays (Nos)	2905
Bays added during the year (Nos)	24
Opening line length (Circuit Km)	10633.90
Line length added during the year (Circuit Km)	166.227
Opening MVA	25344.5
MVA added during the year	785
For assets at the beginning of the year –(A) in Rs. Lakh	61949.59
For assets added during the year – (B) in Rs. Lakh	569.41
Total Normative O&M expenses in Rs. Lakh	62519.32
Total Normative O&M expenses – (A) + (B) Rs. Cr	**625.20**

3.17 It may please be noted that, Hon'ble Commission as per order dated 25.01.2022, has approved 30321 nos as allowable staff strength and observed that the revised numbers will be considered since 01.04.2022. But the tariff regulation for the control period was finalised before the redetermination of staff strength and the release of true up orders for the year 2018-19. As per Tariff Regulation 2021, notified by the KSERC on 16.11.2021, KSERC had determined the norms based on the O&M cost approved in the ARR for 2018-19 along with a sum of Rs. 248.47 Cr towards annual pay revision (due from July/August 2018) impact. Therefore, the provisional norms determined as per Tariff Regulation, 2021 is to be finalized for claiming the eligible O&M expenses in the truing up petition for 2023-24.

3.18 The expected O&M expenses based on the revised norms and actual escalation factor as detailed above is given below. Truing up petition for the financial year is prepared based on the norms specified in the Tariff regulation 2021 with escalation factor as submitted under para 3.15. However, it is submitted that, Hon'ble Commission may revise the O&M norms based on

the revised employee strength & trued up values for 2018-19 with actual pay revision expenses. A calculation of O&M cost based on the norms so arrived is submitted below. Hon'ble commission may approve the parameters for determining SBU T based on the revised norms.

Table 17: SBU T O&M Charges – Revised Norms	
Item	**2023-24**
O&M cost for Bay (Rs lakh/Bay)	8.23
O&M cost (Rs lakh/circuit km)	1.66
O&M cost (Rs lakh/MVA)	0.91
Opening bays (Nos)	2905.00
Bays added during the year (Nos)	24.00
Opening line length (Circuit Km)	10633.90
Line length added during the year (Circuit Km)	166.23
Opening MVA	25344.50
MVA added during the year	785.00
For assets at the beginning of the year –(A) in Rs. Lakh	64618.19
For assets added during the year – (B) in Rs. Lakh	593.91
Total Normative O&M expenses in Rs. Lakh	65212.10
Total Normative O&M expenses – (A) + (B) Rs. Cr	**652.12**

3.19 **Depreciation:** Hon'ble Commission approved depreciation for the assets at the beginning of the year 2023-24 at Rs. 255.15 Cr and for the asset added during the year 2023-24 at Rs 31.30 Cr. Thus, the total depreciation approved for the year 2023-24 is Rs. 286.45 Cr against this actual depreciation as per audited accounts was Rs.416.87 Cr as detailed in **Form T3.** The difference in depreciation is because of the methodology as per Ind AS and Tariff Regulations

3.20 Depreciation eligible as per MYT Regulation for the year 2023-24 has been worked out at Rs. 309.31 Cr as detailed in chapter 5. Comparison of approval, actual and true up requirement is furnished below:

Table-T 18: Depreciation (Rs Cr)				
No	**Particulars**	**Approved**	**Actual**	**True-up requirement**
1	Depreciation	286.45	416.87	309.17

3.21 **Return on Equity:** Hon'ble Commission approved Rs.119.99 Cr towards RoE of SBU-T. Same amount is sought for true up in this petition.

Table-T19: Return on Equity (Rs Cr)				
No	**Particulars**	**Approved**	**Actual**	**True-up requirement**
1	Return on Equity	119.99	119.99	119.99

Amortization of intangible assets:

A. Edamon - Kochi line:

3.22 The Government of Kerala, vide G.O.(Ms) No.29/ 2015/ PD dated 30/07/2015 of Power (C) Department, approved a revised Special Compensation Package, meant exclusively for the construction of 400 kV transmission line in the Edmon – Kochi sector, as a special case.

3.23 Hon'ble Commission, as per order dated 09.08.2019 in OP 58/2018, has provisionally approved 50% of the additional financial liability due to the special compensation package of the 400 KV Edmon - Kochi transmission line as KSEBL's share as per the orders of the Government. KSEBL share of compensation actually paid was allowed to be recovered in 12 years with interest as part of truing up. It is submitted that during the financial year 2023-24, no payment has been affected for Edmon - Kochi Compensation.

3.24 It is respectfully submitted that a sum of Rs.18 Cr was paid during 2019-20 out of which, KSEBL share was Rs.9 Cr. A sum of Rs. 0.75 Cr was proposed for amortization during 2019-20. A sum of Rs.22 Cr was paid during 2020-21 out of which, KSEBL share was Rs.11 Cr. A sum of Rs. 40.65 Cr was paid during 2021-22 against this, KSEBL share was Rs. 20.33 Cr. During the year 2022-23, a sum of Rs. 25.78 Cr was paid and out of which KSEBL share was Rs. 12.89 Cr. During the year 2021-22, Rs. 3.36 Cr was proposed for amortization. During the FY 2022-23 an amount of Rs. 8.08 Cr was proposed for the amortization along with interest. Now for the year 2023-24, an amount of Rs. 8.06 is proposed for the amortization along with interest.

3.25 As per orders of the Hon'ble Commission interest (at average interest rate for 2023-24 at 9.39%) and amortization amounts to Rs.8.06 Cr (Amortization + Interest) for 2023-24 as shown below.

Table T20 - Amortization & interest on compensation for Edamon - Kochi line

				2022-23			2023-24		
Sl. No	**Amount (Rs Cr)**	**Year of disbursement**	**KSEBL share (Rs Cr)**	**Amortization (Rs Cr)**	**Balance (Rs Cr)**	**Interest @ 8.49%**	**Amortization (Rs Cr)**	**Balance (Rs Cr)**	**Interest (Rs Cr)**
1	5.2	2019-20	2.6	0.22	1.73	0.15	0.22	1.51	0.14
2	0.8	2019-20	0.4	0.03	0.27	0.02	0.03	0.24	0.02
3	12	2019-20	6	0.50	4.00	0.34	0.50	3.50	0.33
4	22	2020-21	11	0.92	8.24	0.70	0.92	7.32	0.69
5	40.65	2021-22	20.33	1.69	16.94	1.44	1.69	15.26	1.43
6	25.78	2022-23	12.89	1.07	11.82	1.00	1.07	10.74	1.01
7							0.00	0.00	0.00
8		**Total**		**4.43**	**43.00**	**3.65**	**4.44**	**38.57**	**3.62**

B. Pugalur-Thrissur line

3.26 Government of Kerala vide G.O(MS)No.7/2019/PD dated 04.05.2019 and G.O.(MS) No.15/2019/Power dated 19-7-2019 have issued special compensation package for construction of Thrissur to Pugalur +-320 KV HVDC transmission line from Thrissur to Pugalur

forming part of the Raigarh - Pugalur-Thrissur HVDC scheme and for the Line in Line out construction of Thrissur to Kochi. Government vide order dated 18-5-2021 has ordered that the financial liability due to special compensation package shall be shared equally between Government of Kerala and KSEB Ltd.

3.27 The approximate estimate of compensation to be paid in accordance with the orders is worked out as Rs.45 Crore and KSEB Ltd share is estimated as Rs.25.11 Cr at 50:50 sharing of expense by KSEB Ltd/GoK.

3.28 The Hon'ble Commission vide order dated 01.12.2023 in OP No. 42/2023, provisionally approved 50% of the additional financial liability due to the 'special compensation package' of the 320 kV HVDC Pugalur-Thrissur Transmission line, as KSEB Ltd's share as per the orders of the State Government. KSEBL share of compensation actually paid was allowed to be recovered in 12 years with interest as part of truing up.

3.29 An amount of Rs. 0.15 Crore has been transferred to the District Collector, Ernakulam during the year 2023-24 as detailed below. The same has been accounted under intangible asset in accounts

Table T21- Details of compensation released during 2023-24

Sl No	Amount Paid (Rs)	Payment Details	Sanctioned vide BO	Remarks
1	15,36,407	Cheque No.809881 dtd 12.12.2023.	BO.(FTD) No. 450/2023/D(T&SO)/T4/Pugalur-Thrissur compensation/2020-21 dt. Tvpm. 16.10.2023	Equivalent share of Govt. of Kerala is pending

3.30 In the case of Pugalur-Thrissur line, KSEBL remitted its first share of compensation of Rs. 6,02,638/- towards Tower Standing Area (TSA) in 2021-22, based on the request made by the Special Tahsildar, LA unit, Palakkad. This included the GoK share of Rs. 3,01,319/- also. GoK was requested to reimburse the amount of Rs.3,01,319/- to KSEBL, which is not yet done.

3.31 Since then, KSEBL has been remitting its share of compensation (TSA and line corridor) only to the LA unit, Palakkad, based on the DVS approved by the District Collector, Palakkad and forwarded by the Special Tahsildar, LA unit, Palakkad. GoK has been requested to transfer its share directly to the LA unit, Palakkad.

3.32 Amortization & interest on compensation for Pugalur -Thrissur line is tabulated below.

Sl. No	Year of disbursement	KSEBL share (Rs Cr)	2021-22		2022-23		2023-24		
			Amortization (Rs Cr)	Balance (Rs Cr)	Amortization (Rs Cr)	Balance (Rs Cr)	Amortization (Rs Cr)	Balance Rs(Cr)	Interest Amount
1	2021-22	0.3015	0.025	0.276	0.025	0.251	0.025	0.226	0.021
2	2021-22	2.5	0.208	2.292	0.208	2.083	0.208	1.875	0.176
3	2022-23	4.83			0.403	4.428	0.403	4.025	0.378
4	2023-24	0.154					0.013	0.141	0.013
	Total				0.403	4.428	0.649	4.166	0.588
	Total amount (Amortization & Interest) for 2023-24						**1.24**		

3.33 A sum of Rs. 1.24 Cr (Rs. 0.649 Cr +Rs. 0.588 Cr) is proposed for amortization during 2023-24.

3.34 Total amortization of intangible assets: The total amortization of intangible assets and interest for the year amounts to Rs. **9.30 Cr** as submitted below may kindly be approved.

Table T23- Total amortization of intangible assets and interest for 2023-24

Sl. No.	Transmission Line	Amortization amount (Rs. Cr.)
1	Edamon-Kochi	8.06
2	Pugalur-Thrissur	1.24
	Total	**9.30**

3.35 Transmission Availability: The transmission system of KSEBL could achieve an availability of 99.17 %. The certificate of SLDC and detailed calculations are enclosed as **Annexure-2.** As per Regulation 56(2) of the KSERC (Terms and Conditions for determination of Tariff) Regulations, 2021, the transmission licensee shall be entitled to incentive on achieving annual availability above the target availability, in accordance with the following formula: -

$$\text{Incentive} = \frac{\text{Aggregate Revenue Requirements} \times [\text{Annual availability achieved} - \text{Target Availability}]}{\text{Target Availability}}$$

The Regulation further stipulates that the computation of incentive shall be undertaken during truing up for each financial year. Accordingly, SBU-T prays approval of Rs. 10.49 Cr as calculated below.

Table –T24 : Incentive on transmission availability (Rs Cr)	
ARR excluding incentive (Rs. Cr.)	1542.64
Annual availability achieved	99.17%
Target availability	98.5%
Incentive	**10.49**

It is humbly requested that the above incentive amounting to **Rs. 10.49 Cr** may kindly be approved as incentive for SBU-T.

3.36 Non-Tariff Income: Hon'ble Commission approved Rs. 54.86 Cr under this head. As per trifurcated accounts, a sum of Rs. 86.62 Cr has been accounted under SBU T and a sum of Rs. 75.96 Cr is claimed in the truing up petition as detailed under chapter 5, may kindly be trued up.

3.37 Net ARR (Transfer Cost) of SBU-T and SLDC: As per true up for the year 2023-24 ARR for Transmission Business Unit is Rs. 1629.10 Cr and non-tariff income claimed is Rs. 75.96 Cr. Therefore, Net ARR of SBU-T has been Rs. 1553.14 Cr. This cost may kindly be approved as the Cost of intra-state Transmission in the ARR of SBU-D.

3.38 Transmission loss: The total energy input to the transmission system was 31406.32 MU in FY 2023-24. The losses in the transmission system at different voltage levels are given in the Table below. The peak demand of the State for the year 2023-24 was 5301 MW. The losses are segregated based on Load flow studies done up to 33 kV level using computer simulation model in "Mi-Power" Software The methodology adopted for study is similar to that suggested by CEA and Adopted by Forum of regulators (FOR) for loss assessment. Balance losses are attributable to the HT and LT Network. Thus, the actual energy losses in the transmission system (up to 66KV) were 819.23 MU. The actual transmission losses (Quarterly and overall) are submitted below.

Table T25 - 5204 MW for 2023-24 Q1							
Voltage level	**Flow in MW**	**Transformer loss(MW)**	**Line loss (MW)**	**Loss (MW)**	**% loss corresponding to maximum demand**	**Loss in MU corresponding to LLF of 0.553**	**Loss up to each voltage level (MU)**
400 kV	2907	12.715	6.62	19.335	0.665	23.35	23.352
220 kV	3017.194	21	65.18	86.18	2.856	104.08	127.436
110 kV	4877.994	14.64	57.584	72.224	1.481	87.23	214.665
66 kV	916.698	2.435	11.969	14.404	1.571	17.40	232.061
33 kV	925.495	2.104	16.953	19.057	2.059	23.02	255.08

Table T26- 4366 MW for 2023-24 Q2							
Voltage level	Flow in MW	Transformer loss(MW)	Line loss (MW)	Loss (MW)	% loss corresponding to maximum demand	Loss in MU corresponding to LLF of 0.553	Loss up to each voltage level (MU)
400 kV	2782.867	9.582	7.714	17.296	0.622	21.12	21.119
220 kV	3394.867	11.84	35.23	47.07	1.387	57.47	78.592
110 kV	4264.695	11.398	52.857	64.255	1.507	78.46	157.049
66 kV	883.248	2.204	9.29	11.494	1.301	14.03	171.084
33 kV	853.904	1.778	13.089	14.867	1.741	18.15	189.24

Table T27- 4300 MW for 2023-24 Q3							
Voltage level	Flow in MW	Transformer loss(MW)	Line loss (MW)	Loss (MW)	% loss corresponding to maximum demand	Loss in MU corresponding to LLF of 0.553	Loss up to each voltage level (MU)
400 kV	2829.77	8.751	0.96	9.711	0.343	11.86	11.857
220 kV	3456.286	11.47	42.53	54	1.562	65.94	77.793
110 kV	4173.499	10.969	51.54	62.509	1.498	76.32	154.118
66 kV	757.238	2.09	8.037	10.127	1.337	12.37	166.483
33 kV	827.038	1.637	12.02	13.657	1.651	16.68	183.16

Table T28- 5301 MW for 2023-24 Q4							
Voltage level	Flow in MW	Transformer loss (MW)	Line loss (MW)	Loss (MW)	% loss corresponding to maximum demand	Loss in MU corresponding to LLF of 0.553	Loss up to each voltage level(MU)
400 kV	3354.612	13.594	1.981	15.575	0.464	19.02	19.017
220 kV	3831.519	20.31	58.91	79.22	2.068	96.73	115.747
110 kV	4547.889	16.188	81.087	97.275	2.139	118.78	234.522
66 kV	807.468	2.478	9.869	12.347	1.529	15.08	249.598
33 kV	1003.125	2.712	23.575	26.287	2.621	32.10	281.70

	Table T29- Overall Transmission losses					
Voltage Level	Q1	Q2	Q3	Q4	TOTAL	% Loss
400 kV	23.35	21.12	11.86	19.02	75.35	0.240
220 kV	104.08	57.47	65.94	96.73	324.22	1.032
110 kV	87.23	78.46	76.32	118.78	360.79	1.149
66 kV	17.40	14.03	12.37	15.08	58.87	0.187
					819.23	2.608

3.39 Auxiliary consumption of substations of Transmission SBU:

The actual auxiliary consumption for the year 2023-24 was **27.114 MU** as tabulated below. The auxiliary consumption in substations comprise of energy required for lighting, air compressors, battery charging, switch gears operating mechanism etc.

Table-T26: Auxiliary consumption of substations for FY 2023-24 (MU)

Month	Sub Total
April	2.273
May	2.308
June	2.056
July	2.063
August	2.224
September	2.044
October	2.219
November	2.117
December	2.539
January	2.242
February	2.366
March	2.664
Sub Total	**27.114**

This energy loss is not captured in the transmission losses determined by simulation using load flow software. This energy consumption is actually measured by meters in the respective substations.

3.40 The formats for Truing up of Transmission Business Units as per the Tariff Regulation 2021 are provided hereunder.

Annexure - 1

SUBSTATIONS COMMISSIONED DURING 2023-24										
Sl No	**Name of Substation**	**Volage level (kV)**	**New/Upgn.**	**Start Date**		**Commissioning date**		**AS amount in Lakhs**	**Expenditure in Lakhs**	**Remarks (Reason for Cost and Time overrun if any)**
				Scheduled	**Actual**	**Scheduled**	**Actual**			
1	PALAKKAL	110KV	Upgn	01.07.2021	01.07.2021	31.03.2024	03.04.2023	830	587	
2	KANNAMPULLY	110KV	Upgn	13.8.2021	13.08.2021	31.03.2024	26.10.2023	584	258	
3	ADOOR	110KV	Upgn	2016	2016	15.12.2023	18.12.2023	998	826.467	
4	EANATH	110KV	Upgn	01.04.2023	01.04.2023	15.12.2023	19.01.2024	200	158.2	
5	THAMARASSERY	110KV	Upgn	27-11-2020	27-11-2020	27-11-2022	12.01.2024	960	935.7	Restrictions imposed by govt in view of covid 19 and lock down
6	KARUNAGAPPALLY	110KV	Upgn	26-07-16	26-07-16	30-08-22	21.02.2024	1562	922	
7	KOTTAYAM	400KV	NEW	27.7.2020	27.7.2020	26.01.2022	13.05.2023	28500	12625	
LINES COMMISSIONED DURING 2023-24										
Sl No	**Name of line**	**Volage level(kV)**	**New/Upgn.**	**Start Date**		**Commissioning date**		**AS amount in Lakhs**	**Expenditure in Lakhs**	**Remarks (Reason for Cost and Time overrun if any)**
				Scheduled	**Actual**	**Scheduled**	**Actual**			
1	Upgradation of Sasthamcotta - Karunagapally 66kV SC feeder to 110kV DC (26.4km)	110kV	Upgn	28-02-19	28-02-19	30-08-22	21.02.2024	Included in Substation AS Amount	1232.8	Delay in getting sanction from railway
2	Interlinking 110kV Substation, Kollam with 110kV Substation Kottiyam (10.5km)	110kV	New	18-02-21	18-02-21	31-12-23	13.05.2023	3350	2067	Delay in getting sanction from NHAI and PWD. Delay in handing over utility corridor by NHAI.

3	Vatanappilly-Blangad 33kV SC line(13.5km)	33kV	New	19.09.22	19.09.22	19.01.24	18.12.2023	491	447	
4	Kallettumkara-Parappukkara 33kV (5.1km)	33kV	New	20.10.2022	20.10.2022	31.12.23	10.01.2024	350	266.428	
5	Pazhayannur-Chelakkara 33 kV SC line(12.37km)	33kV	New	05.09.22	05.09.22	05.01.24	17.01.2024	496	468	Due to road widening by KSTP along line route, delay in fixing the ROW
6	Upgradation of Kannampully-Nenmara 66kV line to 110kV (14.6km)	110kV	Upgn	28-12-2021	28-12-2021	22.12.2022	26.10.2023	793	365	Tower erection delayed due to crop harvesting
7	Chakkittappara-Peruvannamuzhi 33 kV SC line(3.6km)	33kV	New	13-04-2023	17-04-2023	12-05-2023	14-07-2023	850	326.74	Heavy rain and water lgging affected the work
8	Upgradation of 66 kV Thamarassery-Kunnamangalam SC line to 110 kV DC (35.4km)	110kV	Upgn	02.12.2021	02.12.2021	28.02.2023	04.11.2023	1750	927.45	Public protest against inserting KLB Tower at Location 158/B
9	LILO of Thirunelveli-Kochi line(1.23km)	400kV	New	15.7.2019	02.08.2019	22.05.2022	13.05.2023	505	303	
10	From Kanjirod to Mundayad (18.12km)	220kV	New	13.07.2019	23.07.2019	12.01.2022	25.01.2023 & 01.11.2023	2816	2435	
11	Mundayad-Azhikode - 5.91km SC	110kV	New	13.07.2019	23.07.2019	12.01.2022	08.12.2022	1849	1554	
12	Payyannur-Cheruvathur - 14.4km DC	110kV	New	13.07.2019	23.07.2019	12.01.2022	23.07.2021 & 14.07.2023	2253	1860	
13	Cheruvathur-Ambalathara tap - 26km DC	110kV	New	13.07.2019	23.07.2019	12.01.2022	17.06.2023 & 14.07.2023	4954	2639	
14	Mangad-Thalipparamba - 14.6km DC	110kV	New	13.07.2019	23.07.2019	12.01.2022	13.12.2023	2284	1395	

15	Koodal-Pathanamthitta 29km DC	110kV	New	29.9.2020	29.9.2020	28.03.2022	05.10.2023.	534	46	
16	Adoor-Pathanamthitta 25.4km DC	110kV	New	29.9.2020	29.9.2020	28.03.2022	03.12.2023.	506	37	
	Total 400kV line									
	Total 220kv line									
	Total 110kv line									

CAPACITY ADDITION DURING 2023-24

Sl No	Name of Substation	Volage level(kV)	New/Upgn/Cap addn/Cap enhmt	Start Date		Commissioning date		AS amount in Lakhs	Expenditure in Lakhs	Remarks (Reason for Cost and Time overrun if any)
				Scheduled	Actual	Scheduled	Actual			
1	Vilakkulam	33/11	Cap addn	22-02-2022	22-02-2022	31-05-2023	45062	910	6	Second 5 MVA transformer commissioned on 16.5.2023
2	Parassala	110/11	Cap addn	01-04-2023	01-04-2023	30-09-2023	27.03.2024	333	255.34	
3	Kavanad	110/11	Cap enhancement	24.11.2022	24.11.2022	27.04.2023	27.04.2023	400	418	
4	Pala	110/11	Cap enhancement	09.12.2022	07.03.2023	07.01.2023	22.05.2023	475	531	
				21.11.2023	22.11.2023	22.05.2023	08.12.2023			
5	Kodimatha	110/11	Cap enhancement	13.02.2024	15.02.2024	14.03.2024	07.03.2024	200	196.78	
6	Perumbavoor	110/11	Cap enhancement	16.11.2021	16.11.2021	31.01.2023	25.04.2023	260	240	
7	Vengola	33/11	Cap addn	01.04.2021	26.04.2021	31.12.2022	19.04.2023	257	125	
8	Kalloorkadu	33/11	Cap enhancement	17.11.2021	15.12.2021	31.03.2023	10.04.2023	345	298	
9	Mattanchery	110/11	Cap enhancement	31.03.2022	01.12.2022	31.03.2024	01.04.2023	692	635	
10	Mattanchery	110/11	Cap enhancement	31.03.2022	01.12.2022	31.03.2024	17.05.2023			
11	Kandanad	110/11	Cap enhancement	01.11.2022	01.11.2022	31.03.2023	09.06.2023	240	190	
12	Pazhayannur	110/33	Cap addn	01.10.2022	15.10.2022	30.11.2023	19.06.2023	250	30.56	Transformer lifted from Edarikode s/s is used
13	Kunnamkulam	110/11	Cap enhancement	20.06.2022	01.10.2022	31.08.2023	18.08.2023	450	290	Transformer taken from Vennakkara s/s is used

14	Kaippamangalam	33/11	Cap addn	19.08.2023	25.08.2023	18.10.2023	03.02.2024	37	26	
15	Panoor	110/11	Cap addn	12.10.2022	12.10.2022		19.04.2023	290	149.13	
16	Kanjirode	110/11	Cap enhancement	21.06.2022	07.10.2022	26.05.2023	04.05.2023	320	20.94	Old transformer from Thalassery S/s is used
17	Chakkittappara	110/33	Cap addn	02.12.2021	02.12.2021	10-05-2023	03.04.2023	850 (Inc. Line)	247.4	
18	Koduvally	110/11	Cap addn	01-02-2023	01-02-2023	31-12-2023	15.12.2023	365	238.36	
19	Kuttiady	110/33	Cap addn	05.09.2022	28.09.2022	24.02.2023	27.12.2023	783 (Inc. Line)	164.25	
20	Othukkungal	33/11	Cap addn	6.5.23	6.5.23	30.8.24	14.06.2023	250	80	
21	Pothukallu	33/11	Cap addn	29.11.22	11.12.22	8.2.23	05.04.2023	120	24.6	
22	Edavanna	33/11	Cap addn	23.11.22	7.12.22	16.4.23	16.04.2023	103	68.55	
23	Malappuram	110/33	Cap addn	25.7.23	31.7.23	31.8.23	02.09.2023	235	198	
24	Chelari	110/11	Cap addn	11.8.23	11.8.23	20.10.24	27.02.2024	460	389	
25	Kozhencherry	110/33	Cap addn				31.05.2023			

Annexure - 2

Consolidated availability for 2023-24															
400kV															
Feeder	**Apr-23**	**May-23**	**Jun-23**	**Jul-23**	**Aug-23**	**Sep-23**	**Oct-23**	**Nov-23**	**Dec-23**	**Jan-24**	**Feb-24**	**Mar-24**			
weightage factor	99248.7540	99248.75	193329.40	193329.40	193329.40	287410.05	287410.05	287410.05	287410.05	287410.05	287410.05	287410.05	2790356.0900		
wi*Tav	99248.7540	99248.75	192598.22	193329.40	192495.33	287410.05	286977.62	286927.59	287410.05	287410.05	287410.05	287410.05	2787875.9399	**99.91**	
	1.0000	**1.0000**	**0.9962**	**1.0000**	**0.9957**	**1.0000**	**0.9985**	**0.9983**	**1.0000**	1.0000	1.0000	1.0000	**0.9991**		
Transformer															
W.F	945.00	945	1260	1260	1575	1575	1575	1575	1575	1575	1575	1575	17010.0000		
Wi*Tav	943.99	928.12	1248.66	1260.00	1574.65	1574.37	1572.05	1552.37	1564.80	1570.87	1555.64	1573.88	16919.4004	**99.47**	
	0.9989	**0.9821**	**0.9910**	**1.0000**	**0.9998**	**0.9996**	**0.9981**	**0.9856**	**0.9935**	0.9974	0.9877	0.9993	**0.9947**		**400kV System**
220kV															**99.71**
Feeder															
weightage factor	444260.10	444260.10	448537.73	448537.73	440870.53	441450.37	441450.37	449852.67	449852.67	449852.67	449852.67	449852.67	5358630.27		
wi*Tav	440056.99	434297.31	440068.11	446028.02	437358.15	436566.97	432669.96	443878.59	447087.66	445498.26	446235.65	445763.93	5295509.60	**98.82**	
	0.9905	**0.9776**	**0.9811**	**0.9944**	**0.9920**	**0.9889**	**0.9801**	**0.9867**	**0.9939**	0.9903	0.9920	0.9909	**0.9882**		
Transformer															
W.F	10146	10146	10146	10020	10146	10146	10146	10146	10146	10146.00	10146.00	10146.00	121626.0000		
Wi*Tav	10097.04	10013.14	10108.85	9972.86	10013.75	9920.1100	10045.10	10108.12	10038.87	10076.42	10092.04	10093.91	120580.2228	**99.14**	
	0.9952	**0.9869**	**0.9963**	**0.9953**	**0.9870**	**0.9777**	**0.9901**	**0.9963**	**0.9894**	0.9931	0.9947	0.9949	**0.9914**		**220kV system**
110kV															**98.96**
Feeder															

weightage factor	171349.12	175085.25	176659.42	176408.16	177378.66	175115.23	177672.29	175881.25	176848.32	177431.36	176775.36	177661.76	2114266.17		
wi*Tav	170247.21	173603.76	175746.51	175375.95	176604.16	173723.86	175860.60	173071.13	175376.46	176062.50	175194.80	175194.80	2096061.73	**99.14**	
	0.9936	**0.9915**	**0.9948**	**0.9941**	**0.9956**	**0.9921**	**0.9898**	**0.9840**	**0.9917**	0.9923	0.9911	0.9861	**0.9914**		
Transformer															
W.F	10839	10916.5	10952.5	10956	10932.5	10949	10961.5	10949	10932.5	11030.00	11032.50	11138.50	131589.50		
Wi*Tav	10741.61	10822.15	10891.33	10911.50	10861.41	10909.98	10903.79	10851.34	10795.00	10887.39	10894.50	11018.49	130488.48	**99.16**	**110kV system**
	0.9910	**0.9914**	**0.9944**	**0.9959**	**0.9935**	**0.9964**	**0.9947**	**0.9911**	**0.9874**	0.9871	0.9875	0.9892	**0.9916**		**99.15**
66kV															
Feeder															
weightage factor	15135.75	14672.35	14422.35	14254.15	14422.35	14422.35	14422.35	14488.95	14515.95	14661.51	14503.21	14531.51	174452.78		
wi*Tav	15047.08	14568.45	14303.92	14153.06	14358.70	14286.32	14281.48	14360.42	14433.54	14564.04	14382.03	14405.00	173144.04	**99.25**	**66 kV system**
	0.9941	**0.9929**	**0.9918**	**0.9929**	**0.9956**	**0.9906**	**0.9902**	**0.9911**	**0.9943**	0.9934	0.9916	0.9913	**0.9925**		**99.13**
Transformer															
W.F	1793.1	1783.1	1783.1	1773.1	1743.1	1773.1	1783.1	1773.1	1755.6	1727.60	1727.60	1717.30	21132.90		
Wi*Tav	1771.66	1738.21	1765.35	1765.06	1738.66	1766.15	1765.30	1762.56	1742.96	1717.61	1709.85	1688.88	20932.24	**99.05**	
	0.9880	**0.9748**	**0.9900**	**0.9955**	**0.9975**	**0.9961**	**0.9900**	**0.9941**	**0.9928**	0.9942	0.9897	0.9835	**0.9905**		
													weightage factor	**wi*Tav**	
			400kV	**220kV**	**110kV**	**66kV**	**Total**						10437705.31	10352591.32	**99.18**
		No. of feeders	6	93	393	111	603						291358.40	288920.34	**99.16**
		No of Trfrs	5	73	651	160	889								**99.17**

Annexure – 2 A

Voltage wise consolidated system availability for 2023-24 of Kerala System													
	Apr-23	**May-23**	**Jun-23**	**Jul-23**	**Aug-23**	**Sep-23**	**Oct-23**	**Nov-23**	**Dec-23**	**Jan-24**	**Feb-24**	**Mar-24**	**Total**
400kV	99.94%	98.93%	99.36%	100.00%	99.79%	99.98%	99.83%	99.25%	99.71%	99.88%	99.44%	99.97%	99.71%
220kV	99.26%	98.18%	98.79%	99.48%	98.98%	98.40%	98.45%	99.09%	99.19%	99.16%	99.32%	99.27%	98.96%
110kV	99.20%	99.14%	99.46%	99.53%	99.43%	99.48%	99.29%	98.84%	98.90%	98.90%	98.88%	98.81%	99.15%
66kV	99.05%	98.21%	99.07%	99.44%	99.67%	99.39%	99.01%	99.29%	99.34%	99.39%	99.05%	98.67%	99.13%
System Availability	**99.28%**	**98.64%**	**99.19%**	**99.58%**	**99.24%**	**99.05%**	**99.07%**	**99.18%**	**99.16%**	**99.20%**	**99.21%**	**99.22%**	**99.17%**

Chapter– 4: TRUING UP OF ARR OF SBU – DISTRIBUTION

4.1 An overlook on Kerala Distribution network:

1. Kerala State Electricity Board Limited supplies electricity to about 3.75 Cr of entire population across 14 districts in Kerala. There are about 1.05 crore domestic consumers, 26.19 Lakh commercial/general consumers, 1.51 Lakh industrial consumers, 5.27 Lakh agricultural consumers. The key statistics of distribution network is given below:

Table-D1 Key Statistics of KSEBL Distribution Network as on 31.03.2024		
Sl No	**Particulars**	**Statistics**
1	Area Sq.km.	38863 km^2
2	Districts No's	14
3	Electrical Circle Offices	25
4	Population in Cr	3.75
5	Consumers (Nos)	13864176
6	Distribution transformers (Nos)	89620
7	HT lines (Ckt. Kms)	72,451.88
8	LT lines (Kms)	3,10,191
9	Energy sales by KSEBL including external sales MU	27674.71
10	Energy supplied by KSEBL including external sales MU	28176.34
11	Energy consumption inside the State (incl open access & captive consumption) in MU	28605.93
12	T&D loss in % (including transmission loss)	9.70%
13	Distribution loss in %	7.28%
13	AT&C loss in % as per CEA methodology	7.55%

TRUING UP of 2023-24

Energy supplied by KSEBL for 2023-24:

2. Hon'ble Commission, vide order dated 25.06.2022, had approved energy sale of 25697.59 MU for the year 2023-24 after accounting the impact of solar penetration and e-mobility. The energy sales estimated without accounting solar penetration and impact of e-mobility was 26060.86 MU.
3. The actual energy sale of KSEBL inside the State for FY 2023-24 was 27603.44 MU. 71.27 MU had been sold outside the state through exchanges.
4. In addition to the above captive power producers injected energy into the grid and availed energy through banking at their consumer premises. Further, solar prosumers also inject

energy into the grid and consumes energy from the grid through banking arrangements. The energy injected by the captive consumer /solar prosumer is taken as export to the grid (injection to grid) and energy adjusted against this injection is considered as the energy drawal from grid i.e., energy import from grid for loss calculation purposes. The details are given in table below

5.

Table D2 Consumption against captive generation							
Consumer	**Tariff category**	**Installed capacity**	**Generation**	**Export to grid**	**Consumption adjustments against export**	**Consumption adjustments at generation point**	**Consumption against captive generation**
		MW	MU	MU	MU	MU	MU
			A	B	C	D	E=C+D
SHEP	EHT	33.00	71.30	71.30	69.23	0.00	69.23
Solar	EHT	55.04	68.11	35.32	30.96	32.79	63.74
Solar	HT	79.95	79.12	14.69	10.47	64.43	74.90
Wind	HT	11.00	25.69	25.69	17.99	0.00	17.99
Solar	LT	646.67	634.41	487.09	367.09	147.33	514.41
Solar off grid prosumers	LT	1.16	1.13			1.13	1.13
Total		**826.82**	**879.76**	**634.09**	**495.73**	**245.68**	**741.40**
Soura scheme under Board fund		23.78	24.64	23.11	3.45	1.53	4.98
Total excluding Soura scheme under Board fund		803.04	855.12	610.98	492.2820	244.15	736.42

6. It can be seen from the above table that 634.09 MU was exported to the grid by captive /solar prosumers and 495.73 MU was adjusted against this export through banking. The remaining was settled at the APPC rate of Rs 3.15 per unit approved by Honourable Commission. In addition to this 245.68 MU was consumed at the generation point resulting In a total consumption of 741.40 MU from captive generation. In addition to the above 1.13 MMU was consumed by off grid consumers

7. In addition to the above 5.90 MU was availed by RGCCPP for meeting the auxiliary consumption from the grid during non-operative periods.

8. The category wise energy sale to consumers as well as the total energy supplied by KSEBL inside the State as well as outside the State is given in table below.

Table. D.3. Energy sales /usage from grid details for the year 2023-24

Sl. No	Consumption Category	Number of consumers	Energy sales	Connected load/ Contract Demand
			MU	MW/MVA
I	LT Category			
	LT I Domestic	10494734	13735.43	22121.83
	LT III	0	1.94	0.00
	LT IV A Industrial	144719	1286.18	1954.87
	LT IV B Industrial	2886	12.13	15.03
	LT V A Agriculture	489383	355.43	930.27
	LT V B Agriculture	37043	95.82	107.88
	LT VI A General	131106	262.29	422.84
	LT VI B General	39516	136.39	172.25
	LT VI C General	47452	281.74	328.55
	LT VI D General	35310	29.29	48.34
	LT VI E General	11000	3.97	10.99
	LT VI F General	578128	1210.62	1277.44
	LT VI G General	20944	110.38	157.04
	LT VII A Commercial	1329401	2444.41	2941.72
	LT VII B Commercial	414964	163.53	166.67
	LT VII C Commercial	3229	35.47	46.93
	LT VIII A Public lighting	1605	217.45	116.02
	LT VIII B Public lighting	71255	157.11	97.40
	LT IX Advertisement & Hoardings	1975	2.10	3.12
	LT X Electric vehicle charging stations	1813	8.45	24.05
	LT total	**13856463**	**20550.14**	**30943.25**
II	HT Category			
	HT I A Industrial	3010	2503.49	1040.95
	HT I B Industrial	27	18.27	7.73
	HT II A General	528	284.70	120.92
	HT II B General	1230	766.75	248.00
	HT III A Agriculture	56	5.58	13.61
	HT III B Agriculture	10	2.39	1.53
	HT IV A Commercial	1241	373.11	189.00
	HT IV B Commercial	1368	602.15	254.87
	HT V Domestic	138	21.82	13.23
	HT VI Electric vehicle charging stations	4	2.19	2.07
	HT VII Temporary	3	1.17	1.63
	HT Total	**7615.00**	**4581.62**	**1893.53**
III	**EHT Category**			
	EHT I 66 KV Industrial	13	237.66	78.8
	EHT II 110 KV Industrial	18	770.90	190.1

Sl. No	Consumption Category	Number of consumers	Energy sales	Connected load/ Contract Demand
			MU	MW/MVA
	EHT III 220 KV Industrial	1	279.27	50
	EHT General A	1	5.09	1
	EHT General B	1	39.92	20
	EHT General C	4	35.24	15.6
	EHT Commercial	2	17.88	3.45
	EHT Railway Traction	13	420.96	120
	KMRL	3	14.31	6.1
	EHT Total	**56**	**1821.24**	**485.05**
IV	Bulk Supply consumers / Licensees			
	Defence Installations and Combined usage incl housing colonies(MES)	32	85.03	21.285
	Cochin Port Trust	1	40.12	11.5
	CSEZ	1	58.20	13
	KDHPCL	1	56.69	9
	KPUPL	1	102.55	25
	Karnataka Electricity Department	1	1.09	0.15
	Rubber Park Ltd	1	32.15	6.7
	Techno park	1	76.62	21
	Thrissur Corporation	1	178.23	8
	Info park, Cherthala	1	12.28	2.8
	Smart City	1	7.48	3
	Sub total	**42**	**650.44**	**121.44**
V	**HT, EHT & Licensees**	**7713**	**7053.30**	**2500.02**
VI	**Grand Total**	**13864176**	**27603.44**	**30943.25/2500.02**
VII	Energy sales outside the State		**71.27**	
VIII	Energy supplied by KSEBL for meeting banking requirements of solar prosumers/captive consumers		**495.73**	
VIII	Energy supplied by KSEBL to RGCCPP for meeting their auxiliary consumption during non operative periods		**5.90**	
VIII	Total energy supplied by KSEBL		**28176.34**	

While approving the energy sales for FY 2023-24, Hon'ble Commission considered the energy sales including energy wheeled through open access. Embedded open access consumers imported 266.09 MU at Kerala periphery and energy injected by IPPs into the grid for sale outside the State through open access was 74.52 MU. The embedded open access consumers availed 255.18 MU through open access at consumer end. The details are furnished in table below.

Table –D4 Energy availed and sold through OA by embedded open access consumers in FY 2023-24(MU)			
	Particulars	**Energy at Kerala Periphery**	**Energy at injection/ drawal point**
1	Import	266.09	255.18
2	Export	74.52	78.36

Thus, the total consumption of the State is given in table below

Table D 5. Total energy consumption of State							
	Energy sales by KSEBL (MU)	**Open access import (MU)**	**Sub Total (MU)**	**Captive generation (MU)**			**Total (MU)**
				SHP	**Solar**	**Wind**	
LT	20550.14	0	20550.14		514.41		21064.55
HT, EHT & Bulk	7053.30	255.18	7308.48	69.24	139.78	17.98	7535.48
Total	**27603.44**	**255.18**	**27858.62**	**69.24**	**654.20**	**17.98**	28600.03
RGCCPP aux. consumption	5.90		5.90				5.90
Total	**27609.34**	**255.18**	**27864.52**	**69.24**	**654.20**	**17.98**	**28605.93**

4.7 The actual energy sale inside the State by KSEBL was 1905.84 MU more than approval. Considering the energy availed by embedded open access consumers through open access and consumption adjusted against captive injection for captive consumers/solar prosumers, energy given to RGCCPP, the actual quantum was 2662 MU more than the approved quantum (2.02%). Total consumption of the State including energy from captive generation is 28605.93 MU which is about 11.32 % more than approved quantum.

Table –D6: -Details of Energy sales inside the State , energy wheeled through open-access, energy adjusted against captive injection - comparison with approved (MU)

Sl. No	Category	Approved	Actual	Deviation	% variation of energy sales against approval against appvl
I	LT I Domestic	13896.64	13735.43	-161.21	-1.16
	LT II	3.77		-3.77	-100.00
	LT III	0	0.00	0.00	
	LT IV A Industrial	1114.22	1286.18	171.96	15.43
	LT IV B Industrial	9.14	12.13	2.99	32.76
	LT V A Agriculture	344.78	355.43	10.65	3.09
	LT V B Agriculture	66.4	95.82	29.42	44.31
	LT VI A General	198.72	262.29	63.57	31.99
	LT VI B General	125.42	136.39	10.97	8.75
	LT VI C General	259.93	281.74	21.81	8.39
	LT VI D General	22.2	29.29	7.09	31.93
	LT VI E General	3.19	3.97	0.78	24.33
	LT VI F General	1006.34	1210.62	204.28	20.30

Sl. No	Category	Approved	Actual	Deviation	% variation of energy sales against approval against appvl
	LT VI G General	76.05	110.38	34.33	45.14
	LT VII A Commercial	1625.74	2444.41	818.67	50.36
	LT VII B Commercial	213.91	163.53	-50.38	-23.55
	LT VII C Commercial	22.92	35.47	12.55	54.74
	LT VIII A Public lighting	198.4	217.45	19.05	9.60
	LT VIII B Public lighting	108.83	157.11	48.28	44.37
	LT IX Advertisement & Hoardings	1.08	2.10	1.02	94.86
	LT X Electric vehicle charging stations	0.55	8.45	7.90	1437.00
	LT total	**19298.24**	**20550.14**	**1251.90**	**6.49**
II	HT I A Industrial	2407.52	2503.49	95.97	3.99
	HT I B Industrial	15.13	18.27	3.14	20.76
	HT II A General	208.02	284.70	76.68	36.86
	HT II B General	646.75	766.75	120.00	18.55
	HT III A Agriculture	8.77	5.58	-3.19	-36.42
	HT III B Agriculture	2.32	2.39	0.07	2.87
	HT IV A Commercial	360.64	373.11	12.47	3.46
	HT IV B Commercial	344.76	602.15	257.39	74.66
	HT V Domestic	24.72	21.82	-2.90	-11.72
	HT VI ELECTRIC VEHICLES CHARGING STATIONS	48.27	2.19	-46.08	-95.47
	HT VII Temporary	0	1.17	1.17	
	HT Total	**4066.88**	**4581.62**	**514.74**	**12.66**
III	EHT I 66 KV Industrial	321.22	237.66	-83.56	-26.01
	EHT II 110 KV Industrial	773.24	770.90	-2.34	-0.30
	EHT III 220 KV Industrial	140.77	279.27	138.50	98.39
	EHT General A	13.18	5.09	-8.09	-61.38
	EHT General B	44.72	39.92	-4.80	-10.72
	EHT General C		35.24	35.24	
	EHT Commercial		17.88	17.88	
	EHT Railway Traction	351.82	420.96	69.14	19.65
	KMRL	10.25	14.31	4.06	39.64
	EHT Total	**1655.2**	**1821.24**	**166.04**	**10.03**
IV	Bulk Supply consumers / Licensees	677.27	650.44	-26.83	-3.96
	Sub total	677.27	650.44	-26.83	-3.96
V	**HT, EHT & Licensees**	6399.35	7053.30	653.95	10.22
	Total energy sales by KSEBL	**25697.59**	**27603.44**	**1905.85**	**7.42**
	Energy availed through open access		255.18		
	Energy adjusted against captive/prosumer injection		495.73		
	Energy supplied by KSEBL to RGCCPP for meeting their auxiliary consumption during non-operative periods		**5.90**		
	Energy usage from grid	**25697.59**	**28360.24**	**2662.65**	**10.36**

4.8 The energy sales for 2023-24 shows an increase of 10.75 % compared to FY 2022-23 against 6.06 % in previous year. The domestic energy sales show an increase of 10.95 % against -2.33 % in previous year. The LT energy sales show an increase of 11.39 % against an increase of 3.12 % in previous year. The increase in energy sales for HT EHT and bulk licensees shows an increase of 8.94 % during 2023-24 whereas the corresponding increase during previous year was 15.44 %.The increase in energy sales was mainly due to the failure of monsoon and the sweltering heat during summer.

4.9 Considering the impact of open access and captive generation, total LT consumption shows an increase of 12.54% and HT & EHT consumption shows an increase of 10.23%. Total energy consumption shows an increase of 11.93 % in FY 2024 compared to 5.91 % in FY 2023,

4.10 The energy sales pattern for 2023-24 is given in table below:

Table-D7: Energy Sales pattern for 2023-24			
No	**Category**	**Energy sales**	
		MU	**%**
1	LT Domestic	13735.43	49.76
2	LT Industrial	1298.32	4.70
3	LT Agriculture	451.24	1.63
4	LT Commercial	4690.58	16.99
5	LT Public Lighting	374.57	1.36
6	HT &EHT Total	5967.58	21.62
7	EHT Railway Traction	420.96	1.53
8	KMRL	14.31	0.05
9	Bulk Supply	650.44	2.36
10	**Total sales to consumers (within state)**	**27603.44**	100
11	**External sale**	**71.27**	
12	**Energy given to RGCCPP for meeting the auxiliary consumption during non operative periods**	**5.9**	
13	**Energy supplied by KSEBL for meeting banking requirements of solar prosumers/captive consumers**	**495.73**	
12	**Total energy sale by KSEBL**	**28176.34**	

T & D Loss

4.11 Hon'ble Commission, vide order dated 25.06.2022, had approved the T&D loss target for FY 2023-24 as 10.82%. The targets approved for transmission loss, distribution loss and total AT &C loss was 3.3 % ,7.78 % and 11.71 % respectively with a collection efficiency target of 99 %.

4.12 KSEBL achieved better loss targets when compared with the targets approved by the Hon'ble Commission owing to the measures depicted in table D10 below. The actual T&D loss for the year 2023-24 was 9.70%. The actual transmission loss and distribution loss for the year

2023-24 was 2.61 % and 7.28 % respectively. The AT&C loss for SBU-D for the year FY 2023-24 as per CEA methodology was 7.55%. KSEBL achieved a collection efficiency of 99.72%.

4.13 Hon'ble Commission may kindly note that KSEBL was able to reduce loss by 1.12% over and above the target approved (10.82%), as detailed below:

Table D8: T&D loss for the year 2023-24 (MU)

Sl No	Particulars	2023-24
i	Energy generation by KSEBL excl auxiliary consumption	5652.67
ii	Power Purchase at Kerala periphery	25585.32
iii	Power Purchase /energy injected by captive consumers/solar prosumers at Kerala periphery	610.97
iv	Sub Total (MU) (A+B)	31848.96
v	Energy injected to the grid by private IPPs for sale outside the State through OA	78.36
vi	Energy wheeled by embedded open access consumers from outside the State at KSEB end	266.09
vii	Total energy input at Kerala periphery	32193.40
viii	Energy sales by KSEBL outside the State	71.27
ix	Energy transfer outside the State by KSEBL through banking arrangement	552.68
x	Energy export through Deviation Settlement Mechanism	61.49
xi	Energy sale by private IPPs for sale outside the State through open access	74.52
xii	Total energy sale/transfer through banking/DSM Export outside the State	759.97
xiii	**Net Energy Input at Kerala periphery (MU)(vi-x)**	**31433.44**
xiv	Substation auxiliary consumption (MU)	27.11
A.xv	**Net Energy Input excluding auxiliary consumption (MU) (xi-xii)**	**31406.32**
B	**Energy sales/usage**	
i	Energy sales by KSEBL	27603.44
ii	Energy consumed by embedded open access consumers at consumer end by wheeling from outside the State	255.18
iii	Energy supplied by KSEBL for meeting the banked energy injected by captive prosumers/solar prosumers including solar prosumers under saura scheme by KSEBL	495.73
iv	Energy availed by RGCCPP to meet the auxiliary consumption	5.90
B.v	Energy at consumer end availed from grid (MU)	28360.25
C	**T&D loss (MU) (A.xiii-B.v)**	3046.08
D	**T& D loss for 2023-24 in %(C*100/A.xiii)**	**9.70**
E	**T& D loss target approved for 2023-24 in %**	**10.82**
F	**Achievement in loss reduction in excess of target approved (E-D in %)**	**1.12**

Distribution loss:

4.14 The actual T&D loss during the year 2023-24 was 9.70%. It may please be noted that the total transmission loss up to 66 KV level was 819.21 MU. The voltage wise transmission loss is depicted in chapter 3 of the petition. Based on the above the distribution loss for 2023-24 is assessed as follows:

Table – D9: Distribution losses			
Sl No	**Particulars**	**Quantum**	**Unit**
(i)	Total Generation and Power Purchase at Kerala periphery (item xiii of table D8)	31,406.32	MU
(ii)	Transmission loss	819.21	MU
(iii)	Total energy input into the distribution system (i)-(ii)	30,587.11	MU
(iv)	Total energy output (item B.v of Table D7)	28,360.25	MU
(v)	Distribution loss (iii)-(iv)	2,226.87	MU
(vi)	Distribution loss in % (v)*100/(iii)	7.28	%

4.15 Hon'ble Commission may kindly note that, KSEBL has been continuously reducing the T&D loss since the year 2001-02. Consistent T&D loss reduction was one of the major achievements of the KSEB Ltd. KSEBL was able to reduce the total T&D loss in the KSEBL system including transmission system from the level of 30.76% in 2001- 02 to the level of 9.70% during the year 2023-24. Thus, the losses in the Distribution system, as a percentage of the total energy input is one of the lowest among the distribution utilities in the country.

4.16 The AT&C loss for the year 2023-24 as per CEA methodology was 7.55 %.

4.17 KSEBL has undertaken a series of measures during 2023-24, as detailed below, in order to achieve the above loss reduction.

Table- D 10 Steps taken for T&D loss reduction	
Particulars	**Achievement during 2023-24**
Substations Commissioned	
400 kV	1
220 kV	
110 kV	6
66 kV	
33 kV	
Total	
Lines Commissioned (Ckt km)	**7**
400kV	1.23
220kV	18.12
110 kV	202.51
66 kV	0
33 kV	29.4
Capacity addition/enhancement of substations (MVA)	785
11 KV line constructed(km)	1297
LT line constructed(km)	4078
No. of distribution transformers	2355
Distribution Transformer Capacity (MVA)	1178
HT Re conductoring (km)	384.68
LT Re conductoring (km)	5691
1 phase to 3 phase conversion(km)	1347
Meter Replacement (Nos)	298976

4.18 Even though KSEBL achieved loss reduction of 1.12% above the target of 10.82%, the T&D loss increased from 9.27 % to 9.70%. This is mainly due to the exorbitant rise in the energy demand during 2023-24 of 11 % against the estimated demand growth of 5%.

4.19 The energy requirement for meeting the consumption of State is given in table below.

Table- D11 Comparison of energy availability

Sl No	Particulars	Approved	Actual	Difference
I	**Generation**	**A**	**B**	**B-A**
1	Energy sales including banked energy supplied by KSEBL to solar prosumers/captive consumers in MU	25697.59	28105.07	2407.48
2	Energy input for meeting the energy sales in MU(C-F-G-J) of table D6	28815.42	31163.51	2348.09
	Internal Generation in MU (excl aux)			0.00
3	Hydro	7077.22	5600.26	-1476.96
4	Thermal	0	-0.36	-0.36
5	Solar & wind	72.5	52.76	-19.74
II	**Subtotal generation**	**7149.72**	**5652.67**	**-1497.05**
6	Power Purchase in MU at Kerala periphery i			0.00
7	CGS	11190.36	10951.87	-238.49
8	RGCCPP			0.00
10	IPP-wind, SHPs, solar CPPs & solar prosumers inside the State	1130.18	1273.35	143.17
11	LTA (Maithon, DVC)	3016.36	3274.05	257.69
12	LTA (DBFOO approved contracts)	2169.87	2240.93	71.06
13	SECI - LTA wind	472.8	212.23	-258.48
14	Power purchase approved in lieu of DBFOO unapproved contracts	3363.29	886.83	-2476.46
15	Medium Term JITPL	962.75	935.79	-26.96
16	TP Saurya		193.49	193.49
17	RSAPL – NTPC Anta		2.09	2.09
18	RPO-solar	66.5		-66.50
19	RPO-non solar	72.18		-72.18
20	Short term contracts (MU)	240.55	1275.19	1034.64
21	Power Purchase excluding purchase from exchanges/swap/DSM	22684.82	21245.80	-1439.02
22	Purchase through exchanges/swap		4486.35	4486.35
23	DSM import		464.14	464.14
24	Sub Total Power Purchase including prosumer/captive injection	22684.82	26196.29	3511.47
25	Sub Total generation & Power Purchase by KSEBL (MU)	29834.54	31848.96	2014.42
26	Energy availability (MU)	29834.54	31848.96	2014.42
27	Substation auxiliary consumption (MU)		27.11	27.11
28	**Energy requirement for sale inside the State**	28815.42	31821.84	3006.42
29	External sale/swap return/DSM Export (MU)	1019.12	685.45	-333.67

4.20 It may kindly be seen from the table that the energy supplied by KSEBL for FY 2023-24 was increased by about 2407.48 MU. Hydro generation was less by 1565.31 MU due to the poor monsoon.

4.21 There was a reduction of 238.49 MU from Central Generating Stations against the approval granted by the Hon'ble Commission. There was a total reduction of 2393.64 MU in power purchase (excluding power purchase through contracts and deviation settlement mechanism) from that of the approved quantum. The major reasons for reduction in power purchase quantum was mainly due to the reduction in availability from generating stations under DBFOO unapproved contracts. The reduction in power purchase from the approved quantum and the peak shortage was met through purchase from short term contracts, exchanges, banking arrangements and through increased hydro generation, which were not originally approved in the order.

Availability of Electricity for Supply:

4.22 The petitioner has ensured 100% availability of power for supply in the state. There had been no load shedding enforced in the State even during the extreme summer months of 2023-24, except for emergency supply interruptions caused by uncontrollable factors. The gross availability of electricity for supply for 2023-24 as per regulations was 100 %.

Analysis of various ARR components

Cost of Generation:

4.23 The gross energy generation from own generating stations was 5718.01 MU The energy injected to grid including solar injection and after deducting auxiliary consumption was 5652.67 MU. ARR for SBU G for the year 2023-24 was Rs.666.69 Cr and Non-tariff income earned is Rs. 40.21 Cr. The net expenses of SBU G transferred to SBU-D are Rs. 626.48 Cr. Deviation from approval is explained in chapter dealing with Truing up of SBU G.

Cost of intra state transmission

4.24 ARR for Transmission Business Unit for the year 2023-24 had been Rs. 1629.10 Cr and non-tariff income earned was Rs.75.96 Cr. Therefore, Net ARR of SBU-T has been Rs.1553.14 Cr. as detailed in chapter 3. This cost may kindly be approved as the Cost of intra-state Transmission in the ARR of SBU-D.

Cost of Power Purchase

4.25 As already submitted, the Distribution Business Unit of the KSEBL had sold 27603.44 MU to its consumers during the year 2023-24 and supplied 5.90 MU to meet the auxiliary consumption of RGCCPP, Kayamkulam during non- operative periods. In addition to the above, KSEBL supplied 495.73 MU to captive consumers /solar prosumers against their banked energy. Thus, KSEBL supplied 28105.07 MU during 2023-24. In addition to that 71.27 MU was sold outside the State. This was met through own generation and power purchase from various sources. The total own generation (net of auxiliary consumption) was 5652.67 MU and power purchase by KSEBL was

25585.32 MU at Kerala periphery. Further captive consumers/solar prosumers injected 610.97 MU to the grid. The energy requirement for meeting the energy consumption of 28105.07 MU was 31848.96 MU at Kerala periphery. Out of this, SBU G was able to supply 5652.57 MU (17.75%) only of the total requirement and balance power requirement was procured from CGS and IPPS within and outside the State, power exchanges, banking arrangements etc.

4.26 Hon'ble Commission had approved a cost of Rs. 10564.23 Cr for the purchase of 22684.82 MU from various sources. The actual power purchase quantum was 25585.32 MU excluding injection by captive/solar prosumer at Kerala periphery at a cost of Rs 12982.59Cr as detailed in **Form D3.1.** Thus, the true up requirement is 12982.59 Cr, which is more than the approved cost by Rs 2418.36 Cr. A summary of the approved and actual power purchase and trued up requirement is given in Table below.

Table – D12: Power Purchase for the year 2023-24

No	Particulars	Approved		Actual		Truing up		Variation	
		Energy (MU)*	Cost (Rs Cr)	Energy (MU)*	Cost (Rs Cr)	Energy (MU)*	Cost (Rs Cr)	Energy (MU)*	Cost (Rs Cr)
1	Central Gen. Stations	11190.36	4897.59	10951.87	4731.08	10951.87	4731.08	-238.49	-166.51
2	Long Term Contracts (Maithon, DVC, approved contracts	5186.22	2489.82	5514.97	2367.59	5514.97	2367.59	328.75	-122.23
3	DBFOO unapproved contracts**	3363.29	1372.22	886.83	373.50	886.83	384.99	-2476.46	-998.72
4	Medium Term Contracts	962.75	320.6	935.79	364.19	935.79	364.19	-26.96	43.59
5	Renewable purchase	1741.65	524.22	1070.19	339.58	1070.19	339.58	-671.46	-184.64
6	Short Term purchase	240.55	96.22	1275.19	718.44	1275.19	718.44	1034.64	622.22
7	*RGCCPP, Kayamkulam (net)*	*0*	*100*	*0*	*70.65*	0.00	70.65	0.00	-29.35
8	Exchanges/DSM/Swap			*4950.49*	*2544.05*	4950.49	2544.05	4950.49	2544.05
9	*Captive/Solar prosumer injection****			*125.99*	*25*	125.99	25.00	125.99	25.00
10	Transmission charges		760.55		1448.27	0.00	1448.27	0.00	687.72
11	NLDC charges		3			0.00	0.00	0.00	-3.00
12	Other charges in connection with sale				0.25	0.00	0.25	0.00	0.25
13	Provisions for energy availed through banking transactions from outside the State which was not returned during FY 2023 -24					0.00	0.00	0.00	0.00
14	**Total**	**22684.82**	**10564.23**	**25711.31**	**12982.59**	**25711.31**	**12982.59**	**3026.49**	**2418.36**
	Average power purchase cost vide MYT order dated 25.06.2022 (Rs/unit)	4.66							
	Average Power Purchase cost as per true up(Rs/unit)	5.05							
	* Energy at KSEB Bus ** Approved to purchase through Medium Term contracts								
	**Energy settled during the year 2023-24								

4.27 KSEB Ltd during the year 2023-24 had purchased power from the following sources such as;

(a) Central Generating Stations,
(b) Wind, solar, small hydro and other IPPs within the State,
(c) Power Purchase from Thermal IPPs,
(d) Power purchase from IPPs/Traders outside the State,
(e) Power purchase through short term contracts including Traders, exchanges
(f) Power availed through banking arrangements and power transacted through Deviation Settlement Mechanism (DSM)

4.28 The source wise procurement of power is detailed in **Annexure-D4**. The details of approval are attached as **Annexure D2**. Hon'ble Commission may please note that the petitioner had taken earnest efforts to obtain power at the lowest cost from different sources duly observing merit order dispatch principle.

Power Purchase from Central Generating Stations:

4.29 Honourable Commission vide MYT order dated 25.06.2022 approved 11587.81 MU at exbus for a total power purchase cost of Rs 4897.59 Cr @ Rs 4.23 per unit. KSEBL scheduled 11345.41 MU at exbus for a total power purchase cost of Rs 4731.08 Cr at an average rate of Rs 4.17 per unit.

4.30 MoP has allocated 85% power from Telangana STPP to Telangana state & the balance 15% remained unallocated. The First Unit (800MW) of Telangana STPP, Stage I (2x800MW) declared CoD wef 00:00 Hrs of 28.09.2023. Hence, SRPC allocated 0.465364% out of 15% of Power to Kerala from Unit I (800MW), which comes to 3.72 MW and intimated that beneficiaries are required to sign PPA and enter into commercial arrangements/ maintain LC with the allocated power. Accordingly, NTPC raised monthly bills wef September 23, without executing PPA. NTPC intimated the indicative tariff of this station for the first year as Rs.3.81/unit & the levelized tariff as Rs.3.51/unit. The final tariff is yet to be decided by CERC. KSEB Ltd has provisionally admitted the claimed amount as per the allocation of SRPC. The Second Unit (800MW) of Telangana STPP, Stage I (2x800MW) declared CoD wef 00:00 Hrs of 01.03.2024 and thus the allocation has been increased to 7.44 MW. KSEBL scheduled 14.84 MU for a total power purchase cost of Rs 8.01 Cr. KSEBL will submit the PPA for approval before the Honourable Commission shortly.

4.31 A station wise comparison of approved and actual energy purchased from CGS and their costs are given in the Table below.

Table – D13 Power Purchase from Central Generating Stations for the year 2023-24									
		Approved		Audited accounts		True up requirement		Variation	
Sl No	Station	Energy* (MU)	Cost (Rs Cr)	Energy* (MU)	Cost (Rs Cr)	Energy* (MU)	Cost (Rs Cr)	Energy* (MU)	Cost (Rs Cr)
1	RSTPS I & II	1696.81	680.42	1788.93	839.11	1788.93	839.11	92.12	158.69
2	RSTPS III	425.93	170.2	512.24	229.64	512.24	229.64	86.31	59.44
3	Talcher Stage-II	2980.96	1010.68	3215.98	860.28	3215.98	860.28	235.02	-150.40
4	Simhadri TPS Stg II	600.03	288.09	584.36	331.31	584.36	331.31	-15.67	43.22
5	NLC Stage-I	422.33	181.34	252.87	99.45	252.87	99.45	-169.46	-81.89
6	NLC Stage-II	603.17	260.44	291.71	99.44	291.71	99.44	-311.46	-161.00
7	NLC 1 Expansion	464.13	192.42	397.94	141.07	397.94	141.07	-66.19	-51.35
8	NLC II Expansion	511.51	277.52	275.59	127.42	275.59	127.42	-235.92	-150.10
9	MAPS	122.72	32.19	65.90	17.31	65.90	17.31	-56.82	-14.88
10	KAIGA	389.17	139.71	598.08	215.54	598.08	215.54	208.91	75.83
11	Kudankulam	1507.12	641.55	1642.85	749.53	1642.85	749.53	135.73	107.98
12	Vallur	350.72	193.37	263.52	147.14	263.52	147.14	-87.20	-46.23
13	NTPL	506.33	267.56	431.40	219.68	431.40	219.68	-74.93	-47.88
14	Kudgi	780.71	445.65	778.17	539.23	778.17	539.23	-2.54	93.58
15	NNTPP	226.17	116.45	231.02	99.63	231.02	99.63	4.85	-16.82
	Telangana STPP			14.85	8.01	14.85	8.01	14.85	8.01
16	Aravally Corporation Ltd (IGSTPS Jhajjar)				7.29	0.00	7.29	0.00	7.29
	Total	**11587.81**	**4897.59**	**11345.41**	**4731.08**	**11345.41**	**4731.08**	-242.40	-166.51
	*Energy at exbus								

4.32 Table D12 shows that there was an overall reduction in CGS to the extent of 242.40 MU from the approved quantum during the year 2023-24. The actual power purchase cost from CGS was less by Rs 166.55 Cr from the amount approved by the Hon Commission as detailed in table above.

4.33 Hon'ble Commission may kindly note that KSEBL has been scheduling energy based on merit order despatch basis. Accordingly, during low demand period and period of excess hydel generation to avoid spillage, KSEB Ltd has surrendered power from the stations having high variable cost. The reduction in availability is due to the energy surrender during low demand periods as well as for utilising the low market rates by surrendering high-cost power and also due to outages. KSEBL has surrendered 167.43 MU during 2022-23 from Central Generating Stations as shown in table below. The details of energy availability as well as the details of energy surrender are detailed in table below:

Table D14 Details of energy availability and surrender of CGS during 2023-24 (MU)					
CGS	**Approved**	**Actual**	**Surrender**	**Total**	**Increase/decrease in availability after accounting surrender**
RSTPS I & II	1696.81	1788.93	25.32	1814.25	117.44
RSTPS III	425.93	512.24	3.7	515.94	90.01
Talcher Stage-II	2980.96	3215.98	9.58	3225.56	244.60
Simhadri TPS Stg II	600.03	584.36	38.87	623.23	23.20
NLC Stage-I	422.33	252.87	0.16	253.03	-169.30
NLC Stage-II	603.17	291.71	0.51	292.22	-310.95
NLC 1 Expansion	464.13	397.94	0.97	398.91	-65.22
NLC II Expansion	511.51	275.59	0.35	275.94	-235.57
MAPS	122.72	65.90	0	65.90	-56.82
KAIGA	389.17	598.08		598.08	208.91
Kudankulam	1507.12	1642.85		1642.85	135.73
Vallur	350.72	263.52	8.32	271.84	-78.88
NTPL	506.33	431.40	6.89	438.29	-68.04
Kudgi	780.71	778.17	70.59	848.76	68.05
NNTPP	226.17	231.02	2.17	233.19	7.02
Telengana		14.85		14.85	14.85
Total	**11587.81**	**11345.41**	**167.43**	**11512.84**	-74.97

4.34 From the above table it can be seen that the actual availability from Central Generating stations is higher as compared to the approved figures except for NLC stations, NTPL, NTECL and MAPs.

4.35 The reasons for reduction in availability for NLC stations was mainly due to the fuel shortage and outages and. The reasons for reduction in availability is enclosed as **Annexure D3**. The reduction in availability from NLC stations, NTPL, NTECL and MAPs for FY 2023-24 was 984.78 MU.

	Table D 15 Reduction in availability from NLC stations and MAPs	
Sl No	**CGS**	**Reduction in availability after accounting surrender (MU)**
1	NLC Stage-I	-169.30
2	NLC Stage-II	-310.95
3	NLC 1 Expansion	-65.22
4	NLC II Expansion	-235.57
5	Vallur	-78.88
6	NTPL	-68.04
7	MAPS	-56.82
	Total	**-984.78**

4.36 The variations in power purchase cost from that of the approved cost is mainly due to following reasons.

i. Variations in fixed charges due to the revision in capacity charges in the Tariff orders issued by CERC during the control period.
ii. Variations in energy charges due to the revision in prices of domestic coal, transportation charges, change in blending ratio, calorific value etc
iii. Ash evacuation expenses
iv. Arrears on account of revision in fixed charges consequent to the issuance of truing up orders for the period 2014-19.
v. Inclusion of RLDC charges, Incentive, supplementary claims such as water cess, security charges

4.37 In the case of Central Generating Stations, the power purchase cost includes fixed cost, variable charges RLDC charges, incentive claims and the various supplementary claims approved vide corresponding regulations/CERC orders. The fixed Charges and variable charges payable to a thermal generating station for a calendar month is calculated in accordance with the formula specified in the CERC tariff regulation for the control period. The fixed charges of each Central Generating stations is approved by Hon'ble CERC from time to time. As per the Tariff regulations, the fixed charges payable to the generator is proportional to the availability. i.e., the fixed charges paid depends on the actual declared availability of the generating Station.

4.38 It may please be noted that Tariff orders for most of the Central Generating stations for the control period 2019-24 were not issued at the time of filing of MYT petition. Hence KSEB Ltd estimated the fixed cost of CGS whose tariff is not yet approved @95% of the fixed charge claimed by the generators in their tariff petitions for the years 2022-23 to 2023-24, on the assumption that CERC mayapprove 95% of the fixed cost claimed while issuing the final orders.

4.39 KSEB Ltd in the MYT petition has also claimed additional fixed cost towards fly ashtransportation Rs 0.15/unit. Thus the fixed charges approved by the Honourable Commission for the year 2022-23 vide order dated 25.06.2022 is given in table below

Table D16 Fixed charges approved by the Honourable Commission

No.	Station	Energy availability (MU)	Total fixed charges (Rs Cr)
1	RSPTS Stage I & II	1696.81	195.23
2	RSTPS Stage III	425.93	53.83
3	TALCHER - Stage II	2980.96	409.74
4	Simhadri Exp	600.03	105.8
5	NLC-II- Stage-1	422.33	59.36
6	NLC-II- Stage-2	603.17	86.29
7	NLC- Exp- Stage-1	464.13	70.38
8	NLC - II Exp	511.51	135.42
9	Vallur JV with	350.72	75.15
10	NTPL (Tuticorin JV)	506.33	105.27
11	Kudgi Unit I II & III	780.71	156.01
12	NNTPP	226.17	61.5
	Total	**9568.81**	**1513.98**

4.40 The fixed charges excluding the ash transportation expenses approved by Honorable Commission was Rs 1091.72 Cr. and actual fixed charges for the year 2023-24 was Rs 975.31 Cr excluding the ash transportation expenses and ECS. Honorable CERC vide order dated 28.10.2022 on petition No. 205/MP/2021 directed that fly ash transportation cost incurred by the generator, shall be recovered, in proportion to the coal consumed corresponding to the scheduled generation at normative parameters in accordance with the 2019 Tariff Regulations or at actuals, whichever is lower, for the supply of electricity to the respective Discoms. Hence the same is considered separately and is included in supplementary claims.

4.41 Further the cost due to the installation of the emission control system is also approved vide MYT order dated 25.06.2022. As the installation of emission control system had not materialized during FY 2023-24, the expenses are not included in the actuals.

4.42 As detailed above, as per the CERC Tariff Regulations, 2019, the tariff of the CGS and IPPs whose tariff determined by Honorable CERC comprises of fixed charges for recovering the annual fixed charges approved by CERC and variable cost for recovering the primary and secondary fuel cost. The fixed charges/capacity charges payable to thermal stations for a calendar month is calculated in accordance with the formula specified as per Regulation 42(2) of the CERC Tariff Regulations, 2019, which is applicable for the control period from 2019-20 to 2023-24.

4.43 The normative annual plant availability factor (NAPAF) in percentage for coal-based station is 85% and that for NLC Stage-1 stations are 72%. The CERC Regulation provides for recovering the fixed cost of generation under two segments in an year, for high demand season (for three months) and low demand (period of remaining nine months).

4.44 KSEB has remitted the fixed charges in accordance with the formula specified as per Regulation 42(2) of the CERC Tariff Regulations, 2019, which is applicable for the control period from 2019-20 to 2023-24.

4.45 Honorable CERC issued tariff orders for the following stations wherein the fixed charges were revised as follows

Table **D17 Tariff orders subsequent to MYT Petition**

Station	Date of Tariff order	Annual Fixed charges for the station as per order	Fixed charges to be payable by KSEBL at normative availability for FY 2023-24
		Rs Cr	Rs Cr
RSTPS Stage I & II	05.10.2023	1142.6458	138.39
RSTPS Stage-III	07.11.2021	299.83	37.97
Talcher	27.04.2023	1144.5544	246.43
Simhadri Stage II	08.02.2023	1006.0391	83.24
Kudgi	19-05-2024	2788.0238	134.49
NLC II Stage I	10.03.2023	356.5867	36.92
NLC II Stage II	10.03.2023	489.0001	54.11
NLC TPS I Expansion	23.12.2022	282.7566	41.15
NLC TPS II Expansion	3.11.2022	665.9447	96.93
NTECL Vallur	30.05.2023	1778.8261	64.13
NTPL	01.08.2024	1214.046	93.18
NNTPS	02.08.2024	1342.6043	43.82
Telangana STPS - Unit 1		1808.1688	8.44

4.46 Further as part of Truing up claim of CGS, the prior period charges adjusted towards AFC during the year 2023-24 is given below.

Table D18 AFC paid for previous years during FY 2023-24		
Sl No	**Station**	**Capacity Charges (Rs Cr)**
1	RSTPS Stage I & II	29.41
2	Kudgi	-0.68
3	NLC II Stage I (Provision reversed and prior period charges)	-8.99
4	NLC II Stage II (Provision reversed and prior period charges)	-12.70
5	NLC I TPS I Expansion Stage IProvision reversed	-7.88
6	NLC II STPSII exp	3.28
7	Aravally Power Corporation Ltd	7.29
	Total	**9.72**

4.47 Thus, the fixed charges as per accounts after accounting prior period expenses are detailed in table below

Table D19 Total Fixed Charges for FY 2023-24 as per accounts (Rs Cr)					
Sl No	**Station**	**Capacity charges for FY 2023-24**	**Prior period charges**	**Provision reversal**	**Total Fixed charges for FY 2023-24**
1	RSTPS Stage I & II	138.19	29.41		167.60
2	RSTPS Stage-III	37.96			37.96
3	Talcher	246.43			246.43
4	Simhadri Stage II	83.32			83.32
5	Kudgi	134.88	-0.68		134.20
6	NLC II Stage I	25.76	-0.01	-8.98	16.77
7	NLC II Stage II	25.30	-0.05	-12.65	12.60
8	NLC TPS I Expansion	37.85	-0.02	-7.86	29.97
9	NLC TPS II Expansion	53.73	3.28		57.01
10	NTECL Vallur	44.35			44.35
11	NTPL	68.26			68.26
12	NNTPS	40.77			40.77
13	Telangana	2.49			2.492
13	Aravally Power Corporation Ltd		7.29		7.29
					949.01
***Capacity charges in proportion to availability**					

4.48 Thus, the fixed charges approved by Commission and paid by KSEB Ltd during FY 2023-24 is given in table below

Table D 20 Fixed charges of Central Generating Station,				
No.	**Station**	**Energy availability**	**Fixed charges excluding ECS & ash transportation (Rs Cr)**	
		(MU)	**Approved**	**Actual**
1	RSPTS Stage I & II	1726.74	138.81	167.60
2	RSTPS Stage III	494.61	36.59	37.96
3	TALCHER - Stage II	3103.94	264.42	246.43
4	Simhadri Exp	564.01	85.34	83.32
5	NLC-II- Stage-1	244.16	37.44	16.77
6	NLC-II- Stage-2	281.63	52.82	12.60
7	NLC- Exp- Stage-1	384.25	48.15	29.97
8	NLC - II Exp	266.14	110.92	57.01
9	Vallur JV with	254.52	58.06	44.35
10	NTPL(Tuticorin JV)	416.50	84.25	68.26
11	Kudgi Unit I II & III	751.52	126.99	134.20
12	NNTPP	222.98	47.93	40.77
13	Telangana			2.49
14	Aravally Power Corporation Ltd			7.29
	Sub total	**8710.99**	**1091.72**	**949.01**

4.49 In the MYT order dated 25.06.2022, the variable charges of various stations for the control period was calculated by this Hon'ble Commission by giving an annual escalation of 2% on annual basis, on the actual variable charges for the first half of the year 2021-22. There was considerable variation in energy charges during 2023-24 due to the variations in cost of domestic fuel consequent to the price notifications issued after the issuance of MYT order dated 25.06.2022, blending of imported coal, changes in calorific value etc.

4.50 Variation of energy charges depends on the price notified for each mine which are different from mine to mine, grade of coal, blending ratio of imported coal and domestic coal, receipt of coal from different sources, distance from the mine, mode of transport etc.

4.51 In the case of coal-based stations the linkage of fuel is mainly from Coal India Ltd (CIL) and Singareni Collieries Ltd (SCCL). Due to shortage from linkage sources, the generating stations had to augment coal supply from other domestic sources and imported coal as well. Coal from Eastern Coal Fields (ECL) and, SCCL ,SECL is expensive as compared to coal from Mahanadi coal fields (MCL) under Coal India Ltd due to the variations in grade of coal and other reasons. Price of coal from Singareni Collieries (SCCL) is based on the price notifications of Singareni Collieries Ltd from time to time whereas price of Coal India Ltd (CIL) is based on the notification dated 27.11.2020. Fuel price is based on the coal price and the transportation cost. There are frequent revisions in SCCL price notification compared to CIL. Following price revisions have been notified by SCCL subsequent to MYT petition which has contributed to increase in energy charges.

1. SCCL price notification no.128/2021-22 dtd 31.08.2021 wef 01.09.2021.
2. Corrigendum 1 to price notification no,128/2021-22 dtd 05.09.2021.
3. Corrigendum 2 to price notification no,128/2021-22 dtd 14.09.2021.
4. SCCL price notification no.129/2021-22 dtd 30.09.2021 wef 01.10.2021.
5. SCCL price notification no.130/2021-22 dtd 30.10.2021 wef 01.11.2021
6. Corrigendum 1 to price notification no,130/2021-22 dtd 30.11.2021.
7. SCCL price notification no.131/2021-22 dtd 10.01.2022 wef 11.01.2022
8. SCCL price notification no.132/2021-22 dtd 28.02.2022 wef 01.03.2022
9. SCCL price notification no.133/2021-22 dtd 12.05.2022 wef 13.05.2022.
10. SCCL price notification no.134/2021-22 dtd 31.05.2022 wef 01.06.2022
11. Corrigendum 1 to price notification no,134/2021-22 dtd 04.06.2022.
12. Corrigendum 2 to price notification no,134/2021-22 dtd 28.06.2022.
13. Corrigendum 3 to price notification no,134/2021-22 dtd 30.07.2022.
14. Corrigendum 4 to price notification no,134/2021-22 dtd 1.08.2022.
15. Corrigendum 5 to price notification no,134/2021-22 dtd 27.08.2022
16. Corrigendum 6 to price notification no,134/2021-22 dtd 31.08.2022
17. Corrigendum 7 to price notification no,134/2021-22 dtd 05.09.2022
18. Corrigendum 8 to price notification no,134/2021-22 dtd 28.09.2022

19. Corrigendum 9 to price notification no,134/2021-22 dtd 03.10.2022
20. Corrigendum 10 to price notification no,134/2021-22 dtd 31.10.2022
21. Corrigendum 11 to price notification no,134/2021-22 dtd 04.11.2022
22. Corrigendum 12 to price notification no,134/2021-22 dtd 14.11.2022
23. Corrigendum 13 to price notification no,134/2021-22 dtd 30.11.2022
24. Corrigendum 14 to price notification no,134/2021-22 dtd 02.12.2022
25. Corrigendum 15 to price notification no,134/2021-22 dtd 31.12.2022
26. SCCL price notification no.135/2022-23 dtd 31.01.2023 wef 01.02.2023.
27. SCCL price notification no.136/2022-23 dtd 07.02.2023 wef 08.02.2023
28. SCCL price notification no.136/2022-23 dtd 07.02.2023 wef 08.02.2023
29. SCCL price notification no.137/2022-23 wef 01.04.2023
30. SCCL price notification no.138/2022-23 wef 01.05.2023
31. Corrigendum 1 to price notification no 138/2022-23 wef 23.05.2023
32. SCCL price notification no.139/2022-23 wef 07.06.2023
33. Corrigendum 4 to price notification no 139/2022-23 wef 08.09.2023
34. SCCL price notification no.140/2023-24 wef 01.11.2023
35. Corrigendum 3 to price notification no 140/2023-24 wef 01.12.2023
36. Corrigendum 4 to price notification no 140/2023-24 wef 10.01.2024
37. Corrigendum 5 to price notification no 140/2023-24 wef 24.02.2024
38. Corrigendum 6 to price notification no 140/2023-24 wef 02.03.2024
39. Corrigendum 7 to price notification no 140/2023-24 wef 01.04.2024
40. Corrigendum 8 to price notification no 140/2023-24 wef 23.04.2024
41. Corrigendum 9 to price notification no 140/2023-24 wef 23.05.2024
42. Corrigendum 10 to price notification no 140/2023-24 wef 01.06.2024

4.52 It is respectfully submitted that the following notifications were issued by Coal India Ltd and its subsidiaries, subsequent to MYT petition of FY 2022-27, which has contributed to variation in energy charges.

- Notification dated 19.01.2022 by Bharat Coking Coal Ltd revised surface transportation charges
- Notification dated 31.01.2022 by Mahanadi Coal Ltd revised surface transportation charges
- Notification dated 14.03.2022 by Eastern Coal fields Ltd revised surface transportation charges
- Notification by CIL vide order dated 30.05.2023 by revising price of Coal

4.53 Monthly energy charges depend upon the landed price of fuel and its calorific value. The energy charge rate calculated as per Tariff regulations is based on the following formula:

Energy charge rate = {(SHR-SFC*CVSF) *LPPF/CVPF+(SFC*LPSFi)+(LC*LPL}*100/(100-Aux)) Where,

AUX =Normative auxiliary energy consumption in percentage.

CVPF = Weighted Average Gross calorific value of coal as received, in kCal per kg for coal-based stations less 85 Kcal/Kg on account of variation during storage at generating

station; In case of blending of fuel from different sources, the weighted average Gross calorific value of primary fuel shall be arrived in proportion to blending ratio
CVSF = Calorific value of secondary fuel, in kCal per ml;
ECR = Energy charge rate, in Rupees per kWh sent out;
SHR = Gross station heat rate, in kCal per kWh;
LPL = Weighted average landed cost of limestone in Rupees per kg;
LPPF = Weighted average landed fuel cost of primary fuel, in Rupees per kg, per litre or per standard cubic metre, as applicable, during the month.
LPSFi = Weighted Average Landed Fuel Cost of Secondary Fuel in Rs/ml during the month:

4.54 Energy charges depends on gross calorific value of primary (CVPF) and secondary fuel (CVSF) as well as landed price of primary and secondary fuel (LPPF and LPSFi). The parameters CVPF in the equation of energy charges varies depending upon the grade of fuel which in turn depends on the receipt of coal from different sources and also blending of fuel. Different grades of coal have different prices. Thus, the landed price of fuel again depends on grade of fuel, source of fuel, transportation cost (depends on distance from coal source and mode of transport), price notified for each mine which are different from mine to mine, transportation charges notified from time to time, blending ratio etc. The monthly variation of these parameters leads to variation in monthly energy charges. The price of imported coal is much higher as compared to domestic coal.

4.55 Ministry of Power vide notification dated 09.01.2023 has directed all Central,State Gencos and IPPs to take necessary action and immediately plan to import coal through a transparent competitive procurement for blending at the rate of 6% by weight so as to maintain coal stocks at their power plants for smooth operations till September 2023. The domestic coal supply to those Gencos who do not follow these instruction shall be restricted on prorata basis. A copy of notification is enclosed as **Annexure-D5.**Central Electricity Regulatory Commission vide order dated 14.02.2023 allowed blending upto 6% from alternate sources of fuel supply including imported coal, subject to technical feasibility, for the period upto 30.09.2023. The Commission also ordered that the operation of second and third provisions of Sub Regulation 3 of Regulation 43 of 2019 Tariff Regulation shall be kept in abeyance. Ministry of Power vide notification dated 01.09.2023 has extended the advisory dated 09.01.2023 by modifying the mandatory blending to 4 % for the remaining period of 2023-24. Subsequently, vide notification dated 25.10.2023, Ministry of Power ordered to blend imported coal @6% (by weight) minimum till March 2024.

4.56 Ministry of Power vide notification dated 04.03.2024. has decided to extend the advisory dated 25.10.2024 till June 2024.Further, Ministry of Power vide advisory dated 27.06.2024 has decided to extend the advisory by modifying the blending requirement to 4 %(by weight) till 15.10.2024. A copy of notification is enclosed as **Annexure -D5.**

4.57 Hence the variable charge increased considerably during FY 2023-24. Thus, the actual per unit variable cost of power purchase from various sources varied from the approved level and this matter was well considered by the Honourable Commission, by issuing regulations for claiming fuel surcharge.

4.58 Table below shows the variation in approved and actual energy charges (excluding ECS) is given in table below

Table D21 Energy Charges for Central Generating Station for FY 2023-24					
Sl No	**Particulars**	**Rate (Rs/kWh)**		**Variable Charges (Rs Cr.)**	
		Approved	**Actual***	**Approved**	**Actual**
1	RSTPS I & II	2.86	3.77	485.19	674.35
2	RSTPS III	2.73	3.68	116.37	188.73
3	Talcher Stage-II	2.02	1.74	600.94	558.85
4	Simhadri TPS Stg II	3.04	4.00	182.29	233.75
5	NLC Stage-I	2.89	3.27	121.98	82.59
6	NLC Stage-II	2.89	2.98	174.15	87.03
7	NLC 1 Expansion	2.63	2.71	122.04	107.97
8	NLC II Expansion	2.78	2.68	142.10	73.96
9	MAPS	2.62	2.63	32.19	17.31
10	KAIGA	3.59	3.60	139.71	215.54
11	Kudankulam	4.26		641.55	749.53
12	Vallur	3.37	4.56	118.21	103.08
13	NTPL	3.21	3.53	162.29	152.46
14	Kudgi	3.71	5.20	289.64	404.60
15	NNTPS	2.43	2.53	54.95	58.51
16	Telangana		3.71		5.51
	Sub Total			**3383.62**	**3713.77**

***Weighted average rate**

4.59 An amount of Rs 40.84 Cr. was remitted towards the differential income tax on return on equity for Kudamkulam Station is included in the energy charges.

4.60 In the case of NLC stations, Honourable CERC directed NLCIL to consider the lignite transfer price @ Rs 2132.24/MT pertaining to previous control period during the year as an interim measure. Subsequently Honourable CERC vide order dated 4.05.2023 in Petition No.223/MP/2022 has approved the interim lignite Input price for the period 2019-2024 in respect of Neyveli Mines as per CERC Terms and Conditions of Tariff), Second Amendment, Regulations,2021. The interim Lignite transfer price for the control period based on CERC order dated 04.05.2023 is given in table below

	Table D 22 Impact of revision in lignite transfer price on energy charges		
Sl No	**Year**	**Lignite transfer price considered for the energy charge in MYT petition (Rs/Tonne)**	**Revised Lignite transfer price based on CERC dated 04.05.2023 (Rs/Tonne)**
1	2019-20	Rs2132.24/Tonne	Rs 2266.23/Tonne
2	2020-21	Rs2132.24/Tonne	Rs 2324.02/Tonne
3	2021-22	Rs2132.24/Tonne	Rs 2383.27/Tonne
4	2022-23	Rs2132.24/Tonne	Rs 2407.79/Tonne
5	2023-24	Rs2132.24/Tonne	Rs2506.38/Tonne

4.61 Further ash transportation expenses remitted to various generators are given in table below

Table D 23(a) Ash transportation expenses for the year FY 2023-24	
Station	**Expenses (Rs Cr)**
RSTPS I & II	1.449
RSTPS III	0.415
Simhadri II	7.213
Talcher-II	30.042
Sub Total	**39.12**

4.62 Further the actual power purchase cost includes RLDC charges, incentive claims, and various supplementary claims approved vide corresponding regulations/CERC orders which is not considered while approving the ARR of FY 2023-24 vide order dated 25.06.2022. The details of supplementary claims including ash transportation charges is given in table below

Table D23 (b)Supplementary claims (Rs Cr)									
Sl No	**Station**	**RLDC**	**Incentive**	**Ash Transportation expenses**	**Deviation charges**	**IT**	**FERV**	**Others**	**Total**
1	RSTPS Stage I & II	-0.29	0.35	1.45	0.00	0.02	0.00	-4.38	-2.84
2	RSTPS Sage III	-0.07	2.27	0.41	0.00	0.63	0.00	-0.29	2.95
3	Talcher	-1.45	11.32	30.04	0.00	3.96	0.00	11.13	55.00
4	Simhadri	0.00	0.32	7.21	0.00	0.00	5.22	1.49	14.24
5	Kudgi	0.06	0.15	0.00	0.00	0.00	0.00	0.23	0.44
6	NNTPP	0.03	0.23	0.00	-0.0858	0.0000	0.1362	0.04	0.35
7	NLC- II Stage I	0.06	0.00	0.00	0.01	0.02	0.02	-0.01	0.10
8	NLC- II Stage II	0.08	0.00	0.00	0.05	0.02	0.03	-0.37	-0.19
9	NLC TPS-I Expansion	0.05	0.78	0.00	-0.01	1.50	1.21	-0.41	3.12
10	NLC TPS-II Expansion	0.06	0.00	0.00	-0.09	0.00	0.02	-3.53	-3.55
11	NTECL	0.04	0.21	0.00	0.00	0.00	0.00	-0.54	-0.28
12	NTPL	0.08	0.00	0.00	-0.56	0.00	0.00	-0.56	-1.04
13	TELANGANA	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Sub Total	**-1.35**	**15.64**	**39.12**	**-0.68**	**6.15**	**6.63**	**2.79**	**68.30**

4.63 In the case of Talcher electricity duty is also included in other charges. In addition to the above as part of truing up for the period 2014-19, Rs 7.29 crores towards Aravally Corporation Ltd was paid towards revision of fixed charges .

4.64 All the above factors contribute to variation in the per unit power purchase cost of FY 2023-24 and these are beyond the control of utility. Thus, the total power purchase cost of Central Generating Stations is given in table below

Table D 24 Total Power Purchase Cost of Central Generating Stations (Rs Cr)

Particulars	Capacity Charges	Provision/prior period under the head of capacity charges	Energy Charges	Other supplementary claims	Total
	Rs Cr				
RSTPS I & II	167.60		674.35	-2.84	839.11
RSTPS III	37.96		188.73	2.95	229.64
Talcher Stage-II	246.43		558.85	55.00	860.28
Simhadri TPS Stg II	83.32		233.75	14.24	331.31
NLC Stage-I	25.76	-8.99	82.59	0.10	99.45
NLC Stage-II	25.30	-12.70	87.03	-0.19	99.44
NLC 1 Expansion	37.85	-7.88	107.97	3.12	141.07
NLC II Expansion	53.73	3.28	73.96	-3.55	127.42
MAPS			17.31		17.31
KAIGA			215.54		215.54
Kudankulam			749.53		749.53
Vallur	44.35		103.08	-0.28	147.14
NTPL	68.26		152.46	-1.04	219.68
Kudgi	134.20		404.60	0.44	539.23
NNTPS	40.77		58.51	0.35	99.63
Telangana	2.49		5.51	0.00	8.01
Aravally Power Corporation td	7.29			0.00	7.29
Total	**975.31**	**-26.30**	**3644.16**	**68.30**	**4731.08**

4.65 Thus, the power purchase cost is dependent on Tariff Regulations and variations in coal price from time to time and is an uncontrollable factor. Therefore, Hon'ble Commission may kindly approve the power purchase cost as submitted in the table below. Further KSEBL humbly requests before the Hon'ble Commission to kindly allow the revisions in charges on the issuance of Tariff orders for remaining Central Generating Stations as and when it materialises.

Table – D25 True up claim of Power Purchase from Central Generating Stations for the year 2023-24			
Sl No	**Station**	**Energy* (MU)**	**Cost (Rs Cr)**
1	RSTPS I & II	1788.93	839.11
2	RSTPS III	512.24	229.64
3	Talcher Stage-II	3215.98	860.28
4	Simhadri TPS Stg II	584.36	331.31
5	NLC Stage-I	252.87	99.45
6	NLC Stage-II	291.71	99.44
7	NLC 1 Expansion	397.94	141.07
8	NLC II Expansion	275.59	127.42
9	MAPS	65.90	17.31
10	KAIGA	598.08	215.54
11	Kudankulam	1642.85	749.53
12	Vallur	263.52	147.14
13	NTPL	431.40	219.68
14	Kudgi	778.17	539.23
16	NNTPS	231.02	99.63
17	Telangana	14.85	8.01
18	Aravally Power Corporation td		7.29
	Total	**11345.41**	**4731.08**
	*Energy at exbus		

Renewable Energy

Power Purchase from wind and small hydro and solar IPPs within the State:

4.66 KSEBL had purchased 588.73 MU for Rs.203.61 Cr against the approved quantum of 621.18 MU for Rs 204.02 Cr. The details of approval are furnished in Annexure D2 of this petition. The purchase of power from Independent Power Producers /CPP injection/solar prosumer injection is given in table below.

Table-D26: Power Purchase from IPPs

Sl No		KSERC approval		Audited Accounts		Truing up requirement		Difference	
	Station	Energy (MU)	Cost (Rs Cr)	Energy (MU)	Cost (Rs Cr)	Energy (MU)	Cost (Rs Cr)	Energy (MU)	Cost (Rs Cr)
	IPPS inside the State								
1	Wind	122.98	46.01	115.92	42.73	115.92	42.73	-7.07	-3.27
2	SHEP	141.71	50.74	141.44	58.52	141.44	58.52	-0.25	7.79
3	Solar	324.04	106.87	405.02	129.26	405.02	129.26	80.97	22.39
4	Waste to Energy	19.44	12.27			0	0	-19.44	-12.27

Sl No	Station	KSERC approval		Audited Accounts		Truing up requirement		Difference	
		Energy (MU)	Cost (Rs Cr)	Energy (MU)	Cost (Rs Cr)	Energy (MU)	Cost (Rs Cr)	Energy (MU)	Cost (Rs Cr)
	IPPS inside the State								
Procurement from outside the State									
	SECI-Wind	472.8	133.33	212.23	61.75	212.23	61.75	-260.57	-71.58
	RSAPL -SECI	522.01	138.424	2.09	0	2.09	0	-519.93	138.424
	TP Sourya			193.49	47.33	193.49	47.33	193.49	47.33
	Sub total	**1602.98**	**487.624**	**1070.18**	**339.59**	**1070.18**	**339.59**	**-532.8**	**-148.034**

Wind IPPs

4.67 Hon'ble Commission approved 122.98 MU from various wind IPPs inside the State for a total cost of 46 Cr against which, KSEBL purchased 115.92 MU for a total cost of Rs 42.73 Cr.

4.68 Further KSEBL entered PSA with SECI to purchase 200 MW of wind energy on long Term basis for a period of 25 years. 75 MW of power starts scheduling from March 2021 onwards. Honourable Commission vide order dated 19.02.2020 approved Power Sale Agreement (PSA) for 200 MW wind power with SECI dated 14.06.2019 and its amendment dated 12.12.2019, for a period of 25 years from the Scheduled Commissioning Date (SCD) or from the date of full commissioning of the projects, whichever is earlier, at the tariff of Rs 2.89 per unit. KSEBL procured 212.23 MU @ 2.89 per unit during the year 2023-24 for a power purchase cost of Rs 61.75 Cr which is inclusive of LC charges amounting to Rs 121291/-

Table - D27: Power Purchase from Wind IPPs

Station	IC	KSERC approval		Audited Accounts		TU requirement		Difference	
	MW	(MU)	(Rs Cr)	(MU)	(Rs Cr)	(MU)*	(Rs Cr)	(MU)	(Rs Cr)
1. Wind sources within the State									
Wind-Ramakkalmedu	14.25	27.87	8.75	25.02	7.860	25.02	7.86	-8.59	-2.62
Wind-Koundikkal	13.80	12.56	3.94	9.72	11.54	9.72	11.54		
Wind- Agali	4.80	29.72	9.33	26.82		26.82			
Wind Ahalya	8.40	20.98	10.97	18.29	8.60	18.29	8.60	-2.69	-2.37
INOX	16.00	30.84	12.61	34.59	14.15	34.59	14.15	3.75	1.54
Kosamattom	1.00	1.01	0.40	1.47	0.58	1.47	0.58	0.46	0.18
Wind Subtotal (within state)	58.25	122.98	46.00	115.91	42.73	115.92	42.73	-7.07	-3.27
2.Wind sources outside the State									
Wind Power through SECI	75	472.8	133.33	212.23	61.75	212.23	61.75	-260.57	-71.58
Wind Sub total	**133.25**	**595.78**	**179.33**	**328.14**	**104.48**	**328.14**	**104.48**	**-267.64**	**-74.85**

4.69 Even though KSEB Ltd entered PPA with SECI for 200 MW of wind power, only 75 MW is made available. This results in the reduction in availability.

4.70 The true up claim for FY 2023-24 towards cost of power purchase from wind generating stations amounts to Rs 104.42 crores may kindly be approved.

Small Hydro IPPs

4.71 Hon'ble Commission vide order dated 25.06.2022 approved 141.71 MU from various small hydel IPPs @ of Rs 50.74 Cr. KSEBL has procured 164.54 MU at a cost of Rs 58.52 Cr. Power Purchase from M/s Arippara was not approved vide MYT order dated 25.06.2022. Subsequently Honourable Commission vide order dated 08.03.2023 has approved the levelised tariff for the electricity generated from the 4.5 MW SHEP installed by M/s CIAL Infrastructures Ltd at Arippara, @ Rs 4.30/unit duly considering the benefit of accelerated depreciation. Arippara SHEP (4.5 MW) injected 12.29 MU to the grid during FY 2022-23. New SHEP MUKKUDAM of 4 MW capacity started injecting power to the grid from October 2023 onwards.

4.72 Thus the true up claim of Rs 58 .52 crores towards the power purchase cost of IPPs may kindly be approved. The station wise details of approved and actual power purchase quantum, power purchase cost is shown in table below .

Table-D28: Power Purchase from SHEP IPP

Station	Installed capacity	KSERC approval		Audited Accounts		TU requirement		Difference	
		(MU)	(Rs Cr)	(MU)	(Rs Cr)	(MU)*	(Rs Cr)	(MU)	(Rs Cr)
Ullunkal (EDCL)	7	17.79	4.34	17.78	4.34	17.78	4.34	-0.01	0
Iruttukanam Stage-I & II (Viyyat)	4.5	24.69	6.67	23.61	5.39	23.61	5.39	-1.08	-1.28
Karikkayam HEP (AHPL)	15	37.56	15.62	39.74	16.53	39.74	16.53	2.18	0.91
Meenvallom (PSHCL)	3	8.44	4.12	6.85	3.34	6.85	3.34	-1.59	-0.78
Kallar of Idukki District Panchayat	0.05	0	0	0	0	0	0	0	0
Mankulam of Grama Panchayat	0.11	0	0	0	0	0	0	0	0
Minar Renewable energy project Ltd	8	25.54	8.91	18.13	6.33	18.13	6.33	-7.41	-2.58
Deviar micro	0.05	0	0	0	0	0	0	0	0
Anakkampoil	8	27.67	11.07	22.59	9.03	22.59	9.03	-5.08	-2.04
Arippara	4.5	0	0	10.59	12.64	10.59	12.64	10.59	12.64
Mukkadom	4	0	0	2.14	0.92	2.14	0.92	2.14	0.92
SHEP subtotal	**54.21**	**141.69**	**50.73**	**141.43**	**58.52**	**141.43**	**58.52**	**-0.26**	**7.79**

Solar IPPs

4.73 Hon'ble Commission approved 324.05 MU from solar IPPs inside the State for a total cost of Rs 106.87 crores. The detail of approval is given in Appendix II. KSEBL purchased 405.03 MU from various solar IPPs inside the State as shown in table below for a total power purchase cost of Rs 129.254 crores. Further Honorable Commission approved 522.02 MU

for a total power purchase cost of Rs 138.424 Cr for purchase of solar power from generators outside the State. KSEBL scheduled 195.58 Mu as given in table below for a total power purchase cost of Rs 47.33 Cr. Honourable Commission may please note that RSAPL ANTHA NTPC scheduled power from March 2024 onwards and the corresponding power purchase cost is accounted only during 2024-25. Kas KSEB Ltd is claiming the power purchase as per accounts, Honourable Commission may kindly approve the power purchase cost of the same during truing up of FY 2024-25. In the MYT order dated 25.06.2022, Honourable Commission has not approved power purchase cost of 10 MW solar project owned by M/s CIAL. Subsequently Honourable Commission vide order dated 10.11.2022, approved the short-term procurement of 10MW Solar power from M/s CIAL Infrastructure Limited @Rs 2.37/unit for the period from 15.06.2022 to 31.03.2023 to meet the Solar RPO for the financial year 2022-23. This was further extended for a period of two years from 01-04-2023 to 31-03-2025@ Rs 2.37 per unit vide order 18.01.2024 in OP No 86/2023.KSEBL purchased 12.434 MU for a total power purchase cost of Rs 2.95 crores.

4.74 The PPA with M/s ANERT was signed on 04.12.2019 and submitted before Honourable Commission for approval. As per the order of Honourable Commission dated 28.08.2021 and 08.10.2021 KSEBL has initialled modified PPA with M/s ANERT and submitted the same before Honourable Commission for approval on 30.11.2021 @ Rs 2.69 per unit as ceiling rate, but the tariff of the petition could not be determined for want of details. Honourable Commission vide order dated 25.03.2022 disposed the petition and directed M/s ANERT to file a fresh petition with complete expenditure details relevant for the determination of project specific tariff of the electricity generated from 2 MW Solar plant at Kuzhal mannam of the petitioner as per the provisions of the KSERC (Conduct of Business) Regulations 2003 and its amendments. KSEBL had been remitting the payment till June 2022 @ Rs 2.69 per unit. Subsequently, Honourable Commission approved the tariff of the project @ Rs 2.69 per unit vide order dated 30-01-2024. But the payment for the year FY 2023-24 was released only during FY 2024-25 for an amount of Rs. 1.16 Cr. Hence the same is not accounted during FY 2023-24 and as a result KSEBL will be claiming the amount only in FY 2024-25. KSEBL humbly request Honourable Commission to kindly consider the same at the time of truing up FY 2024-25

4.75 The station wise details of approved and actual power purchase quantum, power purchase cost is shown in table below

Table-D 29 Power Purchase from Solar IPPs

Station	MW	KSERC approval		Audited Accounts		TU requirement		Difference	
		Energy (MU)	PP Cost (Rs Cr)	Energy (MU)	PP Cost (Rs Cr)	Energy (MU)*	PP Cost (Rs Cr)	MU)	(Rs Cr)
Solar Power inside the State									
Solar IREDA	50.00	73.37	28.10	77.81	29.80	77.81	29.80	4.44	1.70
Solar IPP ANERT	2	0	0	2.50		2.50	0	2.50	0.00

Station	MW	KSERC approval		Audited Accounts		TU requirement		Difference	
		Energy	PP Cost	Energy	PP Cost	Energy	PP Cost		
		(MU)	(Rs Cr)	(MU)	(Rs Cr)	(MU)*	(Rs Cr)	MU)	(Rs Cr)
THDCIL	50.00	73.37	22.74	95.77	29.69	95.77	29.69	22.40	6.95
NTPC Solar	92	177.31	56.03	212.91	62.60	212.91	62.60	35.60	6.57
CIAL Ettukudukka	10	0	0	16.03	7.17	16.03	7.17	16.03	7.17
Solar IPP sub total(i/s the State)	**204.00**	**324.05**	**106.87**	**405.03**	**129.254**	**405.03**	**129.254**	**80.98**	**22.38**
Power purchase from outside the State									
RSAPL -ANTHA NTPC		522.02	138.424	2.09	0	2.09	0	-326.44	-91.0973
TP Sourya				193.49	47.32672	193.49	47.32672		
Total		**846.07**	**245.294**	**600.61**	**176.58**	**600.61**	**176.58**	**-245.464**	**-68.7129**

4.76 KSEBL humbly requests the Honourable Commission to approve the true up claim of Rs 176.58 crores towards the power purchase from solar IPPs as given in table above.

Captive Power producers/Solar prosumers

4.77 Captive power producers injected energy into the grid and availed energy through banking at their consumer premises. In addition to the above, solar prosumers also injected energy into the grid and consumed energy from the grid through banking arrangements. The energy injected by the captive consumer /solar prosumer is taken as export to the grid (injection to grid) and energy adjusted against this injection is considered as the energy drawal from grid ie; energy import from grid for loss calculation purposes.

4.78 The details of energy generation export to the grid and consumption adjusted against the injection is depicted in table D4 of the petition.

4.79 The energy in the bank at the end of settlement period is considered for settlement purposes. Hon'ble Commission vide order dated 28.062024 on OP No.85/2023 has approved the APPC @Rs 3.15/unit for the year 2023-24 for settling the surplus banked energy as on 31.03.2024. The remaining energy in bank as on 31.03.2024 for solar prosumers was 121.85 MU and corresponding amount was Rs 38.38 Cr. The settled energy as on 31.03.2024 for captive consumers was 4.14 MU and corresponding settlement amount was Rs 1.31 Crores. The settlement amount was being refunded to the consumers after adjusting against the arrears during the first quarter of FY 2024-25 as per the agreement. Thus the energy settled at the end of settlement period for captive/solar prosumers was 125.99 MU and the total settlement amount at APPC rate of Rs 3.15 per unit is Rs 39.69 Cr. An amount of Rs 25 crores were accounted against this during 2023-24 which was distributed during the year FY 2023-24 towards the settlement charges for FY 2022-23. Table D30 shows the details of settlement amount against solar injection/captive injection.

Table D30 Total settlement amount paid during FY 2023-24 against captive /solar injection			
Sl No	Source	Energy (MU)	Amount (Rs cr)
1	Grid connected -solar(settlement)	121.85	**38.38**
2	Captive consumer settlement	4.14	1.31
3	Total	125.99	39.69
4	Amount accounted during 2023-24		25.00

Thermal IPPs

Power purchase from RGCCPP, Kayamkulam:

4.80 Hon'ble Commission, vide order dated 25.06.2022, did not approve any schedule from RGCCPP for the year 2022-23 and approved Rs 100 crores as the annual fixed charges of RGCCPP for FY 2023-24. KSEBL did not schedule energy from RGCCPP during FY 2023-24. The auxiliary consumption of the station supplied by KSEBL during non-operative periods was 5.90 MU. KSEBL remitted Rs 98.16 crores towards RGCCPP during FY 2022-23 towards various charges as given in table below.

Table :- D 31 Power Purchase cost towards RGCCPP for the year 2023-24					
Particulars	**Capacity Charges**	**Deviation charges**	**IT**	**Others**	**Total**
	Rs Crores				
RGCCPP	100.00	-31.85	2.47	0.03	70.65

4.81 Hon'ble Commission may kindly approve Rs 70.65 Cr as per the accounts of KSEBL for the year 2023-24.

4.82 The details of purchase of energy from IPPs inside the State and renewable energy from outside the State and the truing up requirement is summarized as below:

Table-D32: Power Purchase from IPPs								
	KSERC approval		**Audited Accounts**		**Truing up requirement**		**Difference**	
Station	**Energy (MU)**	**Cost (Rs Cr)**	**Energy (MU)**	**Cost (Rs Cr)**	**Energy (MU)**	**Cost (Rs Cr)**	**Energy (MU)**	**Cost (Rs Cr)**
RGCCPP	0.00	0.00	0.00	70.65	0.00	70.65	0.00	70.65
Wind	595.78	179.33	328.14	104.48	328.14	104.48	-267.64	-74.85
SHEP	141.69	50.73	141.44	58.52	141.44	58.52	-0.25	7.79
Solar	846.07	245.29	600.61	176.58	600.61	176.58	-245.46	-68.71
Waste to Energy	19.44	12.27					-19.44	-12.27
Grid connected solar prosumers /captive (settlement)			125.99	25.00	125.99	25.00	125.99	25.00
Subtotal	1602.98	487.62	1196.18	435.23	1196.18	435.23	-406.80	-52.40

IPPs outside state through approved firm contracts:

4.83 KSEBL, with the approval of Hon'ble Commission, entered into agreements with various independent power producers outside the State. The details of approval are attached as Appendix D2.

Power Purchase from Maithon and DVC stations whose tariff is determined by CERC

4.84 Hon'ble Commission had approved 3123.49 MU for a total amount of 1510.37 Cr from various generators against which KSEBL had purchased 3391.62 MU for a total cost of Rs 1495.45 Cr from various contracts. The details of cost of power purchase through long term contracts as per audited accounts are given in table below.

Table-D33: Power Purchase thru various LTA as per audited accounts approved for FY -24

Source	KSERC approval		Actual		Truing up		Difference	
	Energy* (MU)	Cost (Rs Cr)	Energy* (MU)	Cost (Rs Cr)	Energy* (MU)	Cost (Rs Cr)	Energy* (MU)	Cost (Rs Cr)
Maithon Power Ltd-I	1034.96	483.68	1169.90	493.84	1169.90	493.84	134.94	10.16
Maithon Power Ltd-II	1034.96	483.68	1169.90	493.84	1169.90	493.84	134.94	10.16
DVC Mejia	705.51	362.56	724.24	350.75	724.24	350.75	18.73	-11.81
DVC RTPS	348.06	180.45	327.58	157.01	327.58	157.01	-20.48	-23.44
Subtotal	**3123.49**	**1510.37**	**3391.62**	**1495.45**	**3391.62**	**1495.45**	**268.13**	**-14.92**
* Energy at ex bus								

Maithon Power Ltd: - (150 MW through PPA-I and 150 MW through PPA-II)

4.85 Honorable Commission may please note that KSEBL scheduled 2339.80 MU for a total power purchase cost of Rs 987.68 Cr against approved quantum of 2069.92 MU for a total power purchase cost of Rs 967.36 Cr from Maithon Power Ltd.The details of power purchase from Maithon Power Ltd are given below

Table-D34: Power Purchase thru various LTA as per audited accounts approved for FY -24

Source	KSERC approval		Actual		Rate(Rs/Kwh)	
	Energy* (MU)	Cost (Rs Cr)	Energy* (MU)	Cost (Rs Cr)	Approved	Actual
Maithon Power Ltd-I	1034.96	483.68	1169.90	493.84	4.67	4.22
Maithon Power Ltd-II	1034.96	483.68	1169.90	493.84	4.67	4.22
Total	2069.92	967.36	2339.80	987.68	4.67	4.22

*Energy at exbus

4.86 The details of fixed charge ,energy charge and other charges for Maithon Power Ltd is given in table below.

Table D 35 Power Purchase from Maithon Power Ltd		
Particulars	**Maithon Power Ltd -PSA I**	**Maithon Power Ltd -PSA II**
Energy at exbus (MU)	1169.90	1169.90
Energy at Kerala periphery (MU)	1129.31	1129.31
Capacity Charges (Rs. Cr.)	153.47	153.47
Energy Charges (Rs. Cr.)	327.77	327.77
RLDC Charges (Rs. Cr.)	0.18	0.18
Incentive (Rs. Cr.)	6.30	6.30
Ash transportation expenses (Rs Cr)	4.89	4.89
CERC Fees 2024-25(Rs Cr.)	0.07	0.07
Provision for FY 2023-24	0.88	0.88
LC charges (Rs Cr)	0.28	0.28
Total	493.84	493.84

DVC Mejia and DVC RTPS

4.87 Honourable Commission may please note that KSEBL scheduled 724.24 MU for a total power purchase cost of Rs 350.75 Cr against the approved quantum of 705.51 MU for a total power purchase cost of Rs 362.56 Cr from 100 MW of power contracted from Mejia Power Station . Further, 327.58 MU was scheduled from 50 MW of power contracted from Reghunathpur TPS for a total power purchase cost of Rs 157.01 Cr. against 348.06 MU approved for a total power purchase cost of Rs 180.45 Cr

Table-D36(a): Power Purchase thru various LTA as per audited accounts approved for FY -24						
Source	**KSERC approval**		**Actual**		**Rate (Rs/Kwh)**	
	Energy* (MU)	**Cost (Rs Cr)**	**Energy* (MU)**	**Cost (Rs Cr)**	**Approved**	**Actual**
DVC Mejia	705.51	362.56	724.24	350.75	5.14	4.84
DVC RTPS	348.06	180.45	327.58	157.01	5.18	4.79
Subtotal	1053.57	543.01	1051.82	507.77	5.15	4.83

4.88 It may please be noted that Honorable CERC vide order dated 27-04-2023 approved the tariff of Mejia Thermal Power Station for the control period. The annual fixed charges for the control period for the year 2023-24 was revised to Rs 1027.80 Cr. The variable charges also increased significantly due to the blending of imported coal as per MOP guidelines.

4.89 It may please be noted that Honorable CERC vide order dated 27-04-2023 approved the tariff of Reghunathpur Thermal Power Station for the control period. There was a reduction in availability for Regunathpur TPS during FY 2023-24 which was 77 % only during low demand season. This results in reduction in fixed charges. The annual fixed charges approved by Honourable CERC for the year 2023-24 was Rs 1339.48 crores. In addition to the above, an amount of Rs 4.57 crores for the period from 2014-19 towards revision of AFC in accordance with the issuance of truing up orders. The variation in variable charges is due to the blending of Imported coal. Thus, the variation in coal prices and transportation charges also contributes to variation in energy charges Honourable Commission may

please note that FGD was implemented for DVC RTPS and the corresponding expenditure amounts to Rs 0.56 Cr. The details of power purchase cost of M/s DVC Mejia and DVC RTPS are given in table below.

Table D 36(b) Power purchase cost details for MTPS and RTPS			
SL No	**Particulars**	**DVC MTPS**	**DVC RTPS**
1	Energy at exbus(MU)	724.24	327.58
2	Energy at Kerala periphery (MU)	699.19	316.24
3	Capacity Charges (Rs. Cr.)	102.78	44.58
	Fixed charges for FY 2023-24	138.52	
	Prior period provision reversal	35.74	
4	Energy Charges (Rs. Cr.)	251.23	112.04
5	Incentive (Rs Cr)	0.25	
6	RLDC Charges (Rs. Cr.)	0.14	0.00
7	Ash evacuation expenses (Rs Cr)	2.80	
8	FERV (Rs Cr.)	0.05	
9	LC charges (Rs Cr)	0.16	0.08
10	Other supplementary claims	-6.66	0.30
11	**Total**	**350.75**	**157.01**

Power contracted through approved DBFOO contracts

4.90 Hon'ble Commission approved power procurement of 2246.94 MU at exbus from 200 MW of Jindal Power Ltd Bid I and 100 MW of M/s BALCO under Bid II of DBFOO contracts for a total cost of Rs 979.45Cr from projects under DBFOO basis for the financial year 2023-24. KSEBL procured 2321.45 MU at exbus for a total power purchase cost as per audited accounts of Rs 872.15 Cr.

Table-D37: Power Purchase thru approved DBFOO contracts								
Source	**KSERC approval**		**Actual**		**Truing up**		**Difference**	
	Energy* (MU)	**Cost (Rs Cr)**	**Energy* (MU)**	**Cost (Rs Cr)**	**Energy* (MU)**	**Cost (Rs Cr)**	**Energy* (MU)**	**Cost (Rs Cr)**
Jindal Power Ltd-Bid I	1497.96	636.8	1545.72	567	1545.72	567	47.76	-69.60
BALCO	748.98	342.65	775.73	304.9415	775.73	304.94	26.75	-37.71
Subtotal	**2246.94**	**979.45**	**2321.452**	**872.1461**	**2321.452**	**872.1461**	74.51	-107.30
* Energy at ex bus								

4.91 In the MYT order dated 25.06.2022, Hon'ble Commission provisionally approved fixed charges as per the provisions of PSA by applying a depreciation of 2 % on the base fixed charges of previous year. Additional expenditure towards the installation of ECS and fly ash transportation is also included in the fixed charges. The variable charges is approved by applying the provisions of PSA on the average variable cost of first half of FY 2021- 22. The details of power purchase approved by Honourable Commission vide order dated 25.06.2022 is given in table below

Table D 38 Power purchase Cost approved for Commission approved PPA through DBFOO contracts (Rs Cr)								
Trader/Source	**Energy at exbus**	**Energy at Kerala periphery**	**Fixed charges at exbus**	**Fixed charges**	**Variable charges at exbus**	**Variable charges**	**Other charges**	**Total**
		A		**B**		**C**	**D**	**E**
		Rs Cr.	**Rs/kWh**	**Rs Cr.**	**Rs/kWh**	**Rs Cr.**	**Rs Cr.**	**Rs Cr**
Jindal Power Ltd-Bid I	1,497.96	1446.58	2.94	440.03	1.44	216.08	-19.31	636.8
Balco	748.98	723.29	3.46	258.96	1.29	96.81	-13.12	342.65
Sub Total	2246.94	2169.87		698.99		312.89	-32.43	979.45

4.92 The actual power purchase cost for these stations for FY 2023-24 are as follows

Table D 39 Power purchase Cost for Commission approved PPA (Rs Cr)									
Trader/Source	**Energy at exbus**	**Energy at Kerala periphery**	**Fixed charges at exbus**	**Fixed charges**	**Variable charges at exbus**	**Variable charges**	**Incentive/ Damages**	**Other charges**	**Total**
		A	**B**	**C**		**D**	**E**	**F**	**G**
	MU	**Rs Cr.**	**Rs/kWh**	**Rs Cr.**	**Rs/kWh**	**Rs Cr.**	**Rs Cr.**	**Rs Cr.**	**Rs Cr**
Jindal Power Ltd-Bid I	1,545.72	1,492.10	2.49	371.22	1.55	231.82	11.02	-46.85	567.20
Balco	775.73	748.83	3.25	243.06	1.05	78.74	7.74	-24.59	304.94
Total	**2,321.45**	**2,240.93**		**614.28**		**310.56**	**18.76**	**-71.45**	**872.15**

4.93 The increase in variable charges for M/s Jindal Power Ltd is due to the increase in fuel cost due to the usage of imported coal during 2023-24. Thus, the total cost of Rs 872.15 crores may kindly be approved.

Power contracted through provisionally approved/ unapproved DBFOO contracts

4.94 Honourable Commission vide order dated 25.06.2022 did not permit KSEB Ltd to continue scheduling of power from the following four provisionally approved/ unapproved contracts, (i) Purchase of 115MW of Power from Jhabua Power Ltd of L2 of Bid-1, (ii) 150 MW from Jindal Power Ltd of Bid-2, (iii) 100MWfrom Jindal India Thermal Power Ltd of Bid-2 and (iv) 100 MW of power from Jhabua Power Ltdof Bid-2. In order to avoid any power shortage and other contingencies, the Commission permitted KSEB Ltd to make necessary arrangements for procuring 465 MW of equivalent power on medium term basis through competitive bidding as per Section 63 of the Electricity Act, 2003and directed KSEBL to make necessary arrangements for procuring 465 MW from power plants through competitive bidding route who have assigned coal linkageas per the SHAKTI policy of the Central Government. In the MYT order Honourable Commission estimated the cost of power purchase for the 3363.29MU of power at Kerala periphery in lieu of the four unapproved DBFOO contracts at Rs 4.00/ kWh for the year 2022-23 for a total cost of Rs 1372.22 crores.

The relevant portion is extracted below

"6.112: In view of the foregoing facts and developments, the Commission cannot permitKSEB Ltd to continue scheduling of power from the following four provisionally approved/ unapproved contracts, (i) Purchase of 115MW of Power from Jhabuapower Ltd of L2 of Bid-1, (ii) 150 MW from Jindal Power Ltd of Bid-2, (iii) 100MWfrom Jindal India Power Ltd and (iv) 100 MW of power from Jhabua Power Ltdof Bid-2.

6.113: The Commission noted that, during the MYT period 2022-23 to 2026-27, KSEB Ltd has proposed to schedule 465 MW and 3263.29 MU annually from the above four DBFOO contracts. As per the power requirement projected by KSEB Ltd in the present petition and in order to avoid any power shortage and other contingencies, the Commission hereby permits KSEB Ltd to make necessary arrangements for procuring 465 MW of equivalent power on medium term basis through competitive bidding as per Section 63 of the Electricity Act, 2003. Alternatively, KSEB Ltd may make necessary arrangements for procuring 465 MW from power plants through competitive bidding route who have assigned coal linkage as per the SHAKTI policy of the Central Government."

4.95 In the meantime, KSEBL has invited medium term power procurement tender for 500 MW RTC power for a period of 5 years on FOO basis from generators having linkage coal with commencement of supply from April 2023. The tender was floated in DEEP portal with Bid start date on 04.11.2022 and only two firms submitted their bids in DEEP portal with offered quantum totaling to 250 MW against requisitioned capacity of 500 MW, but failed to qualify the bid due t the reasons given below

Table D40 Outcome of bidding of Medium term during 07.01.2023

Sl No	Name of firms participated in bid	Quantum Offered	Reason for Not Qualifying
	M/s. Jindal Power Limited	100 MW	Not submitted the original bid documents.
	M/s. RKM Powergen Pvt Limited	150 MW	Failed to meet the required criteria of having linkage coal

4.96 The Hon'ble Supreme Court in Civil Appeal No. 41/2021 filed by this Commission on 10.02.2023 has issued a direction to the Commission that the Commission will take a call and decide O.A. No. 5 of 2021 (OP No.5/2021) as expeditiously as possible but in no case later than three months and both the parties shall cooperate in getting expeditious disposal

of the pending O.A. It was also clarified that the present interim arrangement shall continue up to the date of the disposal of O.A. No. 5 of 2021 and for a further period of two weeks thereafter. Honourable Commission, , vide order dated 10/5/2023 has rejected the prayer in OP No.5/2021 filed by KSEB Ltd seeking to issue final orders with respect to drawal of power from generators of the following unapproved PSAs:

1. 115 MW of power from Jhabua Power Ltd (L-2 of Bid-1)
2. 150 MW of power from Jindal Power Ltd (Bid-2)
3. 100 MW of power from Jindal India Thermal Power Ltd (Bid-2)
4. 100 MW of power from Jhabua Power Ltd (Bid-2)

4.97 KSEBL on 25.05.2023 has filed its appeal (DFR 325/2023) against the judgment of KSERC before APTEL since this is adversely affecting the power position of Kerala. KSEBL has filed a stay petition along with Appeal petition (DFR 325/2023) but stay was not obtained. As per the direction of Government, KSEB Ltd filed petition before the Honourable Commission on 02.06.2023, for the continuation of the interim arrangement (drawal of power from unapproved PSA at the L1 rate of Bids) as directed by the Hon'ble Supreme Court , till KSEBL is able to make alternative arrangement for procurement of power or till the decision in the application of the interim relief filed by KSEBL before Hon'ble APTEL whichever is earlier or in the alternate. Honourable Commission vide order dated **07.06.2023 has directed as follows.**

1) KSEB Ltd is permitted to make arrangements for power procurement by continuing the interim arrangement of scheduling power from the four unapproved DBFOO which was in force for a period of two weeks from 10.05.2023 as per the direction of the Hon'ble Supreme Court, for a further period of 75 days from the date of this Order, or till the alternate arrangements of procuring 500MW RTC power on medium term basis, which ever earlier.
2) The payment for the power supply during the interim arrangement shall be as per the interim Orders of the Hon'ble APTEL dated 21.10.2022, 16.12.2022, 10.02.2023 and 17.04.2023, make payment at L1 rate of Bid-2 subject to the final disposals of the pending appeal petitions before the Hon'ble APTEL.
3) The interim arrangement as above, shall be subject to the final decision of the Hon'ble APTEL in IA 1183/2023 filed by KSEB Ltd in DFR No. 325/2023, against the Order of the Commission dated 10.05.2023 in petition OP No. 05/2021.

4.98 In spite of repeated requests from KSEBL to the unapproved DBFOO generators Jindal Power Ltd (150 MW), Jhabua Power Ltd (100 MW), Jhabua Power Ltd (115 MW) and JITPL (100 MW) to provide uninterrupted scheduling of power to KSEBL, The generators were reluctant to restore the scheduling of power. M/s Jhabua power Ltd had stopped scheduling of power from 1/6/2023 onwards, and M/s JITPL and M/s Jindal power Ltd had stopped scheduling from 3/6/2023 and 6/6/2023 respectively.

4.99KSEBL on 10.06.2023 had issued notice to generators through advocate to resume scheduling of power to KSEBL in terms of the order dt. 07.06.2023 of Ld. KSERC as well as the interim orders dt.

21.10.2022, 16.12.2022, 10.02.2023 and 17.04.2023 passed by the Hon'ble APTEL. It was informed that such non-compliance would also attract the legal consequences under the Electricity Act, 2003 including u/s 142, 146 and 149 thereof.

4.100 M/s JITPL filed an appeal no. DFR 370/2023 before APTEL to set aside the order dt 7/6/2023 passed by the Hon'ble KSERC in OP no. 24/2023 and quash the Impugned letter being issued by KSEBL dt 10/6/2023. : After repeated request by KSEBL M/s Jhabua Power limited 215 MW (against Bid 1 & 2) and M/s. Jindal Power Ltd (150 MW –Bid#2) started scheduling power from 21.06.2023. Coal suppliers to above Gencos i.e. South Eastern Coalfields Limited (SECL) & Mahanadi Coalfields Limited (MCL) curtailed fuel supply and the generator stopped scheduling power from 15.07.2023 onwards. Even though KSEBL have intervened in the matter, the coal suppliers were not ready to supply coal to the generators and urged clarification from KSERC for resuming the coal supply. Intimation from KSERC was given to the suppliers on 18.07.2023 and coal suppliers started coal supply. Subsequently the generators except M/s JITPL started rescheduling power from 19.07.2023.

4.101 The appeal of Jindal Thermal Power Limited &Jhabua Power Limited to set aside the order dated 07.06.2023 of the KSERC was heard by the APTEL and closed the same on 24.07.2023 based on the reply of KSERC that the direction issued by the commission by its order dated 07.06.2023 has been misconstrued by the Appellants as mandating them to sell power to KSEBL at L1 rates. Since no such direction were issued compelling them to do so, it is for the Appellant to decide whether or not to supply Power to KSEBL in terms thereof. Hon'ble APTEL conducted final hearing on the stay appeal of KSEBL on 31.07.2023 and denied the interim relief sought on the KSERC order dated 10.05.2023.

4.102 The pending payment details and nonavailability of payment security mechanism (PSM) were uploaded by the generators in PRAAPTI and PSM portals respectively, further led to power market regulations for the State on 16.06.2023 and 17.06.2023. KSEBL could not procure power from market for two days. Matter was taken up with Executive Director, NLDC for urgent intervention in this issue for helping KSEBL from power regulation as per LPS Rules.

4.103 KSEBL has taken immediate steps to obtain stay order from power regulations as per LPSC Rules and interim order obtained from the Hon'ble high Court of Kerala in connection with the petition filed by KSEBL for urgent intervention in the issue of power regulation in PRAAPTI portal as per LPS Rules. High Kerala of Kerala vide order dated 25.09.2023 had issued an interim direction to the NLDC and SRLDC not to restrict the scheduling of power in favour of KSEBL for the nonpayment of the bills of Jhabua Power Limited.

4.104 KSEB Ltd filed petition (OP 44/2023) seeking removal of time limit of 75 days in the above order. However, Honourable Commission vide order dated 14.07.2023 rejected the prayer of KSEB. Subsequently, KSEBL filed another petition OP 50/2023 seeking approval to allow power drawal from unapproved PPA GENCOs at PPA rate till finalisation of medium-term power procurement. Honourable Commission vide order dated 21.08.2023 extended the interim arrangement for the

period up to 31.12.2023 or till KSEB Ltd make alternate arrangements for procuring 500 MW of power through medium term basis, whichever is earlier. However, KSEBL 's request for PPA rate is not allowed by the Honourable Commission. The generators stop supply of power from 21.08.2023 onwards.

4.105 M/s Jhabua Power Limited, supplying 215 MW RTC power (115 MW under Bid 1 and 100 MW under Bid- 2) and Jindal Power Ltd (150 MW) have intimated that since no interim orders have been granted by Hon'ble APTEL and the relief granted by KSERC vide order dated 07.06.2023 has also been read by APTEL as not creating any obligation on the generators to supply power, there would be no power supply to KSEBL beyond **20.08.2023. Jhabua stopped supplying power on 20/8/2023: Jindal Power stopped scheduling of power on 21/8/2023.** Vide order dt **21/8/2023**, KSERC allowed to extend the interim arrangement for procurement of power from the unapproved DBFOO contracts as per the Order dated 07.06.2023 for the period upto 31.12.2023 or till KSEB Ltd make alternate arrangements of procuring 500MW RTC power on medium term basis, which ever earlier. However, the generators did not resume supply of power.

4.106 Meanwhile medium tenders were invited for procurement of 500 MW of power through DEEP portal for a period of five years. Non-financial bid of Medium-Term tender was opened on 19-8-2023. There were 2 participants: Adani Power Ltd & DB Power Ltd. portal. Financial bids of Medium-Term Tender invited through DEEP portal opened 0n 04.09.2023 and the rate discovered was Rs 6.88 per unit. As the discovered rate was high it was decided to cancel the tender.

4.107 Govt letter was issued on 10.10.2023 by the Addl Chief Secretary to Government of the Secretary, KSERC invoking Section 108 of Electricity Act 2003 in which, KSERC has been directed to reconsider/review their orders in OP No. 5/2021filed by KSEBL, in accordance with the policy of the Govt for the best interest of the state and public at large. KSEBL sought permission on 16.10.2023 from Honourable APTEL to file direction of Govt of Kerala before Honourable KSERC. Honourable APTEL on 30.10.2023 issued directions to remand the case to KSERC. Honourable APTEL permitted the Appeal to be withdrawn, with liberty to the KSEBL to invoke the review jurisdiction of the Commission. KSEBL on 10/11/2023 filed review petition RP No. 03/2023 before KSERC to recall/review the KSERC order dated 10.05.2023 with the request for approval for unapproved DBFOO contracts in the light of Government directive dated 10.10.2023 invoking Sec 108. Honourable Commission conducted hearing of the review petition on 01.12.2023 and the generators M/s Jhabua Power Limited & M/s Jindal India Thermal Power Limited (JITPL) intimated they have tied up the power with other DISCOMs/traders in the light of KSERC order dated 10.05.2023 and are unable to resume power supply to KSEBL.

4.108 Honourable Commission vide order dated 29.12.2023 in RP No 03/2023 allowed the Revision Petition of KSEBL based on the directive of the State Government under Section 108(2) of the Electricity Act, 2003 and the legal positions and directed the respondent generators to supply power to the KSEBL in terms of the PSA signed between the generators as above with immediate effect. This order is set aside by Honourable APTEL vide order dated 26.07.2024. KSEBL filed appeal before Honourable Supreme Court and Supreme Court vide order dated 30.09.2024 has directed

KSEBL to approach Honourable APTEL. for revisiting the appeal filed by KSEBL against KSERC order dated 10.05.2023 . The matter is now pending before Honourable APTEL.

4.109 Thus, it can be seen that despite best efforts, KSEBL could not enter medium term contracts for power in lieu of provisionally approved/unapproved DBFOO contracts. Hence in compliance to the direction in MYT order dated 25.06.2022, in order to avoid power contingencies KSEBL continued the arrangement of scheduling power from 115 MW of Jhabua Power Ltd under Bid I,100 MW of Jindal Power Ltd and 100 MW from JITPL under bid-2 during 2023-24 till August 2023 as explained above. Thus, KSEBL has scheduled 886.83 MU power during 2023-24 for a total cost of Rs 373.50 Cr.

4.110 The details of power purchased from Jhabua Power Ltd under Bid- I (115 MW), Jindal Power Ltd (150 MW-Bid II),Jhabua Power Ltd(100 MW-Bid II),JITPL(100 MW- Bid II) is given in table below.

Table 41 Power Purchase from Jhabua Power Ltd under Bid- I (115 MW) ,Jindal Power Ltd (150 MW-Bid II),Jhabua Power Ltd(100 MW-Bid II),JITPL(100 MW- Bid II)

Source	Energy at exbus (MU)	Energy at Kerala periphery (MU)
Jhabua Power Ltd-Bid I*	231.62	224.14
Jindal Power Ltd-Bid II*	369.92	357.90
Jhabua Power Ltd-Bid II*	201.37	194.87
JITPL*Bid II	113.41	109.92
Sub Total	**916.31**	**886.83**

4.111 Honourable Commission may please note that the per unit rate from these stations was Rs 4.21 per unit. Honourable Commission vide MYT order has directed to procure the 3363 MU of power in lieu of DBFOO unapproved contracts through medium term contracts. As said in previous paragraphs the power procurement through medium term contracts was not materialised owing to the high discovered rates. Honourable Commission may please noted that the average power purchase rate from the above stations is much lesser than the power procurement rate through exchanges for FY 2023-24 (Rs 5.83 per unit), the average of the rate discovered through short term contracts for FY 2023-24 (@ Rs 5.63 per unit and also the rate discovered through the tender for medium term contracts(Rs 6.88 per unit) . Hence Honourable Commission may kindly consider the claim of Rs 373.50 Crores at the time of truing up with the liberty to claim the power purchase cost as per the final orders in this matter as and when the same materialises.

4.112 Considering the above, Hon'ble Commission may kindly approve the true-up claim as per this petition as given in table below.

Table D42 True up claim of Power purchase cost towards power contracted through DBFOO guidelines for FY 2023-24			
Source	Energy at exbus	Energy at Kerala periphery	True up Claim of Power Purchase cost
	MU		(Rs Cr)
Jindal Power Ltd-Bid I	1,545.72	1,492.10	567.20
BALCO-Bid II	775.73	748.83	304.94
Jhabua Power Ltd-PSA I	231.62	224.14	373.50
Jindal Power Ltd-Bid II	369.92	357.90	
Jhabua Power Ltd- Bid II	201.37	194.87	
JITPL-Bid II	113.41	109.92	
Total	**3,237.77**	**3,127.75**	**1245.65**

Power purchase considered under short term/medium term contracts

Medium Term contracts

4.113 In order to meet the peak demand during summer months, KSEB Ltd has taken arrangements for procuring 270 MW RTC poweron medium term basis, through PTC, the Aggregator appointed by MoP underPilot Scheme II, for the period from January to June in a year for three years with the date of commencement of supply from 01-01-2022 with the approval of the Honourable Commission (order dated 07.10.2021) at a rate of Rs 3.31/unit. As per the provisions of PSA the variable charges are to be escalated based on inflation indices and an increaseof 2 paise per unit is expected for subsequent years. Honourable Commission in MYT order dated 25.06.2022 approved 962.75 MU at Kerala periphery for a total power purchase cost of Rs 320.60 crores @ Rs 3.33 per unit at exbus. KSEBL scheduled 935.79 MU at Kerala periphery and the corresponding power purchase cost as per accounts is Rs 364.19 crores.

4.114 As per clause 11.6.4 of the PSA- The Base Variable Charge specified in Clause 11.6.2 shall be increased for every Accounting Year following the Base Year so as to reflect 50% (fifty per cent) of the variation in WPI occurring between January 31st immediately preceding the Base Year and January 31st immediately preceding the Accounting Year for which such revision is undertaken.

4.115 As per the PSA, the Fixed Charge is Rs.1.63/kWh which is constant throughout the contract period and the variable charge is calculated in accordance with Clause 11.6.2, 11.6.3 and 11.6.4 of the PSA.

4.116 As per Clause 11.6.4 of the PSA, the variable charge as on Bid date is Rs 1.63 per unit. The variable charge is escalated by the variation in WPI from January 2019to January 2023. Further due to the blending of imported coal an additional amount of Rs 0.86 Cr was remitted towards energy charges.

4.117 The availability from M/s JITPL during April 2023, February 2024 and March 2024 was less due to outages. As per clause 11.5.2, in the event the availability is less than the normative availability, the fixed charge for such month shall be reduced to the extent of shortfall in normative availability and in addition, any reduction below the normative availability shall be multiplied by a factor of 0.25 of the fixed charges which is deducted on account of short fall in availability below normative availability. Thus KSEBL deducted Rs 2.06 crores as damages for the reduction in availability during 2023-24.

4.118 In addition to the above Rs 37.90 Cr. was remitted towards incentive, GNA charges , open access charges and LC charges.

4.119 The details of power purchase through medium term contract is given in table below.

Table D 43 Power Purchase cost through Medium term contracts

Sl No	Particulars	Quantum
1	Energy at exbus (MU)	970.75
2	Energy at Kerala periphery(MU)	935.79
3	Capacity charges (Rs Cr)	147.90
4	Energy Charges (Rs Cr)	180.45
5	Incentive (Rs Cr)	3.36
6	Damages (Rs Cr)	-2.06
7	Others(LC charges, open access charges)(Rs Cr)	34.54
	Total Power Purchase Cost(Rs Cr)	**364.19**

4.120 Honourable Commission may kindly approve the true up claim of Rs 364.19 crores towards the power purchase cost 270 MW RTC poweron medium term basis, through PTC.

Power procured through Short term contracts/Exchanges

4.121 Due to the failure of South West monsoon, there was a reduction In availability from hydro power of about 1477 MU during FY 2023-24. Further there was a reduction in availability of 2467 MU due to cancellation of contracts, increase in demand of 2679.55 MU compared to previous year and 1905 MU compared to approved quantum, during the year due to the sweltering heat and deficit in monsoon . The cumulative effect of reduced hydro inflow and increased consumption was primarily met from the power exchange markets through Day Ahead Market (DAM), Real Time Market (RTM) and Term Ahead Market (TAM) contracts.

4.122 To manage the rising demand and non-availability of Power on account of reduced inflow and cancellation of 465 MW of DBFOO contracts, KSEB resorted to purchasing power through the Day Ahead Market (DAM) and Real Time Market (RTM) from power exchanges. However, as demand escalated nationwide by 7% to 12% over expectations, the rates during morning peak and evening peak surged to ₹10 per unit. The increased demand led to a significant power shortage in the DAM and RTM, even at elevated prices. The unavailability of these four round-the-clock (RTC) sources under DBFOO diminished the energy supply by 12-15% and peak power availability by approximately 10% during the period of high demand and maximum market rates. To mitigate these challenges, KSEB arranged short-term power arrangements through the DEEP portal and SWAP arrangements were made with other utilities. KSEB had also explored possibility of having Long Duration Contracts (LDC) / Monthly contracts of exchange for power procurement. Thus 300 MW was additionally contracted through long duration contracts for March 2024.

Short Term Contracts

4.123 Thus, KSEB Ltd entered into the short term power procurement arrangements through DEEP portal with traders for meeting the power requirement of the State from April 2023 to March 2024, with the approval of Honourable Commission as detailed below

Table D44-Power Purchase through DEEP contracts

SL No	Period	Total Quantum (MW)	Bidder	Generator/ Trader	Quantum on firm basis (MW)	Rate (Rs/ kWh)	Source
KSERC order dated 26.07.2022							
1	Apr-23	100	L1	PTC	!00 MW RTC	7.95	JNSTPP
	State Commission order dated 17.05.2023						
1	21.05. 23 to 31.05.23	200 MW 19.00 to 23.00	L1	NVVN	50 MW 19.00 to 23.00	**9.26**	
KSERC order dated 31.07.2023							
1	Aug-23	**220**	L1	TPTCL	**50**	**5.23**	Adhunik Power & natural Resources Ltd, WB
			L2	Adani Enterprises Ltd	**100**	**5.23**	Adani power Ltd, Raipur
			L3	TPTCL	**45**	**5.23**	NavaBharath Ventures Ltd,Odisha(NBVL)
			L4	PTC India Limited	**25**	**5.23**	NavaBharath Ventures Ltd,Odisha
2	Sep-23	**250**	L1	TPTCL	**75**	**5.38**	JITPL, Odisha
			L2	TPTCL	**50**	**5.38**	Adhunik Power & natural Resources Ltd,WB
			L3	TPTCL	**45**	**5.39**	NBVL,Odisha
			L4	Jindal Power Ltd	**80**	**5.42**	JPL, Chattisgarh

SL No	Period	Total Quantum (MW)	Bidder	Generator/ Trader	Quantum on firm basis (MW)	Rate (Rs/ kWh)	Source
3	Oct-23	**150**	L1	TPTCL	**100**	**5.12**	JITPL, Odisha
			L2	TPTCL	**50**	**5.14**	JITPL, Odisha
4	Nov-23	**250**	L1	TPTCL	**150**	**5.38**	JITPL, Odisha
			L2	GMRETL	**25**	**5.38**	Maadurga Thermal Power Company Ltd,Odisha
			L3	Jindal Power Ltd	**75**	**5.46**	JPL, Chattisgarh
5	Dec-23	**250**	L1	GMRETL	**25**	**5.48**	Maadurga Thermal Power Company Ltd,Odisha
			L2	TPTCL	**150**	**5.48**	JITPL, Odisha
			L3	NVVN	**75**	**5.5**	JaipeeNigree STPPMP
6	Jan-24	175	L1	GMRETL	**25**	**5.49**	Maadurga Thermal Power Company Ltd,Odisha
			L2	NVVN	**100**	**5.49**	JaipeeNigree STPP, MP
			L3	NVVN	**50**	**5.49**	JaipeeBinaa STPP
7	Feb-24	250	L1	NVVN	**100**	**5.58**	JaipeeNigree STPP, MP
			L2	NVVN	**50**	**5.58**	JaipeeBinaa STPP, MP
			L3	GMRETL	**25**	**5.58**	Maadurga Thermal Power Company Ltd ,Odisha
			L4	TPTCL	**45**	**5.75**	Tata Power Haldia
			L5	Adani Enterprises Ltd	**30**	**5.82**	Adani power Ltd, Jaipur
8	Mar-24	225	L1	NVVN	**100**	**6.13**	JayipeeNigree STPP, MP
			L2	NVVN	**50**	**6.13**	JaipeeBinaa STPP, MP
			L3	GMRETL	**25**	**6.13**	Maadurga Thermal Power Company Ltd,Odisha
			L4	Manikaran Power Limited	**25**	**6.25**	BLA Power Private Ltd, UnitII,MP
			L5	Manikaran Power Limited	**25**	**6.34**	BLA Power Private Ltd,Unit-I,MP

4.124 Table 12 shows the details of energy purchased from short term contracts

Table D45 Power Purchase through short term contracts				
SL No	**Particulars**	**SR periphery**	**KSEBL periphery**	**PGCIL loss**
		MU	**MU**	**MU**
1	Purchase from NVVN/IL&FS	2.28	2.20	0.08
2	Purchase from TPTCL/PPGCL	0.00	0.00	0.00
3	Purchase fromPTC/JNSTPP	63.73	61.32	2.41
4	Purchase from TPTCL/APNRL	89.46	86.42	3.04
5	Purchase from TPTCL/NBVLIPP	65.94	63.72	2.22
6	Purchase fromAEL	90.02	86.95	3.07
7	Purchase from PTC	164.68	159.09	5.59
8	Purchase from TPTCL/JITPL	325.05	313.45	11.60
9	Purchase from GMRETL/MBMP	34.24	32.91	1.34
10	Purchase from NVVN/JPNIGRIE_JNSTPP	212.37	204.39	7.98
11	Purchase from NVVN/JPBINA	171.65	165.26	6.39
12	TATA_POWER_HALDIA	32.52	31.32	1.20
13	Purchase from GMRETL/GMR Warora	17.59	16.93	0.66
14	MP(BLAPOWER 1_& 2 MP)	34.71	33.48	1.23
15	EAST(APNRL)	18.41	17.75	0.66
	Sub total	**1322.64**	**1275.19**	**47.46**

4.125 The details of power purchase cost for procurement of power through short term contracts is given in table-13 below

Table D 46 -Power Purchase through DEEP contracts

Sl No	Period	Generator/ Trader	Quantum (MW) on firm basis	Rate (Rs/ kWh)	Source	Quantum at Kerala periphery MU	Power Purchase Cost (Rs Cr)
KSERC order dated 26.07.2022							
1	Apr-23	PTC(RTC)	100	7.95	JNSTPP	61.32	48.74
KSERC order dated 17.05.2023							
1	21.05.23 to 31.05.23 (19.00 to 23.00 pm)	NVVN	50 MW 19.00 to 23.00	9.26	IL &FS	2.20	2.04
KSERC order dated 31.07.2023							
1	Aug-23	TPTCL	**50**	**5.23**	Adhunik Power & natural Resources Ltd, WB	33.21	17.37
		AEL	**100**	**5.23**	Adani power Ltd, Raipur	66.66	34.86
		TPTCL	**45**	**5.23**	NavaBharath Ventures Ltd,	30.14	15.76
		PTC	**25**	**5.23**	NavaBharath Ventures Ltd,	16.70	8.73
2	Sep-23	TPTCL	**75**	**5.38**	JITPL, Odisha	49.82	26.80
		TPTCL	**50**	**5.38**	Adhunik Power & natural Resources Ltd,WB	35.84	19.28

Sl No	Period	Generator/ Trader	Quantum (MW) on firm basis	Rate (Rs/ kWh)	Source	Quantum at Kerala periphery MU	Power Purchase Cost (Rs Cr)
		TPTCL	**45**	**5.39**	NavaBharath Ventures Ltd,	29.56	15.93
		Jindal Power Ltd	**80**	**5.42**	JPL, Chattisgarh	55.64	30.16
3	Oct-23	TPTCL	**100**	**5.12**	JITPL, Odisha	69.27	35.46
		TPTCL	**50**	**5.14**	PPGCL	34.64	17.80
4	Nov-23	TPTCL	**150**	**5.38**	JITPL, Odisha	97.42	52.41
		GMRETL	**25**	**5.38**	Maadurga Thermal Power Company Ltd,	17.37	9.35
		Jindal Power Ltd	**75**	**5.46**	JPL, Chattisgarh	52.12	28.46
5	Dec-23	GMRETL	**25**	**5.48**	Maadurga Thermal Power Company Ltd,	15.56	8.53
		TPTCL	**150**	**5.48**	JITPL, Odisha	100.96	55.33
		NVVN	**75**	**5.5**	JaipeeNigree STPPMP	53.46	29.40
6	Jan-24	GMRETL	**25**	**5.49**	Maadurga Thermal Power Company Ltd	17.34	9.52
		NVVN	**100**	**5.49**	JaipeeNigree STPP, MP	74.39	40.84
		NVVN	**50**	**5.49**	JaipeeBinaa STPP	36.58	20.08
7	Feb-24	NVVN	**100**	**5.58**	JaipeeNigree STPP, MP	63.88	35.65
		NVVN	**50**	**5.58**	JaipeeBinaa STPP, MP	34.80	19.42
		GMRETL	**25**	**5.58**	Maadurga Thermal Power Company Ltd	16.93	9.45
		TPTCL	**45**	**5.75**	Tata Power Haldia	31.32	18.01
		AEL	**30**	**5.82**	Adani power Ltd, Jaipur	20.29	11.81
8	Mar-24	NVVN	**100**	**6.13**	JayipeeNigree STPP, MP	71.02	43.54
		NVVN	**50**	**6.13**	JaipeeBinaa STPP, MP	35.51	21.77
		GMRETL	**25**	**6.13**	Maadurga Thermal Power Company Ltd,	17.76	10.88
		Manikaran Power Ltd	**25**	**6.25**	BLA Power Private Ltd,U-II	16.74	10.46
		Manikaran Power Ltd	**25**	**6.34**	BLA Power Private Ltd, UnitI,MP	16.74	10.61
		Sub Total				**1275.19**	**718.44**
		Rate(Rs/kWh)				**5.63**	

4.126 In addition to the above, KSEBL procured power from exchanges in order to meet the deficit. It may please be noted that KSEBL has purchased from exchanges duly following the merit order principles.

Power Purchase from Exchanges: -

4.127 As said in the preceding paragraphs, in order to meet the rising demand and to manage the non-availability of Power on account of reduced inflow and cancellation of 465 MW of DBFOO

contracts, KSEBL also purchased 3641.70 MU from exchanges during FY 2023-24. Table below shows the details of energy purchased from exchanges (IEX, PXIL and HPX). Average rate of purchase is Rs 5.83 per unit at Kerala periphery.

4.128 Table-14 given below shows the details of energy purchased from exchanges (IEX, PXIL and HPX). KSEBL procured 3641.70 MU at Kerala periphery.

Table D 47 Energy purchased in MU through exchanges for FY 2023-24

Particulars	SR periphery	KSEBL periphery	PGCIL loss
Power exchanges	MU	MU	MU
IEX			
DAM	801.96	774.01	27.95
TAM	147.52	142.51	5.01
RTM	1838.44	1772.86	65.59
GDAM	53.04	51.21	1.83
GTAM	0.75	0.73	0.02
Subtotal	**2841.71**	**2741.32**	**100.39**
PXIL			
DAM	62.47	60.34	2.13
TAM	146.60	141.64	4.96
RTM	0.00	0	0.00
GDAM	0.00	0.00	0.00
GTAM	0.00	0.00	0.00
Subtotal	**209.07**	**201.99**	**7.08**
HPX		0	0
DAM	2.05	1.98	0.07
TAM	711.54	687.23	24.32
RTM	0.00	0	0.00
GDAM	0	0	0.00
GTAM	9.52	9.20	0.32
Subtotal	723.10	698.40	24.70
Total Power purchase	**3773.89**	**3641.70**	**132.19**

4.129 The cost of power purchase from exchanges is given in table-15 given below.

Table D 48 Power Purchase through exchanges

Source	Energy at generator bus	Energy at KSEB periphery	PP Cost
	MU	**MU**	**Rs Cr**
IEX	2841.71	2741.31	1357.29
PXIL	209.07	201.99	175.48
HPX	723.10	698.40	553.71
Transaction charges for Purchase of Power through power exchange		36.42	36.42

Source	Energy at generator bus	Energy at KSEB periphery	PP Cost
	MU	MU	Rs Cr
Application fee for purchase of power through power exchange		0.26	0.26
Subtotal	3773.89	3641.70	2123.16
Rate (Rs/kWh)	**5.83**		

4.130 Honourable Commission may please note that the price discovered in the market during 2023-24 was high owing to the coal blending. The month wise price discovered in day ahead market of IEX is given in table below

4.131

Table D 49 Month wise IEX DAM rates for FY 2023-24					
Month	**RTC**	**Evening**	**Day**	**Night**	**Morning**
April	5.65	**7.15**	3.55	**7.05**	4.62
May	4.74	**6.37**	3.24	**5.70**	3.23
June	5.36	**7.27**	3.52	**6.83**	3.13
July	4.55	**6.91**	2.86	**4.96**	3.23
August	6.89	**8.69**	4.73	**7.77**	6.45
September	6.23	**8.27**	4.73	**6.31**	5.66
October	6.45	**8.02**	5.25	**6.54**	6.06
November	3.99	**5.47**	3.50	**3.23**	3.93
December	4.55	**5.83**	4.07	**2.85**	6.42
January	5.83	**7.28**	5.82	**2.90**	8.81
February	4.93	**6.14**	4.11	**3.16**	7.62
March	3.93	**4.54**	3.28	**3.75**	4.47

Swap arrangement:

4.132 KSEBL with the approval of Honourable Commission entered into the following banking arrangements with various traders

(i) 200 MW for the period from March 2023 to May 2023, with the trader M/s Arunachal Pradesh Power Corporation Pvt Ltd (APPCL) and PSPCL as supplier at a trading margin of 2.9 Paise per unit and return percentage of 107.96 % at Kerala periphery.

(ii) With MP Power Management Company Limited (MPPMCL) for the period from 13.09.2023 to 31.10.2023 with return period on January 2024.

(iii) with SAPL as trader and UPPCPL as supplier for 1200 MW RTC power on firm basis for the period from 15th to 31st March 2024 and return period from 1st June to 30th June 2024.

(iv) 200 MW power with M/s Arunachal Pradesh Power Corporation Pvt Ltd (APPCPL) and PSPCL as supplier for the period from 1st January 2024 to March 2024 and 150 MW power from Ist April to 15th April 2024 for managing the power deficit from Jan 2024 to April 2024

(v) with APPCPL as trader and BYPL as supplier and return period from 16th June 2024 to September 2024 for power of various quantum as detailed in table below for the period from November 2023 to March 2024 (except January 2024 & February 2024)

(vi) 200 MW RTC with the trader M/s Arunachal Pradesh Power Corporation Pvt Ltd (APPCL) and PSPCL as supplier for May 2023

(vii) with M/s. Kreate Energy (I) Pvt Ltd. (Trader) and supplier (BRPL) with supply period February and March 2023 and return period from 15th June 2023 to 15th July 2023.

(viii) with M/s. Kreate Energy (I) Pvt Ltd. (Trader) and supplier M/s BYPL with supply period February and March 2023 and return period from 15th June 2023 to 15th July 2023. The details are as indicated in the next page:

Table D 50 Energy availed through banking arrangements													
Banking arrangement with Arunachal Pradesh Power Corporation Ltd													
	Supply to KSEBL			**Return from KSEBL**									**Trading Margin (Paise/ kWh)**
					Duration (Hrs)							Return (%)	
Utilities	Supply Period	Duration (Hrs)	Quantum (MW)	Return Period	0-3	3--6	6--18	18-22	22-24				
APPCPL-PSPCL Corporation Ltd	Mar-23	RTC	200	16.06.2023 to 30.06.2023	100	100.13	100	50	100				
	Apr-23	RTC	200	01.07.2023 to 15.07.2023	100	150.13	250	50	150				
				16.07.2023 to 31.07.2023	100	150.13	250	50	150				
	01.05.23 to 20.05.23	RTC	200	Aug-23	100	150.13	246	50	150				
				Sep-23	100	150.13	250	50	100			107.91%	2.9
APPCPL-BYPL Corporation Ltd	Nov-23	RTC	100	11.06 to 10.07									
	Mar—24 (1-15)	RTC	50										
APPCPL-PSPCL	Jan-March 24	RTC	200	16th to 30th June'24	50 MW RTC								
APPCPL-PSPCL	April 1st to 16	RTC	150	balance quantum (RTC) 1st July'24 to 30th September '24								105	3
					0-5	05-8.30	8.30-18.30	18.30 to 22.30	22.30 to 24.00				
APPCPL-BYPL	Nov-23	RTC	100		*5*	*8.3*	*18.3*	*22.3*	*24*				
APPCPL-BYPL	1st to 15th March	RTC	50		*70*	*25*	*25*	*45*	*70*				
			50		*70*	*25*	*25*	*45*	*70*				

			50		70	25	25.985	45	70			
MPPCL	13-09-23 to 24.10.23	RTC	200		RTC quantum during 18:45 hrs-22:00 hrs, however							
	25.09.23 to 31-10-23	RTC	200		RTC/ any time slot between 00:00 hrs to 24:00 hrs							
	Jan-24	RTC			06:00 - 18:00							

Banking arrangement with Kreate Energy (I) Pvt Ltd

	Supply to KSEBL			Return from KSEBL							
					Duration (Hrs)						
Utilites	Supply Period	Duration (Hrs)	Quantum (MW)	Return Period	0-3	3--6	6--18	18-22	22-24	Return %	Trading margin
Kreate Energy(I) Pvt Ltd-BRPL	Feb-23	14-22	50	16-06-2023 to 30.06.2023	240	240.11	16.33	16.33	240.11		
		RTC	50				16.33	16.33			
				01.07.2023 to					240.11		
	Mar-23	14-23	50	15.07.2023	240	240.11					
Kreate Energy(I) Pvt Ltd-BYPL	Feb-23	14-24	75	16-06-2023 to 15.072023	86	50	8	15	86	105	
Shubheksha Advisors Private Limited (SAPL)(Supplier-UPPCPL)	15th to 31st March 2024	RTC power on firm basis	200MW	1st to 30th June 2024,	RTC on firm basis						

4.113 KSEB Ltd availed 844.6 MU at Kerala periphery through banking transactions during the year2023-24

Table .D 51 .Power received through banking arrangement for 2023-24					
Utility	**Trader**	**Period**	**Energy at exbus**	**Energy at Kerala periphery**	Loss
			MU	**MU**	**MU**
APPCL	PSPCL	April-May 2023	239.18	230.15	9.03
APPCL	PSPCL	Nov-23	108.00	104.2	3.76
APPCL	PSPCL	Jan- Mar-24	337.68	325.3	12.42
APPCL	BYPL	Dec-23, Mar-24	73.20	70.42	2.78
MPPMCL		as and when available	44.28	42.7	1.55
SAPTL	UPPCL	Mar-24	76.80	71.8	4.96
Total			879.1	844.65	34.49

4.114 KSEBL returned 552.68 MU during 2023-24as part of banking transactions.

Table 52. Power returned by KSEBL through banking arrangements	
Particulars	**Energy (MU)**
Swap return KEIPL -BYPL	-22.05
Swap return KEIPL -BRPL	-65.47
SWAP-Returrn-3(APPCL)	-418.67
SWAP-Returrn-4(MPSEB)	-46.50
Sub Total	-552.68

4.115 An amount of Rs 5.09 Cr was expended towards banking transactions and a provision of Rs 209.13 CR was provided in the accounts towards the cost of power to be returned during 2024-25 as part of banking arrangements. Honorable Commission may kindly approve the true up claim of KSEBL td amounting to Rs 214.22 Cr. towards the power arranged through swap arrangements.

Deviation Settlement Mechanism (DSM):

4.116 KSEB had availed 464.14 MU to the grid and delivered 61.49 MU through deviation settlement mechanism (DSM). **Table below shows** the details of energy transacted through deviation settlement mechanism during the year 2023-24.

Table D 53 Power Transfer through Deviation Settlement Mechanism for the year 2023-24			
Month	**Import in MU**	**Export in MU**	**Net Deviation in MU**
April-23	10.70	-12.37	-1.67
May-23	18.06	-10.03	8.04
June-23	49.81	-4.64	45.17
July-23	54.17	-3.34	50.83
Aug-23	50.91	-2.35	48.56
Sept-23	42.39	-3.28	39.10
Oct-23	43.47	-4.12	39.36
Nov-23	40.60	-5.09	35.51
Dec-23	46.58	-3.09	43.49
Jan-24	30.92	-5.94	24.98
Feb-24	37.29	-3.83	33.45
March-24	39.23	-3.40	35.83
Total import	**464.14**	**-61.49**	**402.64**

4.117 An amount of Rs 206.67 Cr was expended towards power transfer through deviation settlement mechanism and is accounted in power purchase cost. It may please be noted that the DSM import is considered as input to system and is considered in power purchase and DSM export is considered as an energy transferrred to outside the State.

4.118 In view of the details furnished above, it may kindly be seen that the shortfall in energy availability from approved sources had been effectively met through procurement of power from various sources at optimum cost.. The details of purchases through long-term, short-term contracts are given as **Annexure D4.** The summary of power Purchase from power exchanges and power transfer through deviation settlement mechanism against quantum approved towards short term contracts for FY 2023-24 is furnished below:

Table D 54: Details of power purchase against approval for short term procurement								
Source	**Approval**			**Actual**			**Difference**	
	Energy (MU)	**Cost (Rs Cr)**	**Rate Rs/unit)**	**Energy (MU)**	**Cost (Rs Cr)**	**Rate Rs/unit)**	**Energy (MU)**	**Cost (Rs Cr)**
Short Term	240.55	96.22	4	1275.19	718.44	5.63		
IEX/PXIL				3641.7	2123.16	5.83		
Deviation Settlement Mechanism (Import)				464.14	206.67	4.45		
SWAP				844.65	214.22	2.54		
Total	**240.55**	**96.22**	**4**	**6225.68**	**3262.49**	**5.24**	**5985.13**	**3166.27**

Energy sale outside the State: -

4.119 Hon'ble Commission approved sale of surplus power of 1019.09 MU for a total revenue of Rs 382.16 Cr. KSEBL sold 71.27 MU of power through exchanges during 2023-24 for a total revenue of 60.60 Cr as given in table below.

Table D55(a) Energy sale o/s State					
Sl No	Source	Energy at KSEB periphery	Total Amount	Rate	Expenditure
		MU	Rs. Cr.	Rs/Unit	Rs Cr
1	IEX DAM	24.88	18.51	6.32	0.25
2	IEX RTM	44.88	40.80	6.94	
3	**IEX Total**	69.76	**59.31**	**6.57**	
4	PXIL TAM	1.52	1.29	4.8	
5	**PXIL Total**	1.52	**1.29**	**5.13**	
	Total	71.27	60.6	6.46	

4.120 Thus, KSEBL sold 71.27 MU for a total amount of Rs 60.6 Cr. @ Rs 6.46 per unit against the approved quantum of 1019.09 MU for Rs 382.16 Cr and the expenditure associated with sale through power exchanges amounts to Rs 0.25 Cr.

4.121 It may kindly be noted that KSEB Ltd could realise a higher rate from sales through short term market than the rate derived while procuring power through exchanges (Rs 5.83 per unit). Hence Honourable Commission may kindly approve the expenditure associated with the sale amounting to Rs 0.25 Cr.

4.122 The total energy transfer to outside the State is given in table below

D 55(b)Energy transactions outside the State		
SL No	Particulars	Quantum (MU)
1	Energy sale outside the State	71.27
2	Deviation settlement Mechanism (Export)	61.49
3	SWAP Return	552.68
	Sub Total	685.45

Inter-state Transmission charges paid to PGCIL:

4.123 Hon'ble Commission, vide order dated 25.06.2022, approved interstate transmission charges of Rs 760.55 Cr for 2023-24. Actual transmission charges incurred by KSEBL during 2023-24, as per audited accounts, were Rs. 1448.27 Cr. which may kindly be approved.

4.124 Honorable CERC has issued the Central Electricity Regulatory Commission (Sharing of Inter-State Transmission Charges and Losses) Regulations, 2020 on 04.05.2020 with effect from November 2020. MoP introduced a new open access regime General Network Access under the Electricity (Transmission System Planning, Development and Recovery of Inter-State Transmission Charges) Rules 2021 in a gazette notification dated

October 1, 2021, for optimum investment in transmission network and consequent adequate development of transmission system. The Central Electricity Regulatory Commission (CERC) issued the CERC (Connectivity and General Network Access to the Inter State Transmission System) Regulations, 2022 (GNA Regulations) in June 2022.The GNA Regulations replaced the CERC (Grant of Connectivity, Long-term Access and Medium-term Open Access in Inter-State Transmission and related matters) Regulations, 2009 (CERC Regulations 2009). In the earlier regime transmission charges were remitted under the point-to-point connectivity regime, and shell out short-term charges for any deviations from the agreed path. All that changes with GNA, where transmission charges are now based on GNA capacity rather than energy usage.

4.125 On implementation of new regulations, the transmission charges for Kerala increased considerably by about 80% in comparison with that before 2020. Honourable Commission may please note that actual transmission charges incurred by KSEBL during 2022-23, as per audited accounts, was Rs. 1788.90 Cr.

4.126 The increase in transmission charges during 2023-24 compared to approved quantum was mainly due to the impact of the new methodology in CERC (Sharing of Inter-State Transmission Charges and Losses) Regulations, 2020 as well as inclusion of the transmission charges for Raigarh - Pugalur -Thrissur line and other transmission assets. The Power Grid Corporation of India Limited had filed Tariff petitions before Honourable CERC For Determination of Tariff of Raigarh Pugalur Trichur HVDC system under the CERC terms and conditions of the Tariff Regulation 2019 from the Date of COD to 31.03.2024. In the Tariff petitions (685/TT/2020). Honourable CERC vide order dated as 29.09.2022 allowed the annual, fixed charge, (AFC) for 2019-24, considering 30% of the yearly transmission charges (YTC) as National Component and 70% of YTC as Regional Component with effect from 1/11 /2020. The request of SR beneficiaries submitted before CERC during the hearing of the petition to consider the asset as National Component was denied.

4.127 CERC on 27-10-2023 issued an amendment to the CERC (Sharing of Inter-State Transmisison Charges and Losses) Regulations, 2022, wherein a new methodology for sharing of transmission charges of inter-regional HVDC lines evolved. Under this methodology, the transmission charges to be included under 'National Component' to be shared on all India basis will be based on capacity of flow of power in reverse direction. If the reverse power flow capacity is more than 30%, that percentage will be considered for sharing under all India basis. If it is less than 30%, 30% of transmission charges will be shared on all India basis. All the SR constituents had presented their objections before CERC against this amendment Regulations as this percentage basis is not made applicable to the other two lines, but only to Raigarh - Pugalur-Thrissur case.

4.128 Pursuant to the order dt. 18.07.2023 passed by APTEL, the CERC passed a fresh order 30.10.2023 in Petition No. 685 / TT / 2020 (Tariff Petition of Pole-1). As per this order, CERC issued following orders on transmission charges sharing of Pole-1 of Raigarh-Pugalur HVDC line in total disregard to the request of the SR constituents.

1. The asset was commissioned prior to the notification of CCERC Sharing Regulations 2020. Therefore, from CoD of the asset to the notification of 2020 Sharing Regulations (issued in November 2020): the sharing of transmission charges were as per the Sharing Regulations existing prior to 2020ie' 10% on all India basis and 90% on regional basis.
2. From November 2020 to 26-10-2023: 30% on all India basis and 70% on regional basis.
3. From 27-10-2023 onwards: 50% on all India basis and balance depending on the reverse power flow. If reverse power flow is more than 50%, that percentage will be considered for sharing of transmission charges.

4.129 As per the above order of CERC, after 27-10-2023, sharing of transmission charges of HVDC Pole-1 of Raigarh-Pugalur system is such that 50% will be shared on all India basis and 50% on regional basis. The percentage will increase if the reverse flow capacity (SR to WR) is increased.

4.130 The HVDC Raigarh Pugalur system has 4 HVDC poles. The order dated 27-10-2023 is related to -1 alone. APTEL has remanded the appeal petitions of the remaining poles also back to CERC by observing the same issues and the matter is before CERC for re-consideration. Therefore, similar orders are anticipated for the remaining poles also. Since the tariff petitions of the remaining poles are set aside by APTEL and the matter is before CERC, CTUILis not billing the transmission charges of the HVDC poles 2,3 & 4 of Raigarh-Pugalur HVDC system from September 2023 onwards. This causes a slight reduction in transmission charges for FY 2023-24. Honourable Commission may kindly approve the final bill amount as per the final orders in this matter as and when the same materialises.

4.131 KSEBL have filed appeals against the above order (Order dated 27-10-2023) of CERC before Honourable APTEL. However, in the case of Pugalur-Thrissur portion of HVDC line, APTEL has dismissed the appeal, stating that this portion of the HVDC line is not an inter-regional HVDC line and Kerala is the only beneficiary and so it cannot be treated in the manner in which Raigarh-Pugalur portion is treated. By way of this order, the SR beneficiaries including Kerala lost the opportunity of reduction in the monthly transmission charges of Pugalur-Thrissur portion of this asset.

4.132 Thus, the cumulative impact of the above factors resulted in significant increase in the transmission charges to be paid by SR beneficiaries. The capital investment of Raigarh Pugalur Thrissur HVDC alone was Rs 20,000 Cr. Since the capital investment of this scheme is very huge, the transmission tariff impact on SR beneficiaries is huge.

4.133 Honourable Commission may kindly approve the true up claim of transmission charges amounting Rs. 1448.27 Cr.

Total Power Purchase Cost for FY 2023-24

4.134 The total power purchase cost as per annual accounts is Rs 12982.59 Cr. Honourable Commission may kindly note that despite the extremely challenging situations during FY 2023-24, KSEBL could limit the power purchase cost to Rs 5.05 per unit including swap transactions, Hence Hon'ble Commission may kindly true up the total power purchase cost amounting to Rs 12982.59 Cr as detailed in **Annexure D 4,** summary of which is provided in table below.

Table D 56

	Source	2023-24(At Kerala periphery)	
		Quantum (MU)	**Power Purchase Cost (Amount in Rs Cr.)**
1	CGS	10951.87	4731.08
2	Maithon & DVC stations	3274.05	1495.44
3	DBFOO- approved contracts	2240.93	872.15
4	Addl power purchase in lieu of provisionally Approved/ Un approved DBFOO	886.83	373.50
5	Medium term contracts	935.79	364.19
6	RE purchase	1070.19	339.58
7	Power Purchase through short term contracts	1275.19	718.44
8	RGCCPP		70.65
9	Exchanges	3641.70	2123.16
10	Inter State Transmission charges		1448.27
11	Swap	844.65	214.22
12	Deviation import	464.14	206.67
13	Other charges in connection with sale of power through power exchange	125.99	0.25
14	Grid Connected Average Pooled Power Purchase Cost-Solar Energy/captive consumers		25.00
15	Total power purchase	25711.31	12982.59
16	Average Power Purchase Cost(Rs/kWh)	Rs 5.053/kWh	

The source wise details of power purchase was detailed in Form D3.1 of the petition

4.8 O&M Expenses:

4.135 Honourable Commission approved Rs 3605.40 Cr towards O&M expenses which comprised of Rs 3044.56 crores towards Employee expenses and A&G expenses and Rs 560.83 Cr towards R&M expenses for FY 2023-24. Actual O&M expenses for SBU D for the year 2023-24 has been Rs. 4030.39 Cr (net of capitalization) as detailed in Form D3.4.

4.136 Cost Drivers: Tariff Regulation, 2021 considers the following cost drivers for determination of normative O&M expenses. The growth of the drivers is given in the Table below.

Table D 57 Cost drivers of O &M Expenses

Item	FY15	FY16	FY17	FY-18	FY-19	FY-20	FY-21	FY 22	FY 23
Consumers (Nos)	11430895	11668031	11994816	12276321	12552206	12826185	13142910	13422642	13648851
Dist Transformers (Nos)	71199	73460	75759	77724	79876	81470	83465	85594	87911
HT Lines (KM)	57309	59476.66	61398.43	62855	64393.5	64808.75	66648	68972.66	70269.41
LT lines(km)	268753	278574	283811	286784	289804	293280	295921	299538	302626
Energy Sales (MU)	18426	19325.07	20087.55	20998	22369.59	22728.34	22413.03	25602.22	26743.1834

4.137 The parameters of FY-22 are taken for computing normative O&M expenses. The norms for determining the O&M expenses for FY 2023-24 as depicted in Tariff regulations 2021 and normative values of employee cost and A&G expenses is given in table below:

Table D 58: Normative Employee expenses and A&G expenses for 2023-24

No	Parameters	Quantity
	Norms	
1	No. of consumers (Rs. Lakh/1000)	4.539
2	Dist Transformers (Rs. Lakh/DTr)	0.896
3	Length of HT line (Rs. Lakh/km)	0.887
4	Length of LT line (Rs. Lakh/km)	0.194
5	Energy sales (Rs/unit)	0.200
	Parameters for FY 2022-23	
6	No. of consumers for FY 2022-23	13648851
7	No. of Dist Transformers (Nos)	87911
8	Length of HT Line (Km)	70269.41
9	Length of LT line (km)	302626
10	Energy sales in MU	26743.18

4.138 Even though Honourable Commission in the Tariff Regulations considered an escalation rate of 4.454 % for determining the norms for the control period 2022-27, Annexure 7 of Tariff Regulation allows actual escalation rates based on actual inflation existing for each year of

control period at the time of truing up of the considered financial year. The actual inflation rates are given in table below.

Table D. 59 Escalation Rate based on actual inflationary index for FY 2023-24				
Sl No	**Escalation factors/year**	**2021-22**	**2022-23**	**2023-24**
1	CPI Industrial Workers)2001=100	356.06	377.62	397.25
2	Annual CPI Increase (%)	5.13	6.05	5.19
3	WPI(2011-12) series	139.4	152.5	151.4
4	Annual WPI Increase (%)	12.97	9.4	-0.72
5	CPI: WPI (70:30) weightage (%)	7.48	**7.06**	**3.41**

4.139 The norms based on actual escalation rate for FY 2023-24 @ 3.41% and the norms based on actual inflation is given in table below.

Table D 60: Normative O& M expenses for 2023-24						
No	**Parameters**	**Norms for FY 2021-22**	**Actual Escalation Rate from 2022-23**	**Norms for FY 2022-23**	**Actual Escalation Rate for FY 2023-24**	**Norms for FY 2023-24 based on actual inflation**
1	No. of consumers (Rs. Lakh/1000)	4.1	7.06	4.389	3.41	4.539
2	Dist Transformers (Rs. Lakh/DTr)	0.809	7.06	0.866	3.41	0.896
3	Length of HT line (Rs. Lakh/km)	0.801	7.06	0.858	3.41	0.887
4	Length of LT line (Rs. Lakh/km)	0.175	7.06	0.187	3.41	0.194
5	Energy sales (Rs/unit)	0.181	7.06	0.194	3.41	0.200

4.140 The normative O&M expenses based on the norms specified in above table is given below

Table D 61: Normative Employee expenses and A&G expenses for 2022-23		
No	**Parameters**	**Quantity**
	Norms	
1	No. of consumers (Rs. Lakh/1000)	4.539
2	Dist Transformers (Rs. Lakh/DTr)	0.896
3	Length of HT line (Rs. Lakh/km)	0.887
4	Length of LT line (Rs. Lakh/km)	0.194
5	Energy sales (Rs/unit)	0.200
	Parameters	
6	No. of consumers	13648851
7	No. of Distribution Transformers (11& 33 kV)(Nos)	87911
8	Length of HT Line (Km)	70269.41
	Length of LT line (km)	302626
9	Energy sales in MU	26743.18
10	**Employee, A&G Cost (Rs Cr)**	**3152.28**

4.141 As per Annexure 7 of Tariff regulations 2021, R&M expenses of Distribution licensees is:

(i) *The R&M expenses for the distribution licensees is benchmarked at 4% of value of opening Distribution gross fixed assets (excluding the value of land and land under lease) as on Ist April of respective financial year.*

4.142 Based on the above normative R&M expense is calculated as follows:

Table D 62 Normative R&M expenses (Rs Cr.)	
Particulars	**Amount (Rs Cr)**
Opening GFA as on 01.04.2023 (Rs Cr.)	15804.50
Less land	22.52
R& M @ 4% on opening GFA	631.28

4.143 Thus, O&M expenses sought for true up as per the norms prescribed in the Tariff Regulations, 2021 amounts to Rs. 3783.56 Cr. (Rs. 3152.28 Cr+ Rs 631.28 Cr). The actual O&M expenses for the year 2023-24 was Rs 4030.39 crores. A comparison of approved normative and actual O&M expenses is given in table below:

Table – D63 : Components of O&M Expenses for SBU D (Rs Cr)							
No	**Particulars**	**Approved**	**As per accounts**	**Normative**	**True up claim**	**Variation of true up claim from approval**	**Remarks**
1	**O & M Expenses**	3605.40	4030.39	3783.56	3783.56	178.16	**See Form D 3.4**

4.144 It may please be noted that, Hon'ble Commission, as per order dated 25.01.2022, has approved 30321 nos as allowable staff strength and ordered that the revised numbers will be considered since 01.04.2022. But the tariff Regulation for the third control period was finalized before the redetermination of staff strength and the release of true up orders for the base year (2018-19). As per Tariff Regulation, 2021 notified by the KSERC on 16.11.2021, KSERC had determined the norms based on the O&M cost approved in the ARR for 2018-19 (as per order dated 08.07.2019) along with a sum of Rs. 248.47 Cr towards annual pay revision (due from July / August 2018) impact (provision as per accounts RS.176 Cr /8.5 months *12 months) Therefore, the provisional norms determined as per Tariff Regulation, 2021 is to be finalized for claiming the eligible O&M expenses in the truing up petition for 2023-24.

4.145 The expected O& M Expenses based on the revised norms as detailed above is given in table below.

Table D 64 Normative O& M expenses for 2023-24		
No	**Parameters**	**Quantity**
1	No. of consumers (Rs. Lakh/1000)	4.857
2	Dist Transformers (Rs. Lakh/DTr)	0.959
3	Length of HT line (Rs. Lakh/km)	0.949
4	Length of LT line (Rs. Lakh/km)	0.208
5	Energy sales (Rs/unit)	0.214
6	No. of consumers (Nos)	13648851
7	No. of Dist Transformers (Nos)	87911
8	Length of HT Line (Km)	70269.41
9	Length of LT line (km)	302626
9	Energy sales in MU	26743.18
10	**Employee, A&G Cost (Rs Cr)**	**3374.22**

Expenses towards pay revision

4.146 Pay Revision arrears in line with the orders of the Honourable Commission amounts to Rs 7.93 Cr. as detailed in Chapter 5 may kindly be approved over and above norms.

Depreciation:

4.147 Hon'ble Commission had approved the depreciation of SBU-D for FY 23-24 at Rs. 285.00 Cr. Actual Depreciation for the Distribution assets as per accounts for the year has been Rs. 807.68 Cr as detailed in **Form D3.5 (2).**

4.148 Hon'ble Commission may kindly note that depreciation as per accounts has been worked out in line with IND AS and claw back depreciation was determined and credited to other income. But, as per MYT Regulation for the control period, depreciation is allowable as per the provisions contained in the Tariff Regulations, 2021 in which applicable rates as well as methodology to be followed are specified therein. Accordingly, allowable depreciation for SBU D has been worked out at Rs.309.36 Cr. as detailed in Chapter 5. Hon'ble Commission may kindly approve the same.

Interest and Finance charges

4.149 Hon Commission had approved the interest and finance charges for SBU-D for the year 2023-24 at Rs. 1541.56 Cr. However, as per the audited accounts, the actual expenses incurred under interest and finance charges were Rs. 2052.14 Cr (after capitalized portion of interest). True up claim amounts to Rs. 1637.86 Cr as summarized below.

Table D65: Comparison of I&FC for 2023-24 (Rs. Cr)				
No	**Particulars**	**Approved**	**Accounts**	**True up claim**
1	Interest on outstanding Loans	501.91	336.88	483.76
2	Less: Interest capitalized		105.78	
3	Net interest	501.91	231.1	483.76
4	Interest on Security Deposit	156.11	265.92	146.88
5	Interest on GPF	196.44	164.88	164.88
6	Other Interest		44.55	44.07
7	Interest on Master Trust Bond	475.19	477.03	477.03
8	Carrying cost on revenue gap till 2023-24	211.91	0	321.24
9	Interest on working capital		161.77	0
10	Finance cost on fair valuation of KWA arrear		706.89	
10	**Grand Total**	**1541.56**	**2052.14**	**1637.86**

4.150 KSEBL has executed capital works to the tune of Rs. 1463.68 Cr. and added fixed assets of Rs. 1766.73 Cr under SBU D during the year as detailed in Chapter 5. Physical addition to fixed assets is summarized below:

Table – D66: Details of Capital Works during 2023-24		
	Particulars	**Achievement**
1	Service Connections (Nos)	338864
2	HT Line construction (km)	1326.4
3	LT Line construction (km)	4078
4	Transformer installation (Nos)	2355
5	Line conversion (km)	1347
6	HT re-conductoring (conductor km)	384.68
7	LT re-conductoring (conductor km)	5691
8	Faulty meter replacement (Nos)	298976

Interest on normative loan

4.151 Honourable Commission vide MYT order dated 25.06.2022 provisionally approved Rs 501.91 Cr. as interest on normative basis. The normative loan at the beginning of the year ie; on 01.04.2023 has been determined as per Regulation by considering GFA, Approved depreciation, Consumer contribution and grants and Equity. Thus, addition, repayment and closing normative loan as on 31.03.2024, interest on normative loan along with SBU wise segregation are determined and furnished in chapter 5. Based on the computation, KSEBL is entitled to Rs. 483.76 Cr as interest on normative loan for the year 2023-24, which may kindly be approved.

Interest on security deposit:

4.152 Hon'ble Commission approved interest on SD for the year at Rs. 156.11 Cr. Actual disbursement in the FY 2023-24 amounted to Rs. 265.92 Cr (disbursement made against provision amounting to Rs 128.74 Cr created @ 4.25% on SD for FY 2022-23) as per provisions in Supply Code, 2014. Hon'ble Commission may kindly approve the actual disbursement in 2023-24 amounting to Rs. 146.88 Cr under SBUD.

Interest on working capital:

4.153 The Normative Working Capital for SBU-D**,** as per Regulation 32, includes O&M Cost for one month, cost of maintenance spares as one twelfth of average of the book value of stores for that financial year, receivables for two months excluding security deposits. KSEB Ltd has computed the Interest on Working Capital at a rate of 11.15% (9.15% (base rate) + 2%) as per Regulation 32(2). The parameters adopted for computation of Interest on Working capital for the control period are furnished below.

	Table D67 Parameters for estimation of working capital (Rs Cr)		
Sl No	**Particulars**	**Amount (Rs Cr)**	
1	Normative O&M cost for FY 2023-24	3783.56	
2	Pay revision expenses	7.93	
3	Additional contribution to Master Trust	333.42	
4	Total O&M Cost		4124.90
5	Revenue for the year 2023-24	**19818.44**	**19818.44**
6	Opening stock of inventory	943.97	
7	Closing stock of inventory	1025.62	
8	Average inventory		984.80
9	Opening Security deposit (Note 21)	3941.69	3941.69

4.154 Based on the above, net working capital computed is negative as detailed below

	Table D 68 Interest on working capital (Rs Cr)	
Sl No	**Particulars**	**Rs Cr**
1	O&M Exp for 1 month	343.74
2	Maintenance spares @ One twelfth of average of opening and closing balance of Inventory	82.07
3	Revenue (2 months)	3303.07
4	Total	3728.88
5	Less: Security deposit	3941.69
6	**Net Working Capital**	**-212.81**

Carrying cost on approved revenue gap:

4.155 Hon'ble Commission approved carrying cost of Rs. 211.91 Cr for the year 2023-24 after deducting average GPF balance from the approved gap to determine the revenue gap for which carrying cost is eligible. Hon'ble Commission had trued up revenue gap of Rs. 6408.37 Cr till 2022-23 (excluding 2014-15, for which orders are yet to be issued). Thus, carrying cost for the revenue gap of Rs. 6408.37 Cr is claimed in this petition. In addition to the above, the carrying cost of current year gap amounting to Rs 1002.51 Cr (excluding carrying cost) is also claimed in this petition.

4.156 Based on the methodology adopted by the Hon'ble Commission, average GPF balance of 2926.81 Cr (Rs.2891.85 Cr for 2022-23) and security deposit over normative working capital (Rs 212.81 Cr) have been excluded from the approved gap to determine the revenue gap eligible for carrying cost in line with Regulation 29 (9) of Tariff Regulation, 2021.

4.157 Actual average interest on loans during 2023-24 for SBU D had been 8.52 %. Accordingly, a sum of Rs.278.87 Cr is claimed as carrying cost up to 2022-23 in respect of trued up revenue gap. In addition to the above, the carrying cost amounting to Rs 42.71 Cr of current year gap amounting to Rs 1002.51 Cr is also claimed in this petition as detailed below:

Table D 69: Computation of carrying cost on approved revenue gap (Rs Cr)		
1	Trued up revenue gap till 31.03.2023	6408.37
2	Less: Average PF balance for 2023-24 (3000.09+2852.48)/2	2926.28
3	Less: Security deposit as per Reg 29 (9)	212.81
4	Net revenue gap (1-2-3)	3269.28
5	**Carrying cost at average interest rate of 8.52 % on above(A)**	278.54
6	Current Year revenue Gap excluding carrying cost	1,002.51
7	**Carrying cost on current year revenue gap at average interest rate of 8.52 % on above(B)**	42.71
8	**Total carrying cost(A) +(B)**	321.24

4.158 Hon'ble Commission may kindly approve Rs. 321.24 Cr towards carrying cost for 2023-24 for SBU D.

Interest on PF:

4.159 Hon'ble Commission approved Rs 196.44 Cr towards interest on PF for 2023-24. But as per the audited accounts, the actual interest on PF was Rs. 164.88Cr, which is less by Rs. 31.56 Cr. Therefore Rs. 164.88 Cr is sought for true up as detailed in chapter 5, which may kindly be approved.

Interest on Master Trust Bonds:

4.160 State Government, as per notifications dated 31.10.2013 and 28.01.2015, ordered creation of a Master Trust for meeting the unfunded liability of pension, gratuity and leave surrender as on 31.10.2013, in respect of the personnel transferred from erstwhile KSEB to KSEBL. The total liability as on 31.10.2013 was estimated at Rs.12418.72 Cr and necessary funding arrangements put in place through issue of 2 series of Bonds. Hon Commission recognized the unfunded pension liabilities as above and approved recovery of interest on KSEBL share of Bonds as per Tariff Regulations, 2021. Thus, Hon'ble Commission approved Rs.475.19 Cr towards interest on

Master Trust bonds against share of SBU D. Actual share of expenses for SBU D has been Rs. 477.03Cr as detailed in chapter 5, which may kindly be approved.

Other Charges:

4.161 Hon'ble Commission has not approved any amount towards other interest. Actual expenses incurred as per accounts amounts to Rs.750.96 Cr as detailed in chapter 5, which may kindly be approved.

Table D.70 Details of other charges under Interest and Finance Charges (Rs. Cr)		
No	**Particulars**	**SBU D**
1	Generation based incentive	
2	Other Bank Charges	0.81
3	Interest on Power purchase	43.26
4	Finance cost on fair valuation of amount received from KWA arrears	706.89
5	**Total**	**750.96**

4.162 KSEBL is claiming only Rs 44.07 Cr towards other expenses as detailed in Chapter 5 of this petition.

4.163 In view of the above submission, Hon'ble Commission may kindly approve interest and finance charges as summarized below:

Table D71: Comparison of I& F Charges-Approved, Actual and TU requirement for 2023-24 (Rs. Cr)					
No	**Particulars**	**Approved**	**Accounts**	**TU Sought**	**Difference over approval**
1	Interest on outstanding Loans	501.91	336.88	483.76	-18.15
2	Interest on Security Deposit	156.11	265.92	146.88	-9.23
3	Interest on GPF	196.44	164.88	164.88	-31.56
4	Other Interest	0	44.55	44.07	44.07
5	Interest on Master Trust Bond	475.19	477.03	477.03	1.84
6	Carrying cost on revenue gap till	211.91	0	278.54	66.63
	Carrying cost on current year revenue			42.71	42.71
7	Interest on working capital	0	161.77	0	0.00
8	Less: Interest capitalized	0	105.78	0	0.00
9	Finance cost on fair valuation of KWA	0	706.89	0	0.00
10	**Grand Total**	**1541.56**	**2052.14**	**1637.86**	**96.30**

Additional contribution to Master Trust

4.164 Hon'ble Commission was pleased to provisionally approve Rs. 333.42 Cr towards additional contribution to Master Trust under SBU-D. KSEBL seeks truing up of Rs. 333.42 Cr also as detailed in chapter 5.

Repayment of Bonds issued to Master Trust

4.165 Hon'ble Commission, in line with Regulation 34 (iv) of Tariff Regulation, 2021, had approved Rs. 407.20 Cr for the year 2023-24 in the MYT order dated 25.06.2022. Hon'ble High Court of Kerala, as per judgment dated 07.09.2023 in WP (C) 19205/2023, set aside the enabling Tariff Regulation 2021 (Regulation 34 (iv)) owing to fault with the procedure. Hon'ble Court however observed that by following the procedure contemplated for framing regulations, even the aspects contained in Regulation 34(iv) can be included. In view of the order of the Hon'ble High Court on the above petition, KSERC has published the draft Second Amendment notification on 1.12. 2023. After conducting public hearing and considering the suggestions and objections of the stakeholders, the final notification viz. Kerala State Electricity Regulatory Commission (Terms and conditions for determination of Tariff) (Second Amendment) Regulation, 2024 was published on 27th February 2024. The State Commission has also issued the Truing up order for the year 2022-23 on 28.06.2024 duly allowing the additional contribution of the Master Trust.

4.166 In view of the above Honourable Commission may kindly approve the aim of Rs. 339.42Cr for truing up for the year 2023-24, under bond repayment.

Return on Equity:

4.167 Hon'ble Commission vide order dated 25.06.2022 approved Return on equity at Rs 253.50 Cr, which, the Hon'ble Commission may kindly true up.

Other expenses and exceptional items:

4.168 As detailed in chapter 5, a sum of Rs. 22.19 Cr is sought for true up, under other expenses and exceptional items Rs. 15.20 Cr, which may kindly be trued up.

Amortisation of Intangible assets towards software development cost

4.169 KSEBL has developed its own software in major functional areas such as human resources, accounting, Oruma and Oruma net, etc. The employee cost attributable to the software developments were capitalised and then amortised the software development cost during 2023-24 to the tune of Rs. 9.64 Cr.

Non-tariff income and other income:

4.170 Hon'ble Commission approved Rs.801.27 Cr under non-tariff income. The share of non-tariff income and other income of SBU D as per accounts amounting to Rs. 1240.90 Cr is detailed in Form D 2.4. Rs. 920.18 Cr is sought for true up as detailed in chapter 5 of the petition. Hon'ble Commission may kindly approve the same.

Tariff income:

4.171 Hon'ble Commission approved revenue from sale of power as Rs 15903.34 Cr and net revenue as Rs 15873.80 Cr after deducting Rs 29.54 Cr as power factor incentive. An amount of Rs 382.16 Cr was also approved towards revenue from surplus energy sales outside the State. Further Honourable Commission vide order dated 25,06.2022 has approved to recover an additional revenue of about Rs 1010.94 crores through revision in Tariff. Further Honourable Commission

vide interim order dated 31.10.2023 has approved to recover an additional revenue of Rs. 532.50Cr through revision in tariff. Further Honourable Commission vide order dated 25.01.2023 has approved to recover fuel surcharge amount for the period of April to September 2022 of Rs 87.07 crores and allowed KSEBL to recover it from all consumers including consumers of other distribution licensees except consumers having monthly consumption less than 40 units and connected load less than 1000 W. from the consumption from February 2022 for a period of 4 months. . Further Honourable Commission vide order dated 31.05.2023 has approved to recover fuel surcharge amount for the period of July to December 2022 of Rs 248.04 crores and allowed KSEBL to recover it from all consumers including consumers of other distribution licensees except consumers having monthly consumption less than 40 units and connected load less than 1000 W. from the consumption from June 2023 till the recovery of the amount. In addition to the above, KSEBL collected Fuel Surcharge through Automatic Recovery Mechanism, limited to a ceiling limit of 10paise per unit.

4.172 Thus, the KSEBL has realised revenue based on the prevailing Schedule of Tariff and prevailing fuel surcharge. Actual revenue from tariff during the year 2023-24 which amounted to Rs. 19757.84 Cr. Hon'ble Commission may kindly note that reconciliation of revenue sale as per this submission and as per audited accounts is furnished in chapter 5. A summary of the revenue from sale of power inside the State as well as outside State is given below.

Table – D72: Revenue from Sale of Power within the State

No	Category	Energy sales		Revenue	
		MU	%	Rs Cr	%
1	Domestic incl colonies	13735.43	49.75%	7527.85	38.10%
2	Commercial	4690.58	16.99%	4837.65	24.48%
3	Public lighting	374.57	1.36%	195.74	0.99%
4	Irrigation & Dewatering	451.24	1.63%	152.23	0.77%
5	Industrial LT	1298.32	4.70%	1027.34	5.20%
6	Railway Traction	435.27	1.58%	293.99	1.49%
7	Bulk Supply /CESCK	565.41	2.05%	490.29	2.48%
9	HT	4666.65	16.90%	4168.95	21.10%
10	EHT	1385.96	5.02%	1063.8	5.38%
11	Energy supplied to RGCCPP to meet its auxiliary consumption during non-operative periods	5.9			
12	**Total sales (within state)**	**27609.33**	100.00%	**19757.84**	100.00%
13	Interstate sale	71.27		60.6	
14	**Total**	**27680.6**		**19818.44**	

4.173 Hon'ble Commission while approving the revenue from sale of power had deducted Rs. 29.54 Cr. as power factor incentive against which incentive actually extended during the year was Rs 56.49 Cr, (booked under A&G expenses) as explained in chapter 5. Accordingly, actual power factor incentive to the tune of Rs. 56.49 Cr is deducted from the revenue from SoP. The revenue from sale of Power includes fixed charges/demand charges, energy charges, the subsidy allowed

by the Government in the case of domestic consumers with monthly consumption up to 120 units and LT Agricultural consumers and cross subsidy surcharge. It may be seen that 49.75% of total consumption was domestic which contribute 38.10% of revenue. Actual revenue was more by Rs. 3854.5 Cr from approval. A comparison of approved vs actual sale is given in table below.

Table – D75: Revenue comparison Approved Vs Actuals							
No	**Category**	**Energy sales (MU)**			**Revenue (Rs. Cr)**		
		Approved	**Actual**	**Deviation**	**Approved**	**Actual***	**Deviation**
1	Domestic	13900.41	13735.43	-164.98	6696.97	7527.85	830.88
2	Industrial	1123.36	1298.32	174.96	835.07	1027.34	192.27
3	Agriculture	411.18	451.24	40.06	112.72	152.23	39.51
4	Commercial	3556.05	4690.58	1134.53	3487.38	4837.65	1350.27
5	Public Lighting	307.23	374.57	67.34	148.6	195.74	47.14
6	HT Total	4066.88	4666.65	599.77	3179.17	4168.95	989.78
7	EHT Total	1293.13	1385.96	92.83	785.98	1063.8	277.82
8	Railway Traction	362.07	435.27	73.2	226.02	293.99	67.97
9	Bulk Supply	677.27	565.41	-111.86	431.43	490.29	58.86
10	Energy given to RGCCPP to meet their auxiliary consumption		5.9				
11	**Total sales (within state)**	**25697.59**	**27609.33**	**1911.74**	**15903.34**	**19757.84**	**3854.5**
12	Interstate sale	1019.09	71.27	-947.82	382.16	60.6	-321.56
13	**Total**	**26716.68**	**27680.6**	**963.92**	**16285.5**	**19818.44**	**3532.94**
14	Less Power Factor incentive			0	29.54	56.49	26.95
15	**Revenue from Sale of power**	**26716.68**	**27680.6**	**963.92**	**16255.96**	**19761.95**	**3505.99**

4.174 Hon'ble Commission may kindly true up actual revenue from sale of power (including external sale) as above.

4.175 It is respectfully submitted that, as per audited accounts, KSEBL has suffered a loss of Rs.534.91 Cr. But, as per regulatory accounts, revenue gap during 2023-24 is Rs. 1323.75 Cr only as deliberated in detail in this petition. Even though Hon'ble Commission had amortized past approved revenue gap of to the tune of Rs. 850 Cr while approving ARR for the year 2023-24, in the absence of regulatory surplus during the year, KSEBL is not able to amortize the past revenue gap during the year as approved by Honourable Commission. Summary of ARR & ERC of SBU D for the year 2023-24 is furnished below:

TABLE – D76: ARR & ERC OF DISTRIBUTION BUSINESS UNIT (Rs Cr)			
Particulars	**Approved**	**True up**	**Difference**
Cost of Generation (SBU-G)	690.46	626.48	-63.98
Cost of Power Purchase incl RLDC charges	10564.23	12,982.59	2418.36
Cost of Intra-State Transmission (SBU-T)	1533.34	1553.14	19.8

Particulars	Approved	True up	Difference
Interest & Finance Charges	1541.56	1637.86	96.3
Additional contribution to Master Trust	333.42	333.42	0.00
Depreciation	285	309.36	24.36
Normative O&M Expenses	3605.4	3783.56	178.16
Pay revision arrears	0.00	7.93	7.93
Return on equity (14%)	253.5	253.5	0.00
Other Expenses	0.00	22.19	22.19
Exceptional items	0.00	15.2	15.2
Sharing of gains on account of higher T&D loss reduction	0.00	131.59	131.59
Recovery of past gap	850	0.00	-850
Amortisation of intangible assets	0.00	9.64	9.64
Repayment of Bonds	339.42	339.42	0.00
Total ARR	19996.33	22,005.88	2009.55
Tariff Income incl fuel surcharge	15903.34	19,757.84	3854.5
Less Power factor incentive	29.54	56.49	26.95
Revenue from external sale	382.14	60.6	-321.54
Non-Tariff Income	801.27	920.18	118.91
Total ERC	17057.21	20,682.13	3624.92
Net Revenue Gap (-)/ Surplus (+)	**(-)2939.12**	**(-)1,323.75**	**(-)1,615.37**

4.176 Annexures described in paragraphs above are attached here under. The forms are attached thereafter. The next chapter contains explanations relating to various expenses and income in general.

Annexure-D 1	
Power Availability for 2023-24	
Month	**Gross Availability of electricity for Supply**
April	100%
May	100%
June	100%
July	100%
August	100%
September	100%
October	100%
November	102%
December	101%
January	101%
February	101%
March	100.18%
Average	100.52%

Appendix –D2: Details of approval of long term /short term contracts for procuring power during 2022-23			
Source	**Allocated capacity (MW)**	**Commission Approval Letter/0rder**	**Approved Tariff**
Central generating Stations			
RSTPS I & II	**254.1**	**Deemed Approval vide Tariff Regulations 2014**	**CERC Tariff**
RSTPS III	**63.2**		**CERC Tariff**
Talcher Stage-II	**430.4**		**CERC Tariff**
Simhadri TPS Stg II	**83**		**CERC Tariff**
NLC Stage-I	**65.142**		**CERC Tariff**
NLC Stage-II	**92.82**		**CERC Tariff**
NLC 1 Expansion	**61.026**		**CERC Tariff**
NLC II Expansion	**72.65**		**CERC Tariff**
MAPS	**23.628**		**CERC Tariff**
KAIGA Stage I to IV	**77.704**		**CERC Tariff**
Kudankulam Unit I & II	**267.8**		**CERC Tariff**
Vallur	**53.85**		**CERC Tariff**
NTPL	**76.7**		**CERC Tariff**
Kudgi(Unit I to III)	**115.44**	**KSERC order dated 22.02.2023**	**CERC Tariff**
NNTPP	**32.6**	**KSERC order dated 22.02.2023**	**CERC Tariff**
continued			

Annexure D 2 continued

Source	Contracted Capacity (MW)	Commission Approval Letter/0rder	Approved Tariff
Long Term contracts			
Maithon Power	150	No.2158/C.Engg/Maithon/2013/1398 dated 26.12.2013	Approved under Section 62 of Electricity Act 2003
Maithon Power	150	Order dated 08 - 07- 2015	
DVC Mejia	100	No.500/C.Engg/DVC/2014/348 dated 28.03.2014 & Order dated 21.01.2016	
DVC RTPS	50		
Jindal Power Ltd-PSA I (DBFOO)	200	Order dated 30.8.2016	Approved under Section 63 of Electricity Act 2003.
BALCO	100	Order dated 30.8.2016	
Jhabua Power Ltd-PSA-I (DBFOO)	115	Pending for approval	Rate yet to be approved
Jindal Power Ltd-PSA I (DBFOO)*	150		
Jhabua Power Ltd-PSA-I (DBFOO)*	100		
JITPL*	100		
SECI Wind	200	Order dated 19.09.2020 on OP 53/19	2.89
IPPs inside the State			
RGCCPP, Kayamkulam	359.28	Deemed Approval vide Tariff Regulations 2014	
Wind- Ramakkalmedu	14.25		3.14
Wind-Koundikkal	4.8		3.14
Wind- Agali	13.8	Order dated 07.08.2008	3.14
Wind Ahalya	8.4	Order dated 22-02-2017/ Order dated 10.10.2022	5.34/4.70
INOX	16	Order dated 03.10.2018.	4.09
Kosamattom	1	Order dated 7.12.2019	3.93
Ullunkal(EDCL)	7	Order dated 13.11.2005	2.44
Iruttukanam Stage-I	*4.5*	Order dated 15.12.2006	2.07
Iruttukanam Stage-		Order dated 30.09.2011	2.94
Karikkayam HEP(AHPL)	15	Order dated 01.11.2014	4.16
Meenvallom(PSHCL)	3	Order dated 12.01.2015	4.88
Minar Renewable energy	8	06.09.2019 & 14.11.2019	3.96

Source	Contracted Capacity (MW)	Commission Approval Letter/0rder	Approved Tariff
Anakkampoil	8	Interim order dated 25.10.2021	4
Arippara	4.5	Order dated 08.03.2023	4.3
IPPs inside the State			
Source	**Contracted Capacity (MW)**	**Commission Approval Letter/0rder**	**Approved Tariff**
Mukkadom	4	Interim Order on OP NO 64/2023 dated 26.04.2024	4.15
Solar IREDA	50	Order dated 06.02.2019	3.83
THDCIL	50	Order dated 29.10.2021 on OP No 33/2021	3.1
CIAL Ettukudukka	10	Order dated 10.11.2022	2.37
NTPC Solar Float	92	CERC order dated 09.05.2022	2.94
TP Sourya	110	Order on OP NO 73.2021 dated 5.01.2022	2.44
RSAPL ANTA	90	Order on OP NO 11.2021 dated 28.01.2022	2,44
SECI-Wind	75	Order dated 19.02.2020	2.89
Medium Term			
JITPL through PTC		Order dated 07.10.2021	Approved under Section 63 of Electricity Act 2003.

Power Purchase through DEEP contracts

Sl No	Period	Generator/ Trader	Quantum (MW) on firm basis	Rate (Rs/ kWh)	Source	Quantum at Kerala periphery MU	Power Purchase Cost (Rs Cr)
KSERC order dated 26.07.2022							
1	Apr-23	PTC(RTC)	100	7.95	JNSTPP	61.32	48.74
KSERC order dated 17.05.2023							
1	21.05.23 to 31.05.23 (19.00 to 23.00 pm)	NVVN	50 MW 19.00 to 23.00	9.26	IL &FS	2.20	2.04

Sl No	Period	Generator/ Trader	Quantum (MW) on firm basis	Rate (Rs/ kWh)	Source	Quantum at Kerala periphery MU	Power Purchase Cost (Rs Cr)
KSERC order dated 31.07.2023							
1	Aug-23	TPTCL	**50**	**5.23**	Adhunik Power & natural Resources Ltd, WB	33.21	17.37
		AEL	**100**	**5.23**	Adani power Ltd, Raipur	66.66	34.86
		TPTCL	**45**	**5.23**	NavaBharath Ventures Ltd,	30.14	15.76
		PTC	**25**	**5.23**	NavaBharath Ventures Ltd,	16.70	8.73
2	Sep-23	TPTCL	**75**	**5.38**	JITPL, Odisha	49.82	26.80
		TPTCL	**50**	**5.38**	Adhunik Power & natural Resources Ltd, WB	35.84	19.28
		TPTCL	**45**	**5.39**	NavaBharath Ventures Ltd,	29.56	15.93
		Jindal Power Ltd	**80**	**5.42**	JPL, Chattisgarh	55.64	30.16
3	Oct-23	TPTCL	**100**	**5.12**	JITPL, Odisha	69.27	35.46
		TPTCL	**50**	**5.14**	PPGCL	34.64	17.80
4	Nov-23	TPTCL	**150**	**5.38**	JITPL, Odisha	97.42	52.41
		GMRETL	**25**	**5.38**	Maadurga Thermal Power Company Ltd,	17.37	9.35
		Jindal Power Ltd	**75**	**5.46**	JPL, Chattisgarh	52.12	28.46
5	Dec-23	GMRETL	**25**	**5.48**	Maadurga Thermal Power Company Ltd,	15.56	8.53
		TPTCL	**150**	**5.48**	JITPL, Odisha	100.96	55.33
		NVVN	**75**	**5.5**	JaipeeNigree STPPMP	53.46	29.40
6	Jan-24	GMRETL	**25**	**5.49**	Maadurga Thermal Power Company Ltd	17.34	9.52
		NVVN	**100**	**5.49**	JaipeeNigree STPP, MP	74.39	40.84
		NVVN	**50**	**5.49**	JaipeeBinaa STPP	36.58	20.08
7	Feb-24	NVVN	**100**	**5.58**	JaipeeNigree STPP, MP	63.88	35.65
		NVVN	**50**	**5.58**	JaipeeBinaa STPP, MP	34.80	19.42
		GMRETL	**25**	**5.58**	Maadurga Thermal Power Company Ltd	16.93	9.45
		TPTCL	**45**	**5.75**	Tata Power Haldia	31.32	18.01
		AEL	**30**	**5.82**	Adani power Ltd, Jaipur	20.29	17.78
8	Mar-24	NVVN	**100**	**6.13**	JayipeeNigree STPP, MP	71.02	43.54
		NVVN	**50**	**6.13**	JaipeeBinaa STPP, MP	35.51	21.77
		GMRETL	**25**	**6.13**	Maadurga Thermal Power Company Ltd,	17.76	10.88
		Manikaran Power Ltd	**25**	**6.25**	BLA Power Private Ltd,U-II	16.74	10.46
		Manikaran Power Ltd	**25**	**6.34**	BLA Power Private Ltd, UnitI,MP	16.74	10.61

Annexure D3 Details of energy availability and surrender of CGS during 2023-24(MU)						
CGS	**Approved**	**Actual**	**Surrender**	**Total**	**Increase/decrease in availability after accounting surrender**	**Reasons for reduction in availability**
NLC Stage-I	422.33	252.87	0.16	253.03	-169.30	Forced Outage of units and reduced DC due to space constraint for lignite conservation. Lignite conservation is seen in all months. NLC STG22 AOH of U1 from 6.10.23 to 30.10.24, Unit1 FSD due to Stator E/f 1.1.24 to 30.1.24
NLC Stage-II	603.17	291.71	0.51	292.22	-310.95	
NLC 1 Expansion	464.13	397.94	0.97	398.91	-65.22	1.4.23 to 20.5.23 Dc reduced due to coal issue. Unit 1 under FSD from 23.5.23 to 22.6.23 .DC reduced 2.7.23 to 18.8.23 due to fuel shortage. AOH of Unit2 1.10.23 to 21.10.23 , Unit 2 AOH 21.11.23 to 26.11.23. AOH unit 1 from 1.12.23 to 26.12.23
NLC II Expansion	511.51	275.59	0.35	275.94	-235.57	Unit 2 FSD 14.11.23 to 23.11.23 and 8.12.23 to 21.12.23 due to FBHE four coil puncture. Unit 1 RAPH choke 1.2.24 to 15.2.24. 8.10.23 to 21.10.23 cyclone puncture Unit 2 and 17.10.23 to 27.10.23 Unit 1
MAPS	122.72	65.90	0	65.90	-56.82	Unit1 continuing under mace work and not revived
Vallur	350.72	263.52	8.32	271.84	-78.88	FSD of unit 1 under LP turbine blade failure 1.9.23 to 1.03.24,
NTPL	506.33	431.40	6.89	438.29	-68.04	AOH of Unit1 19.7.23 to 4.9.23, DC reduced to imported coal issue.
Total	**2980.91**	**1978.93**	**17.2**	**1996.13**	**-984.78**	

Appendix – D4: Source wise details of Power Purchase for 2023-24

No	Source	Approved Energy (MU)	Approved Cost (Rs Cr)	Actual Energy (MU)	Actual Power purchase cost (Rs Cr)	True up claim (Rs Cr)	Deviaton	
							Energy (MU)	Cost (Rs Cr)
1	**CENTRAL GENERATING STATIONS**							
	RSTPS I & II	1696.81	680.42	1788.93	839.11	839.11	92.12	158.69
	RSTPS III	425.93	170.20	512.24	229.64	229.64	86.31	59.44
	Talcher Stage-II	2980.96	1010.68	3215.98	860.28	860.28	235.02	-150.40
	Simhadri TPS Stg II	600.03	288.09	584.36	331.31	331.31	-15.67	43.22
	NLC Stage-I	422.33	181.34	252.87	99.45	99.45	-169.46	-81.89
	NLC Stage-II	603.17	260.44	291.71	99.44	99.44	-311.46	-161.00
	NLC 1 Expansion	464.13	192.42	397.94	141.07	141.07	-66.19	-51.35
	NLC II Expansion	511.51	277.52	275.59	127.42	127.42	-235.92	-150.10
	MAPS	122.72	32.19	65.90	17.31	17.31	-56.82	-14.88
	KAIGA	389.17	139.71	598.08	215.54	215.54	208.91	75.83
	Kudankulam	1507.12	641.55	1642.85	749.53	749.53	135.73	107.98
	Vallur	350.72	193.37	263.52	147.14	147.14	-87.20	-46.23
	NTPL	506.33	267.56	431.40	219.68	219.68	-74.93	-47.88
	Kudgi	780.71	445.65	778.17	539.23	539.23	-2.54	93.58
	NNTPP	226.17	116.45	231.02	99.63	99.63	4.85	-16.82
	Telangana STPP	0.00	0.00	14.85	8.01	8.01	14.85	8.01
	Aravally Corporation Ltd(IGSTPS Jhajjar)	0.00	0.00	0.00	7.29	7.29	0.00	7.29
	PGCIL losses	397.46		393.54		0.00	-3.92	0.00
	Sub Total	**11190.35**	**4897.60**	**10951.87**	**4731.08**	**4731.08**	**-238.48**	**-166.52**

No	Source	Approved Energy (MU)	Approved Cost (Rs Cr)	Actual Energy (MU)	Actual Power purchase cost (Rs Cr)	True up claim (Rs Cr)	Deviaton	
							Energy (MU)	Cost (Rs Cr)
	Thermal IPP							
	RGCCPP, Kayamkulam		100.00	0.00	70.65	70.65		-29.35
	Renewable Energy							
	Wind IPP							
	Wind- Ramakkalmedu	27.87	8.75	25.02			-2.85	
	Wind-Koundikkal	12.56	3.94	9.72	19.41	19.41	-2.84	
	Wind- Agali	29.72	9.33	26.82			-2.90	-2.61
	Wind Ahalya	20.98	10.97	18.29	8.6	8.60	-2.69	-2.37
	INOX	30.84	12.61	34.59	14.15	14.15	3.75	1.54
	Kosamattom	1.01	0.4	1.47	0.58	0.58	0.46	0.18
	Wind Sub total	122.98	46.01	115.91	42.73	42.73	-7.07	-3.28
	SHEP							
	Ullunkal(EDCL)	17.79	4.34	17.78	4.34	4.34	-0.01	0.00
	Iruttukanam Stage-I & II (Viyyat)	24.69	6.67	23.61	5.39	5.39	-1.08	-1.28
	Karikkayam HEP (AHPL)	37.56	15.62	39.74	16.53	16.53	2.18	0.91
	Meenvallom (PSHCL)	8.44	4.12	6.85	3.34	3.34	-1.59	-0.78
	Minar Renewable energy project Ltd	25.54	8.91	18.13	6.33	6.33	-7.41	-2.58
	Deviar micro	0	0	0	0	0.00	0.00	0.00
	Anakkampoil	27.67	11.07	22.59	9.03	9.03	-5.08	-2.04
	Arippara	0	0	10.59	12.64	12.64	10.59	12.64
	Mukkadom	0	0	2.14	0.92	0.92	2.14	0.92
	SHEP subtotal	141.71	50.74	141.43	58.52	58.52	-0.28	7.78

No	Source	Approved Energy (MU)	Approved Cost (Rs Cr)	Actual Energy (MU)	Actual Power purchase cost (Rs Cr)	True up claim (Rs Cr)	Deviaton	
							Energy (MU)	Cost (Rs Cr)
	Solar							
	Solar IREDA	73.370	28.100	77.815	29.803	29.803	1.703	1.703
	Solar IPP ANERT	0.000	0.000	2.500	0.000	0.000	2.500	0.000
	THDCIL	73.370	22.740	95.769	29.688	29.688	22.399	6.948
	NTPC Solar	177.310	56.030	212.912	62.596	62.596	35.602	6.566
	CIAL Ettukudukka	0.000	0.000	16.031	7.167	7.167	16.031	7.167
	Solar IPP sub total	324.050	106.870	405.026	129.254	129.254	80.976	22.384
	Sub Total IPPs	588.740	203.620	662.366	230.504	230.504	73.626	26.884
	Grid connected -captive prosumer/solar prosumer (settlement)*			125.99	25.00	25.000	125.99	25.00
	Wind Power through SECI	472.8	133.33	212.23	61.75	61.750	-260.57	-71.58
	RSAPL-Antha	522.01	138.424	2.09	0.00	0.000	-326.43	
	TP Saurya			193.49	47.33	47.327		-91.10
	RPO-Solar/Nonsolar	994.81	271.75	407.81	109.08	109.08	-587.00	-162.68
	Waste to Energy	19.44	12.27					
	Additional RPO solar requirement	72.18	20.35					
	Additional RPO non- solar requirement	66.50	16.23					
	Sub Total-RE	1741.65	524.22	1196.17	364.58	364.58	-387.38	-110.79
3	**Long Term Contracts outside the State**							
	Maithon Power Ltd-I	1034.96	483.68	1169.90	493.84	493.84	134.94	10.16
	Maithon Power Ltd-II	1034.96	483.68	1169.90	493.84	493.84	134.94	10.16
	DVC Mejia	705.51	362.56	724.24	350.75	350.75	18.73	-11.81
	DVC RTPS	348.06	180.45	327.58	157.02	157.02	-20.48	-23.43
	Jindal Power Ltd-Bid I*	1,497.96	636.80	1,545.72	567.20	567.20	47.76	-69.60
	Jhabua Power Ltd-Bid I	3482.76	1372.22	916.31	373.50	373.50	-2566.45	-998.72

No	Source	Approved Energy (MU)	Approved Cost (Rs Cr)	Actual Energy (MU)	Actual Power purchase cost (Rs Cr)	True up claim (Rs Cr)	Deviaton	
							Energy (MU)	Cost (Rs Cr)
	Jindal Power Ltd-Bid II*							
	Jhabua Power Ltd-Bid II							
	JITPL							
	BALCO	748.98	342.65	775.73	304.94	304.94	26.75	-37.71
	Total exbus	8853.19	3862.04	6629.39	2741.10	2741.10	-2223.80	-1120.94
	PGCIL losses	303.68		227.58			-76.10	0.00
	Sub TotalLTA	8549.51	3862.04	6401.81	2741.10	2741.10	-2147.70	-1120.94
4	**Medium term Short term/Exchanges/swap**							
	Purchase from JITPL through PTC	962.75	320.6	935.7875	364.1942	364.1942	-26.96	43.59
5	**Short Term*(Energy at Kerala periphery)**							
	PTC-JNSTPP			61.32	48.74	48.74		
	NVVN-IL&FS			2.20	2.04	2.04		
	TPTCL-APNRL			33.21	17.37	17.37		
	ADANI Enterprises Ltd			66.66	34.86	34.86		
	TPTCL-NBVNL			30.14	15.76	15.76		
	PTC India Limited -NBVL			16.7	8.73	8.73		
	TPTCL -JITPL			49.82	26.80	26.80		
	TPTCL -APNRL			35.84	19.28	19.28		
	TPTCL -NBVL			29.56	15.93	15.93		
	Jindal Power Ltd			55.64	30.16	30.16		
	TPTCL -JITPL			69.27	35.46	35.46		
	TPTCL -PPGCL			34.64	17.80	17.80		
	TPTCL -JITPL	**240.55**	**96.22**	97.42	52.41	52.41		
	GMRETL-MTPCL			17.37	9.35	9.35		

No	Source	Approved Energy (MU)	Approved Cost (Rs Cr)	Actual Energy (MU)	Actual Power purchase cost (Rs Cr)	True up claim (Rs Cr)	Deviaton	
							Energy (MU)	Cost (Rs Cr)
	Jindal Power Ltd			52.12	28.46	28.46		
	GMRETL-MTPCL			15.56	8.53	8.53		
	TPTCL -JITPL			100.96	55.33	55.33		
	NVVN-Jaypee			53.46	29.40	29.40		
	GMRETL-MTPCL			17.34	9.52	9.52		
	NVVN--Jaypee Nigree			74.39	40.84	40.84		
	NVVN-JaypeeBinaa			36.58	20.08	20.08		
	NVVN -JaypeeNigree			63.88	35.65	35.65		
	NVVN -JaypeeBinaa			34.8	19.42	19.42		
	GMRETL-MTPCL			16.93	9.45	9.45		
	TPTCL-TATA Power Haldia			31.32	18.01	18.01		
	Adani Enterprises Ltd			20.29	11.81	17.78		
	NVVN -JaypeeNigree			71.02	43.54	43.54		
	NVVN -JaypeeBinaa			35.51	21.77	21.77		
	GMRETL -MPTCL			17.76	10.88	10.88		
	Manikaran Power Ltd-BLA Unit II			16.74	10.46	10.46		
	Manikaran Power Ltd-BLA Unit I			16.74	10.61	10.61		
	Sub Total			1275.19	718.4400	718.44		
	Exchanges			3641.70	2123.16	2123.16		
	Deviation Settlement Mechanism*(net)			464.14	206.67	206.67		
	Swap Purchase			844.65	214.22	214.22	844.65	214.22
	Sub Total	**240.55**	**96.22**	**6225.68**	**3262.49**	**3262.49**	**844.65**	**214.22**
	Total LTA /MTOA/STOA	9752.81	4278.86	13563.27	6367.78	6367.78	-1330.02	-863.13
	Transmission charges payable to PGCIL		760.55		1448.27	1448.27	0.00	687.72
	RLDC charges		3					-3.00

No	Source	Approved Energy (MU)	Approved Cost (Rs Cr)	Actual Energy (MU)	Actual Power purchase cost (Rs Cr)	True up claim (Rs Cr)	Deviaton	
							Energy (MU)	Cost (Rs Cr)
	Total Power Purchase	**22684.81**	**10564.23**	**25711.30**	**12982.35**	**12982.35**	**-1982.85**	**-455.73**
	Power returned outside the State through banking trannsactions			552.68		0.00	552.68	0.00
	DSM Export			61.49				
	Sale through IEX	1946.22		69.76	0.25	0.25	-1876.46	0.25
	Sale through PXIL			1.52				
	Net Power Purchase	20738.59	10564.23	25025.85	12982.59	12982.59		
	Total Power Purchase	**22684.82**	**10564.23**	**25711.30**	**12982.59**	**12982.59**	**3026.48**	**2418.36**

*Settled energy

Annexure D 5A

File No: FU-21/2020-FSC(Vol-IV)
Government of India
Ministry of Power

Shram Shakti Bhawan, Rafi Marg,
New Delhi, dated 9.1.2023

To,

1. Principal Secretaries/Secretary (Power/Energy) of all State Governments/ UTs
2. CMDs, Thermal Generating Stations (Central, State GENCOs/IPPs)

Subject: Direction to all Gencos including Independent Power Producers (IPPs) for timely Import of Coal for blending purposes and maximising production in captive coal mines – reg.

Due to recent surge in demand and consumption of electricity, the share of coal-based generation has increased. Although the supply of coal from all sources has increased, it is not commensurate with the requirements of thermal power plants (TPPs).

2. Ministry of Power (MoP) is very closely monitoring the coal availability for TPPs. Efforts have been made to increase coal availability from all sources (CIL, SECL, Captive) in coordination with Ministry of Coal (MoC). Although coal supply has increased during Q4, it is not adequate to meet the unprecedented increase in the demand of electricity. Similarly efforts have been made to address logistic issues in cooperation with Ministry of Railways (MoR). In order to ease out the logistic pressure on Railways, MoP has planned to transport available coal from MCL region through RSR mode to TPPs located in Northern & Western parts of the country. Accordingly directions have been issued to source 10 to 15% of their requirement through RSR mode to the States of Gujarat, Rajasthan, Maharashtra and Punjab as well as NTPC. Efforts have also been made to address the constraints in transportation of coal (RCR, Road, ARR) through all modes. Despites the above mentioned efforts, it has been noted that the infrastructure related logistic constraints of Railways will take some time to get fully addressed.

3. Grid India (POSOCO) has reported that the energy demand has increased sharply and it is expected to remain at increased level during H1 of 2023-2024. CEA has accordingly revised the target for coal requirement for Q4 of current FY and H1 of next FY. MoC has committed to supply coal to meet the revised target of 197.7 MT, however it is subject to availability of rakes and other logistic issues. As per the past trend, the likely supply of domestic coal during the H1, FY 2023-24 would be around 392 MT. Thus, around 24 MT would be the anticipated shortfall during the H1 of FY 24. It is noted that the shortfall between the daily coal consumption and the daily arrival of domestic coal is ranging from 3 Lakh Tonnes per day to 1 Lakh Tonnes per day. This shortfall is made up by blending with imported coal. It is assessed that coal stock position without blending of imported coal in DCBs will progressively decline to zero which will severely impact power supply position in the country.

4. Import of coal for blending purpose needs to be done in a streamlined manner in order to avoid clogging of logistics. During Q1 & Q2 of FY 23, imported coal stocks at ports piled up due to logistics constraints.

5. In the light of above scenario where energy demand is increasing and increase in supply of coal is not commensurate with the domestic coal requirement, the need has arisen to continue use imported coal for blending purpose. MoP, after consultation with CEA, MoC, MoR and Association of Power Producers (APP) has decided to direct all Gencos to import coal for blending 6% (by weight) for the remaining period of this current FY and H1 of the next FY (upto September, 2023). The domestic coal supply of those Gencos who do not follow these instructions shall be restricted to pro-rata basis. The Central, State Gencos and IPPs are directed to take necessary action and immediately plan to import coal through a transparent competitive procurement for blending at the rate of 6% by weight so as to have coal stocks at their power plants for smooth operations till September 2023.

6. This issues with the approval of Hon'ble Minister of Power & NRE.

(Subhash Chand)
Under Secretary to the Govt. of India

Copy to:

1. Secretary, Ministry of Coal
2. Secretary, CERC/ All SERCs
3. Chairman, Railway Board
4. Chairman, Central Electricity Authority
5. DG, Association of Power Producers (APP)

Copy also to NIC with the request to upload the same on MoP Portal.

NIC

9.1.2023

Annexure D 5B

File No. FU-21/2020-FSC (Vol-IV)
Government of India
(भारत सरकार)
Ministry of Power
(िवद्युत मंत्रालय)

नई िदल्ली, िदनांक 25 अ³टूबर, २०२३
New Delhi, dated the 25th October, 2023

To

I. Principal Secretaries/Secretary (Power/Energy) of all State Governments/ UTs

II. CMDs, Thermal Generating Stations (Central, State GENCOs/lPPs)

Subject: Direction to all GENCOs including Independent Power Producers (IPPs) for timely Import of Coal for blending purposes and maximizing production in captive coal mines — reg.

Sir/Madam,

As you are already aware, Ministry of Power issued an advisory dated 01.09.2023 regarding blending of imported coal @ 4% by weight till March, 2024.

2. The status of Power Supply position has been reviewed by Govt and it has been observed that there is consistent rising trend in the Power demand in the country coupled with inadequate supply of domestic coal which has resulted in rapid depletion of coal stocks at Domestic Coal Based (DCB) Plants across the country. The gap between receipt of domestic coal and consumption of coal (domestic + equivalent domestic of imported coal) during 01st September to 09th October, 2023 was to the tune of 12 MT.

3. About 11% fall of Hydro Generation has been recorded in H1 of FY 24 compared to the corresponding period of FY 23 due to variable monsoon rainfall. Approximately 2 GW of hydro capacity is out because of recent floods in Sikkim. The Reservoir levels in Northern, Eastern and Southern regions are less compared to the previous year as on 9th Oct' 2023, which has resulted in lower reservoir energy content at pan India level.Thishas put additional burden on Coal Based Thermal Generation.

4. Therefore, in order to ensure uninterrupted power supply across the country, after careful consideration and in consultation with Central Electricity Authority (CEA) and Grid India, it has been decided blending of imported coal @ 6% (by weight) minimum

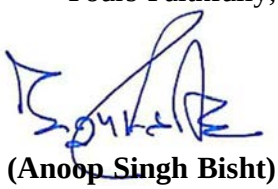
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may be continued till March 2024. The Gencos may continuously review their stock position and opt for blending as per the requirements if the shortfall in domestic coal supply is more than 6 per cent.

5. The other conditions mentioned at Para 3 & 4 of advisory dated 01.09.2023 will remain the same.

6. This issues with the approval of Hon'ble Minister of Power & NRE.

Yours Faithfully,

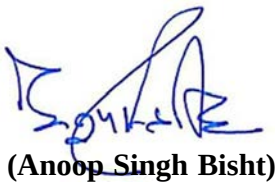


(Anoop Singh Bisht)
Deputy Secretary to the Government of India
Ph. 011-23752497
Email: anoopsingh.bisht@nic.in

@Endorsement
No. FU-21/2020-FSC (Vol-IV)

Copy to,

1. **Secretary, Ministry of Coal**
2. **Chairman, Railway Board**
3. **Chairman, Central Electricity Authority**
4. **Secretary (CERC)/All SERCs**



(Anoop Singh Bisht)
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Copy also to: NIC with the request to upload the same on MoP Portal

Annexure D 5C

सत्यमेव जयते

File No.FU-21/2020-FSC (Vol-IV)
Government of India
(भारतसरकार)
Ministry of Power
(विद्युतमंत्रालय)

Shram Shakti Bhawan, Rafi Marg,
New Delhi, the 4th March, 2024

To,

I. **Principal Secretaries/Secretary (Power/Energy) of all State Governments/ UTs**
II. **CMDs, Thermal Generating Stations (Central, State GENCOs/IPPs)**

Subject: Advisory to all GENCOs including Independent Power Producers (IPPs) for timely Import of Coal for blending purposes and maximizing production in captive coal mines — reg.

Sir/Madam,

As you are already aware, Ministry of Power issued an advisory dated 25.10.2023 regarding blending of imported coal @ 6% (by weight) till March' 2024(copy enclosed).

2. The status of Power Supply position has been reviewed by Ministry and as per the projections, the peak demand is likely to reach upto 250 GW in summer season (April-June'2024). It has been further observed that despite the increase in loading of domestic coal rakes, the supplies of domestic coal will remain constrained due to various logistical issues associated with railway network.

3. In order to meet the power demand during the crucial summer months and to ensure un-interrupted power supply across the country, adequate coal reserves in domestic coal based plants (DCBs) need to be maintained by all the Central/State GENCOs and IPPs. **The Ministry of Power has therefore decided to extend the advisory dated 25.10.2023 till June'2024.**

4. Accordingly, all the GENCOs (Central/State and IPPs) are to firm up their imported coal contracts for ensuring supplies till June'2024. Further, GENCOs must also continuously review the stock positions of their domestic coal based (DCBs) plants and opt for blending as per the requirements so that the adequate coal stocks are maintained at the thermal power plant level.

-2-

5. This issues with the approval of Hon'ble Minister of Power & New and Renewable Energy.



(Anoop Singh Bisht/ अनूप सिंह बिष्ट)
Deputy Secretary to the Government of India
उपसचिव, भारतसरकार
Ph. 011-23752497
Email: anoopsingh.bisht@nic.in

Copy to:

1. **Secretary, Ministry of Coal,**
2. **Chairman, Railway Board,**
3. **Chairman, Central Electricity Authority,**
4. **Secretary (CERC)/ All SERCs**



(Anoop Singh Bisht/ अनूपसिंहबिष्ट)
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CHAPTER – 5: GENERAL EXPLANATION ON VARIOUS EXPENSES

5.1. Interest and finance charges: An amount of Rs.2141.69 Cr had been approved by the Hon'ble Commission under this head. The actual expense incurred was Rs.2409.63 (net of capitalized interest Rs.693.15 Cr). A comparison summary of I & F charges under various heads, as approved by the Hon'ble Commission and actually incurred as per audited accounts, is given below.

Table 5.1: Interests and finance Charges (Rs Cr)

	Approval				Actual				Difference over approval			
Item	**SBU G**	**SBU T**	**SBU D**	**Total**	**SBU G**	**SBU T**	**SBU D**	**Total**	**SBU G**	**SBU T**	**SBU D**	**Total**
Interest on Term Loan	118.52	316.61	501.91	937.04	109.66	694.86	336.88	1141.39	-8.86	-378.25	-165.03	-204.35
Security Deposit Interest			156.11	156.11			265.92	265.92			-109.81	-109.81
Master Trust Bond Interest	30.78	64.13	475.19	570.10	31.88	61.17	477.03	570.08	1.10	-2.96	-1.84	0.02
WC/OD interest	6.87	23.99		30.86	5.25	11.85	161.77	178.87	-1.62	-12.14	-152.51	148.01
PF interest	12.73	26.50	196.44	235.67	9.94	18.85	164.88	193.67	-2.79	-7.65	-31.56	-42.00
Other charges					0.59	0.81	44.55	45.95			-44.55	-45.95
Carrying cost			211.91	211.91							-211.91	-211.91
Finance cost on fair valuation on receivable from KWA							706.89	706.89			706.89	706.89
GROSS TOTAL	168.90	431.23	1541.56	2141.69	157.32	787.54	2157.92	3102.77	-11.58	-292.00	616.36	961.08
Less: Interest Capitalized					37.29	550.08	105.78	693.15	-37.29	-550.08	-105.75	-693.15
Total I&FC	**168.90**	**431.23**	**1541.56**	**2141.69**	**120.03**	**237.46**	**2052.14**	**2409.62**	**-48.87**	**-193.77**	**510.58**	**(-) 267.93**

5.2. Interest on loans: The normative interest on loan provisionally approved amounted to Rs.937.04 Cr but actually incurred Rs.1141.39 Cr (inclusive of moratorium period interest on KIIFBI

loan mounting to Rs.410.87 Cr), which was higher than the sum approved by Rs.204.35 Cr. The variation from approved expenses is mainly because of the normative approach adopted to assess the extent of loan on which interest is allowable and the weighted average interest rate. While approving ARR, Hon'ble Commission considered trued up normative loan for 2020-21 and provisional GFA addition for 2021-22 and 2022-23. Thereafter approved GFA addition, depreciation, and grants for the year 2023-24 was considered to arrive at the normative loan as on 31.03.2024. Actual rate of interest (9.87%) during the year 2020-21 was considered and normative interest approved at Rs.937.04 Cr as summarized below:

Table 5.2: Normative loan as on 31.03.2024 and interest as per MYT order dated 25.06.2022 (Rs Cr)

Sl. No	Description	SBU G	SBU T	SBU D	Total
1	Opening levels of normative Loan (as on 1-4-2023)	1037.24	2737.55	4978.50	8753.29
2	Provisional Asset Addition for 2023-24	493.54	1593.33	667.68	2754.55
3	Contribution & Grants		340.23	200.00	540.23
4	Net Addition to Normative loan 2023-24 (2-3)	493.94	1253.10	467.68	2214.32
5	Repayment for 2023-24 (Depreciation)	166.31	286.45	285.00	737.76
6	Closing provisional Normative Loan 31.03.2024	1364.47	3704.19	5161.18	10229.84
7	Average normative loan	1200.85	3220.87	5069.84	9491.56
8	Rate of interest %	9.87	9.83	9.90	9.87
9	Interest charges	118.52	316.61	501.91	937.04

5.3. However, the actual outstanding loan as on 31.03.2024 was Rs.9352.96 Cr as against the assessed normative loan of Rs.10,229.84 Cr by the Hon'ble Commission in the MYT order. Actual loan was lower than the assessed levels by Rs.876.88 Cr. The actual opening loan as on 31.3.2023 was less than the approved opening normative loan by Rs.1420.29 Cr. Average interest for the year was 8.93% against 9.87% considered for ARR approval. These have contributed to the difference of actual interest over approval. Hon'ble Commission may kindly note that closing loan as per accounts does not include Rs.154.83 (Rs.123.69 Cr for 2022-23) Cr received from DRIP, which is disclosed under Note 21 of the audited accounts.

5.4. Summary of loan details are furnished below. Source-wise borrowings, repayments and interests are furnished in Appendix-1. SBU wise details are furnished in Appendix 1A, 1B and 1C respectively for SBU G, SBU T and SBU D.

Table 5.3: Summary of loans, interest, and average rate of interest (Rs. Cr)

SBU	Opening (01/04/2023)	Add: additions during the Year	Less: Repayments during the year	Closing (31/03/2024)	Average loan	Interest on loan	Average interest rate (%)
SBU G	1253.58	280.52	260.42	1273.68	1263.63	109.66	8.68
SBU T	1860.14	4769.53	2231.66	4398.01	3129.08	694.86 *	9.39*
SBU D	4219.28	752.20	1290.21	3681.27	3950.28	336.88	8.52
Total	7333.00	5802.23	3782.28	9352.96	8342.98	1141.39*	8.93*

*** Note:** It was earlier reported in true-up petition for the 2022-23 that, as terms and condition

of the loan from KIIFB was not finalized and the same was classified as 'Other Financial Liability'. Hence, neither finance cost nor the maturities of dues were considered as loan in the financial statement of KSEBL. The Government of Kerala vide its GO(Rt). No.92/2023/POWER, dated 27.5.2023 has ordered that the fund provided by KIIFB shall be considered as the loan availed by KSEBL and the ownership of the asset created shall remain with KSEBL. Accordingly, the revised Tripartite Agreement and Repayment Loan Agreement were executed, and the outstanding loan amount and interest has been duly disclosed in the accounts during 2023-24. The original borrowing from KIIFB was for Rs. 1680.81 Cr, to which the accumulated interest during the moratorium period Rs.410.87Cr was added to arrive the opening outstanding balance of the loan of SBU T. The interest cost during the moratorium period amounting to Rs.410.87 Cr is also recognized as interest expense for the year as disclosed in Note.35 of the audited accounts. Thus, the interest pertaining to the earlier years up to 31.03.2023 amounting to Rs.302.74 Cr also included in the current year interest amount of Rs.694.86 Cr. This has to be excluded to arrive the current year interest and normal average rate of interest of the SBU-T. Likewise, the opening loan amount of KIIFB was added to the opening loan amount of SBU T. Accordingly the total opening loan of SBUT is worked out as Rs.3951.82 Cr. (Rs.1860.14 Cr + Rs.1680.81 Cr+Rs.410.87). The average of the opening and closing balance of the loan is calculated as Rs.4174.91 Cr (Rs.3951.82+4398.01 Cr/2). The net current year interest in respect of SBUT on excluding prior period interest of Rs.302.74 Cr, is Rs.392.12 Cr (Rs.694.86-Rs.302.74 Cr). Hence the average rate of interest for SBUT is worked out as 9.39 % (Rs.392.12x100/4174.91). On considering the above, the interest amount in respect of the three SBUs of KSEBL during the year is worked out as Rs.838.65 (Rs.1141.39 Cr-Rs.302.74 Cr), the opening loan amount as Rs.9424.68 Cr and the average rate of interest as 8.93% (Rs.838.65 Cr x 100 / Rs.9388.82)

5.5. Basis of apportionment loans among SBUs in accounts. The basis adopted in apportioning loan among SBUs in accounts are tabulated below:

Table-5.4: Basis of apportionment of Loans and Interests (2022-23)

No	Institution	Basis
A	**SECURED LOANS-TERM LOANS**	
1	L I C	GFA+CWIP
2	REC ON VARIOUS SCHEMES	GFA+CWIP
3	REC R-APDRP PART-B	Distribution
4	R E C – RGGVY	Distribution
5	REC – Medium Term Loan	GFA+CWIP
6	PFC-Pallivasal	Generation
7	PFC R-APDRP	Distribution
8	SOUTH INDIAN BANK-Barapole	Generation
9	PFC GEL KAKKAYAM	Generation
10	REC-Distribution-23 circle scheme	Distribution
11	REC-Distribution-Meter scheme	Distribution
12	REC-Thottiyar	Generation
13	REC-Transmission-Kattakkada-Pothencode	Transmission
14	REC-Transmission-Group 1	Transmission
15	Special loan assistance REC	GFA+CWIP

No	Institution	Basis
16	DRIP	Generation
17	R E C-DDG Scheme	Distribution
18	RIDF of NABARD-Upper Kallar & Banasurasagar	Generation
19	Special loan assistance –PFC	GFA+CWIP
20	PFC GEL Perunthenaruvi	Generation
B	**UNSECURED LOANS TERM LOANS**	
	REC	GFA+CWIP

5.6. Hon'ble Commission may kindly note that since short term loans could be obtained at comparatively lower rate than long term loans, KSEBL had earlier availed short-term loans for meeting capital liabilities in the past. Considering the risk involved in financing capital project through short term loans, KSEBL started availing long and medium-term loans for this purpose by restricting STL borrowings to the possible extent. STL as on 31.03.2024 has been Rs.700 Cr. A summary of Short-Term loans is given in Appendix-2. Interest on STL for the year 2023-24 had been Rs.10.91 Cr (Rs.18.69 Cr for 2022-23). The details are given in Appendix-3. Overdrafts, Short loan & Demand loans as on 31.03.2024 amounted to Rs.2057.78 Cr (Rs.1566.80 Cr in 2022-23) with an interest to the tune of Rs.178.88 Cr (Rs 133.22 Cr for 2022-23). Thus, Long term loan and STL increased by Rs.2019.96 Cr and decreased in OD balance by Rs.209.02 Cr during the year. There was an increase in receipt from DRIP to the tune of Rs.31.14 Cr (Note No. 21 of Audited accounts). The overall borrowings were higher by Rs.1842.28 Cr in comparison with 2022-23.

5.7. During 2023-24, KSEBL executed capital works to the tune of Rs.3161.85 Cr and works capitalized to the tune of Rs.2923.48 Cr including the intangible assets. The GFA addition claim is for an amount of Rs.2868.92 Cr. Project wise details of CWIP are furnished in Appendix 4. A summary of capital works is furnished below:

Sl. No	SBU	CWIP as on 01/04/2023 (₹ Cr)	Works executed during the year (₹ Cr)	GFA Addition including intangible assets (₹ Cr)	GFA Addition (₹ Cr)	CWIP as on 31/03/2024 (₹ Cr)
1	Generation	1917.71	392.33	478.53	476.78	1831.51
2	Transmission	860.86	1305.84	630.88	625.41	1535.82
3	Distribution	865.80	1463.68	1814.07	1766.73	515.41
4	GRAND TOTAL	3644.37	3161.85	2923.48	*2868.92	3882.74

5.8. It is respectfully submitted that Hon'ble Commission has determined normative loan as on 01.04.2015 (at the beginning of the 3-year control period from 2015-16 to 2017-18). Aggrieved by the determination of normative loan balance as on 01.04.2015 (Rs.2276.22 Cr) by the Hon'ble Commission in the truing up order for 2015-16, KSEBL sought review for an additional claim of Rs.467.60 Cr, which was turned down as per order dated 16.05.2019. Considering the long-lasting financial implication, KSEBL has filed appeal petitions before the Hon'ble APTEL for each succeeding year (from 2015-16 to 2022-23). In view of the pendency of the above appeals, KSEBL retains the disputed claim in true up petition for 2023-24 also as has been done in the earlier years. Trued up normative loan for 2021-22 was Rs.7966.49 Cr. As per the trued up order for 2022-23 dated 28.06.2024 the Commission has adjusted an amount of Rs.295.09 Cr from the

normative loan amount, which was provided in fact to compensate the cost of asset lost due to natural calamities. This has resulted in understatement of normative loan and hence, KSEBL was constrained to approach Hon'ble APTEL as an appeal petition. In view of the pendency of the dispute, the claim is retained in this petition. Therefore, Normative loan at the beginning of the year 2023-24 has been considered at Rs.10,473.98 Cr, inclusive of disputed claim, as summarized below:

Table 5.6 Computation of Normative loan as on 01.04.2023

Sl. No	Description	SBU G (₹ Cr)	SBU T (₹ Cr)	SBU D (₹ Cr)	Total (₹ Cr)
1	Closing balance of Normative loan as per True up order for 2022-23 (OP 85/2023 dated 28.06.2024)	1030.84	3738.56	4941.89	9711.29
2	Add: Disputed claim pending before APTEL	135.23	131.21	201.16	467.60
3	Add: Disputed claim pending before APTEL against order dated 28.6.2024 on OP.No.85/2023			295.09	295.09
3	Normative loan as on 01.04.2023 considered in this petition	1166.07	3869.77	5438.14	10473.98

Additional capitalization during 2023-24 and normative loan thereon:

5.9. GFA addition during 2023-24 was Rs. 2868.92 Cr. The asset class wise details of fixed assets addition under the strategic business units of SBU G, SBU T and SBU D during 2022-23 as per accounts is furnished below:

Table 5.7 SBU wise -Asset class wise GFA addition for 2023-24 (Rs Cr)

Sl. No	Description	SBU G	SBU T	SBU D	Total
1	Land	30.94	6.73	0.21	37.88
2	Building	21.22	30.74	9.75	61.71
3	Hydraulic works	152.92	0.19	0.05	153.16
4	Other Civil works	82.72	29.29	25.20	137.21
5	Plant & Machinery	184.79	317.61	116.78	619.18
6	Lines, cable networks etc.	1.93	238.84	1588.99	1829.76
7	Vehicle	0.40	0.15	2.11	2.66
8	Furniture & Fixtures	0.68	0.59	1.55	2.82
9	Office equipment	1.06	1.15	21.66	23.87
10	Total	476.66	625.29	1766.30	2868.25
11	Add: Decommissioning liability 2023-24	0.12	0.12	0.43	0.67
12	GFA addition during the year-2023-24	476.78	625.41	1766.73	2868.92

5.10. SBU wise details of GFA addition considered for truing up with explanation is furnished below:

5.11. SBU GENERATION:

5.11.1. GFA worth Rs.55.78 Cr represents part capitalization within the meaning of Tariff Regulations. Asset category wise details of part capitalized assets are furnished in Appendix 4A.

5.11.2. Hon'ble Commission did not approve decommissioning liability in GFA addition, accounted by virtue of IND AS provisions, during the course of truing up for the year 2016-17. Therefore, decommissioning liability of SBU G is not considered in GFA addition during the year for truing up purpose. Further, Rs.3.13 Cr worth of part capitalized assets in earlier years has been considered as GFA addition during 2023-24, details of which are furnished in Appendix 4B.

5.11.3. After adjusting part capitalized assets and de commissioning liability from the GFA addition as per accounts, the asset addition during the year eligible as per Tariff Regulation amounted to Rs.424 Cr as detailed in Table below.

5.12. Item wise break up of GFA of SBU G are summarized below:

Table 5.8: GFA addition for 2023-24 eligible as per Tariff Regulation for SBU G (Rs. Cr)

		Gross fixed assets for 2023-24 as per Accounts			Part capitalized in 2023-24	Part capitalized in earlier years but considered in 2023-24									Decom. liability	TU Claim
No.	Particulars	Year beginning	Additions	Year End		2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23		
1	Land & land rights	1519.17	30.94	1550.11	29.74			2.264	.00001		0.812					4.276
2	Hydraulic works	1702.80	152.92	1855.72	1.883											151.04
3	Building & Civil works of Power plant	741.36	103.94	845.30	7.642								0.02			96.31
4	Plant & machinery including sub-station Equipment	13808.51	184.79	13993.3	16.491											168.29
5	Line, cable network Etc	10.84	1.93	12.77												1.93
6	Vehicles	16.26	0.40	16.66							.005					0.41
7	Furniture & fixtures	13.58	0.68	14.26	.007				.001							0.67
8	Office Equipment	15..76	1.06	16.82	0.02	0.004		0.004	0.00028	0.004	0.019		0.002			1.07

		Gross fixed assets for 2023-24 as per Accounts			Part capitalized in 2023-24	Part capitalized in earlier years but considered in 2023-24									Decom. liability	TU Claim
No.	Particulars	Year beginning	Additions	Year End		2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23		
9	Decommissioning liability		0.12	0.12											-0.12	0
	Total (1) to (9)	17828.28	476.78	18305.06	55.79	0.004	0	2.268	.000038	0.005	0.836	0	0.022	0		424.00
	Enhancement in value due to revaluation	11988.98		11988.98												
	Net of enhanced inValue	**5839.3**	**4762.78**	**6316.08**												

5.12.1. Decommissioning liability amounting to Rs.0.12 Cr included in the GFA addition for the year has not been considered in the computation of GFA addition eligible as per Tariff regulation. Hon'ble Commission may kindly note that Rs.13.80 Cr was received as capital subsidy and grants during 2023-24. Hon'ble Commission may be pleased to approve GFA addition under SBU G at Rs.410.20 Cr (Rs.424 Cr – Rs.13.80 Cr) for the year 2023-24 for the purpose of normative loan and Rs.405.92 (Excluding land value of Rs.4.28 Cr) for depreciation.

5.13. SBU Transmission:

5.13.1. As per audited accounts, GFA addition during the year has been Rs.625.41 Cr. Details are furnished in Appendix 4. Out of the GFA addition during the year, assets worth Rs.18.98 Cr represent part capitalization within the meaning of Tariff Regulations. Asset category wise break up in respect of part capitalized assets are furnished in Appendix 4C.

5.13.2. KSEBL, as per truing up petition for 2016-17 to 2022-23, has excluded part capitalized projects in respect of SBU T as envisaged in the Tariff Regulations. Out of the projects part capitalized in earlier years, projects aggregating to Rs.55.91 Cr were commissioned in 2023-24. Details are furnished as per Appendix 4 D to 4 F. Project wise GFA addition details are furnished in Appendix 4 G.

5.13.3. Thus, part capitalized assets and decommissioning liability (Rs.0.12 Cr) during the year were deducted from the GFA addition as per accounts. Thereafter, part commissioned assets in earlier years but commissioned in 2023-24 were considered to arrive at the asset addition eligible as per Tariff Regulation. This amounted to Rs.662.22 Cr as detailed in Table below.

Table 5.9 GFA addition for 2023-24 eligible as per Tariff Regulations (SBU T) (Rs Cr)									
No	**Assets Group**	**Gross fixed assets**			**Adjustments**	**Part capitalized in earlier years but commissioned during 2023-24**			**Addition as per Regulation**
		Year beginning	**Additions**	**Year End**	**Part capitalized**	**2020-21**	**2021-22**	**2022-23**	
1	Land & land rights	517.23	6.73	523.96	5.004			0.03	1.76
2	Buildings	514.13	30.74	544.87	0.962		0.101	1.471	31.35
3	Hydraulic Works	11.88	0.19	12.07					0.19
4	Other Civil works	541.43	29.29	570.72	2.005	0.009	0.374	4.634	32.30
5	Plant & Machinery	3622.59	317.61	3940.20	2.962	4.492	13.89	20.318	353.35
6	Lines, cable networks etc.	4244.12	238.84	4482.96	8.051	0.377	4.057	6.157	241.38
7	Vehicle	12.44	0.15	12.59					0.15
8	Furniture and Fixtures	10.76	0.59	11.35		0.002		0.001	0.59
9	Office Equipment	19.98	1.15	21.13				0.001	1.15
10	Decommissioning Liability		0.12	0.12					
11	Gross Asset (Total (1) to (10))	9494.81	625.41	10119.97	18.984	4.88	18.422	32.612	662.22

Decommissioning liability amounting to Rs.0.12 Cr is included in the GFA addition for the year and the same has not been considered in the computation of GFA addition eligible as per Tariff regulation.

5.13.4. Hon'ble Commission may kindly approve GFA addition of Rs.662.22 Cr for 2023-24. Consumer contribution, grants and subsidies received during the year amounted to Rs.46.10 Cr as detailed Appendix 5. GFA addition to the tune of Rs.616.12 Cr (Rs.662.22 Cr-Rs.46.10 Cr) may kindly be approved as GFA addition eligible for normative loan and Rs.614.36 Cr (excluding land value Rs.1.76 Cr) as GFA eligible for depreciation.

5.14. GFA addition for SBU D:

5.14.1. As per audited accounts, assets capitalized during the year has been to the tune of Rs.1766.73 Cr after excluding decommissioning liability of Rs 0.43 Cr GFA addition eligible as per true up worked out is Rs 1766.30 Cr as detailed below:

Table 5.10 GFA addition for 2022-23 eligible as per Tariff Regulations (SBU D) (Rs Cr)

No	**Assets Group**	**Gross fixed assets**			**Decommissioning liability**	**Addition as per Regulation**
		Year beginning	**Additions**	**Year End**		
1	Land & land rights	22.52	0.21	22.73		0.21
2	Buildings	141.10	9.75	150.85		9.75
3	Hydraulic Works	0.08	0.05	0.13		0.05

No	Assets Group	Gross fixed assets			Decommissioning liability	Addition as per Regulation
		Year beginning	Additions	Year End		
4	Other Civil works	12.24	25.20	37.44		25.20
5	Plant & Machinery	3112.45	116.78	3229.23		116.78
6	Lines, cable networks etc.	12407.84	1588.99	13996.83		1588.99
7	Vehicle	11.92	2.11	14.03		2.11
8	Furniture and Fixtures	52.31	1.55	53.86		1.55
9	Office Equipment	44.04	21.66	65.70		21.66
10	Decommissioning liability		0.43	0.43	0.43	0
11	Gross Asset (Total (1) to (14))	15804.5	1766.73	17571.23		1766.30

5.14.2. Decommissioning liability amounting to Rs. 0.43 Cr is included in the GFA addition for the year. Consumer contribution, grants and subsidies received during the year 2023-24 has been Rs.977.46 Cr, details of which are given in Appendix 5. After excluding contribution etc GFA addition eligible for normative loan amounts to Rs.788.84 Cr. (Rs.1766.30 Cr-Rs.977.46). GFA addition eligible for depreciation amounts to Rs.788.63 Cr after excluding land value of Rs.0.21 Cr.

5.15. Capitalization of expenses: It is respectfully submitted that KSEBL earmarks amounts meant for capitalization of various expenses incurred on capital works carried out during the year. Such amount earmarked for subsequent capitalization is deducted from the actual employee cost, A&G expenses, and interest on loan annually and directs the ARUs to include the same under CWIP itself. Expenses so earmarked for capitalization during subsequent years amounted to Rs.1193.74 Cr. It may kindly be noted that expenses earmarked annually would continue to be disclosed under CWIP till completion of related works and transfer to GFA. Apportionment of expenses earmarked for capitalization among SBUs in 2023-24 is furnished below:

Table5.11: Details of expenses capitalized in 2023-24 (Rs Cr)

Sl. No	Description	SBU G	SBU T	SBU D	Total
1	Interest and Finance charges	37.29	550.08 *	105.78	693.15
2	Employee cost	47.85	97.40	344.09	489.34
3	A&G expenses	4.81	0.68	5.76	11.26
4	Total	89.95	648.16	455.63	1193.74

***Note:** Higher amount of the capitalization of interest and finance charge for the transmission SBU is due to the recognition of the interest cost of the KIIFBI loan during the moratorium period amounting to Rs.410.87 Cr. Since the loan availed from KIIFBI pertains to the specific transmission project, the entire interest cost of the moratorium period amount to Rs.410.87 Cr was capitalized during the financial year 2023-24.

5.15.1. In respect of the capitalized portion of expenses, Hon'ble Commission may kindly note that the capitalized portion of expenses comprises of employee cost, interest, and A&G expenses. KSEBL is following the rules, policies and standards prescribed in Electricity (Supply) Annual Accounts Rules (ESAAR) 1985 saved as per section 185 (2) if the Electricity Act, 2003 for Capitalization of expenditure. Accounting standards for capitalization of expenditure is specified in Annexure III - Basic Accounting Policies and principles. It is respectfully submitted that KSEBL is consistently following the above prescribed accounting policies. The employee cost attributable to the creation of fixed assets is being capitalized as cost of fixed asset by following the prescribed accounting standards. The highlights are as follows:

-All employee costs in respect of the construction units are fully charged as cost of capital assets.

-Staff costs which are chargeable to capital works shall be:

1) Identified to specific capital job wherever possible.
2) Failing which identified to a specific group of capital jobs wherever possible (and within the group allocated on an ad-valorem basis).
3) Failing which identified to a project wherever possible (and allocated on an ad-valorem basis over various jobs within the project).
4) Failing which, allocated on an ad-valorem basis over various projects and various jobs within each project.

- Identification to one or more jobs should be done only, if possible, to identify without any allocation. In all other cases, ad-valorem allocation shall be adopted.

5.15.2. As the employees of the SBUs, Generation (Civil), Transmission and Distribution are also engaged in the capital expenditure nature of works, every year a portion of such employee cost capitalized in a pre-set percentage as below.

Generation (Civil)-100%
Transmission-25%
Distribution-14 %
Head office-5%

5.15.3. It is respectfully submitted that the portion of expenses capitalized at HO level are communicated to the ARUs for inclusion in accounts. Such expenses are booked under CWIP (various capital projects in progress) by the ARUs and therefore capitalized portion of expenses are carried over under the respective CWIP head of account till the asset is capitalized transferred to the GFA account. Hon'ble Commission may kindly note that the expenses so communicated for capitalization at field level will not be reflected under GFA addition during the year in its entirety as the amount is distributed among all the works-both completed during the year as well as ongoing projects.

5.15.4. Interest and finance charges are capitalized at the Head office as prescribed in the accounting policies. These amounts are also allocated to the ARUs based on actual capital expenditure incurred during the year for charging in the concerned project/assets. Thus, capitalization of interest (Interest during construction-IDC) is done based on relevant standards applicable. Capitalization of borrowing cost is computed based on the

weighted average cost of general borrowing that are outstanding during the period and used for the acquisition/construction/erection of capital asset. As per the method, interest applicable for loans attributable to capital works in progress is segregated from the total interest burden and only the interest portion applicable to already capitalized assets alone is charged to P&L account.

5.15.5. The accounting treatment followed in annual accounts towards capitalization of expenses differs from that adopted for Regulatory purposes. Regulatory approval of normative interest is granted only for the commissioned assets and no allowance is permissible for CWIP. Similarly, O&M expenses regarding GFA are allowed at normative basis on assets put to use. The norms were derived based on net trued up expenses of previous years. Therefore, in the truing up petition, normative interest as well as normative O&M expenses are considered as a charge in ARR. Expense portion earmarked towards future capitalization could be claimed only as part of GFA value from the date the relevant asset is put to use.

5.16. Contribution and Grants:

5.16.1. SBU wise break up of Capital subsidies and Grants received during the year as per accounts and considered in true up are as follows:

Table 5.12 SBU wise break up of Contribution & grants received during 2023-24 (Rs Cr)

SL No.	PARTICULARS	GEN	TRANS	DIST	TOTAL
1	Consumer contribution	0.03	17.68	341.61	359.32
2	Subsidies	8.31		0.77	9.09
2	Grants Recd. Towards Cost of Fixed Assets	5.46	28.20	598.58	632.23
3	Others	0	0.22	36.50	36.72
4	Gross Total	13.80	46.10	977.46	1037.36

5.17. Based on the above submission, GFA addition eligible as per Tariff Regulation, 2021 amounts to Rs.2856.89 Cr against Rs.2868.92 Cr as per audited accounts. GFA addition eligible for depreciation amounts to Rs.2412.40 Cr and additional normative loan Rs.2422.08 Cr. Details are furnished below:

Table 5.13: Breakup of additional capitalization as per Accounts and as per KSERC Regulations for 2023-24

Sl. No	Particulars	SBU G	SBU T	SBU D	TOTAL
1	Addition as per Accounts	476.78	625.41	1766.73	2868.92
2	Less: Part capitalization	55.79	18.98		74.77
3	Less: Decommissioning liability	0.12	0.12	0.43	0.67
4	Add: Part Capitalization in 2014-15 but commissioned during 2023-24	0.004			0.004
5	Add: Part Capitalization in 2015-16 but commissioned during 2023-24	0			0
6	Add: Part Capitalization in 2016-17 but	2.268			2.268

Sl. No	Particulars	SBU G	SBU T	SBU D	TOTAL
	commissioned during 2023-24				
7	Add: Part capitalization in 2017-18 but commissioned during 2023-24	0			0
8	Add: Part capitalization in 2018-19 but commissioned during 2023-24	0.005			0.005
9	Add: Part capitalization in 2019-20 but commissioned during 2023-24	0.836			0.836
10	Add: Part capitalization in 2020-21 but commissioned during 2023-24	0	4.88		4.88
11	Add: Part capitalization in 2021-22 but commissioned during 2023-24	0.022	18.422		18.444
12	Add: Part capitalization in 2022-23 but commissioned during 2023-24	0	32.612		32.612
	GFA addition eligible as per Regulation	424.00	662.22	1766.30	2852.52
	Less: Consumer contribution & grants	13.80	46.10	977.46	1037.36
	GFA addition eligible for normative loan as per Regulation	410.20	616.12	788.84	1815.16
	Less: Land value addition (excluding the portion under part capitalization)	4.276	1.76	0.21	6.246
	GFA addition eligible for depreciation	405.92	614.36	788.63	1808.91

5.18. Normative loan as on 31.03.2024 and normative interest for the year works out to Rs.11510.95 Cr and Rs.939.04 Cr respectively as detailed below may kindly be approved.

Table 5.14 Computation of Normative loan as on 31.03.2024 and normative interest for 2023-24 (Rs Cr)

Sl. No	Description	SBU G	SBU T	SBU D	Total
1	Closing balance of Normative loan as per True up order for 2022-23 (OP.85/2023 dated 28.06.2024)	1030.84	3738.56	4941.89	9711.29
2	Add: Disputed claim pending before APTEL	135.23	131.21	201.16	467.60
3	Add: Disputed claim pending before APTEL against order dated 28.6.2024 on OP.No.85/2023			295.09	295.09
4	Opening normative loan for 2023-24 as per KSEBL	1166.07	3869.77	5438.14	10473.98
5	Additional normative loan for 2023-24	410.20	616.12	788.84	1815.16
6	Normative repayment 2023-24	159.67	309.17	309.36	778.19
7= (4+5)-6	Closing normative loan 31.03.2024	1416.60	4176.72	5917.62	11510.95
8= (4+7)/2	Average Normative loan for the year 2023-24	1291.33	4023.24	5677.88	10980.14
9	Weighted average interest rate for 2022-23 (%)	8.09	8.72	8.52	8.54
10	Normative interest for 2023-24	104.46	350.82	483.76	939.04

5.19. Interest on Overdraft: A sum of Rs.178.88Cr (Rs. 133.22 Cr in 2022-23) was incurred during the year under this head. Expenses under this head could not be claimed as per Tariff Regulation, 2021. Hon'ble Commission has been approving carrying cost on approved and un bridged revenue gap along with working capital interest. In view of the above, KSEBL is not seeking true up of actual interest paid on overdraft during the year 2023-24.

5.20. Carrying cost on approved revenue gap:

5.20.1. Hon'ble Commission had trued up revenue gap of Rs.6,408.37 Cr till 2022-23 (excluding 2014- 15, for which orders are yet to be issued). Carrying cost for the revenue gap of Rs.6,408.37 Cr is claimed in this petition.

5.20.2. Based on the methodology adopted by the Hon'ble Commission, average GPF balance of Rs. 2926.28 Cr (Rs.2891.85 Cr for 2022-23) and security deposit over normative working capital (Rs.212.81 Cr) have been excluded from the approved gap to determine the revenue gap eligible for carrying cost in line with Regulation 29 (9) of Tariff Regulation, 2021.

5.20.3. Actual average interest on loans during 2023-24 for SBU D had been 8.52 %. Accordingly, a sum of Rs.278.54 Cr is claimed as carrying cost in 2023-24 in respect of trued up revenue gap till 31.03.2023 as detailed below:

Table 5.15: Computation of carrying cost on approved revenue gap (Rs Cr)		
1	Trued up revenue gap till 31.03.2023	6408.37
2	Less: Average PF balance for 2023-24 (3000.09+2852.48)/2	2926.28
3	Less: Security deposit as per Reg 29 (9)	212.81
4	Net revenue gap (1-2-3)	3269.28
5	Carrying cost at average interest rate of 8.52% on above	278.54

5.20.4. It is further submitted that the Hon'ble APTEL, as per judgment dated 08.04.2015 in appeal 160 of 2012 and batch has laid down the principle based on which carrying cost is to be allowed. The decision was reiterated in judgment dated 22.04.2015 in appeal 174 of 2013 as well. The same is reproduced below:

42.*We find that for carrying cost, the State Commission has considered the revenue gap to be applicable from the end of the year of the occurrence of the revenue gap up to the middle of the year in which the same is proposed to be recovered. This is not correct. The interest to be calculated for the period from the middle of the financial year in which the revenue gap had occurred up to the middle of the financial year in which the recovery has been proposed…This is because the expenditure is incurred throughout the year and its recovery is also spread out throughout the year. Admittedly, the revenue gap will be determined at the end of the financial year in which the expenditure is incurred. However, under or over recovery is the resultant of the cost and revenue spread out throughout the year. Similarly, the revenue gap of the past year will be recovered throughout the financial year in which its recovery is allowed. Therefore, interest on revenue gap as a result of true up for a financial year should be calculated from the mid of that year till the middle of the financial year in which such revenue gap is allowed to be recovered.*

43*.To explain this point let us assume that there is a revenue gap of 12 crores in the true up of FY 2010-11. If the cost and the revenue and the permitted expenditure had been properly balanced, this gap of 12 crores would have been recovered throughout the 12 months of FY 2010-11. Now, this revenue gap is allowed to be recovered in tariff during FY 2013-14. The recovery of gap of Rs. 12 crores from the distribution licensee consumers will be spread over the*

12 months period of 2013-14. Therefore, carrying cost would be calculated from the middle of FY 2010-11 to middle of FY 2013-14 i.e. 3 years.

5.20.5. Revenue gap for the year 2023-24 sought for trued up is Rs.1002.51 Cr (excluding carrying cost on revenue gap till 31.03.2023 as per Table 5.20.3). In view of the APTEL order, Hon'ble Commission may kindly approve carrying cost for the revenue gap for 2023-24 from the middle of the year at Rs.42.70 Cr (Rs.1002.51 Cr x 8.52 % x 1/2).

5.20.6. The Hon'ble Commission may kindly note that KSEBL has managed the overall finances through borrowings and deferred liabilities. Therefore, Hon'ble Commission may please approve Rs.321.24 Cr towards carrying cost for 2023-24 under SBU D as detailed below:

Table 5.15 A: Computation of carrying cost for 2023-24 (Rs. Cr)		
1	On approved revenue gap till 31.03.2023	278.54
2	On current year revenue gap Rs.1002.51 Cr	42.70
3	**Total**	**321.24**

5.21. Interest on working capital

5.21.1. Hon'ble Commission may kindly approve Rs.9.21 Cr and Rs.30.86 Cr respectively for SBU G and SBU T as detailed in chapter 2 & 3 of this petition.

5.22. Interest on security deposit:

Hon'ble Commission approved interest on Security Deposit for the year at Rs.156.11 Cr. The Actual Security Deposit balance as per accounts was Rs.3941.69 Cr at the beginning of the year (01-04-2023) and Rs.4352.01 Cr at the year end. Interest at 6.75 % on SD is provided in accounts for FY 2023-24 (Rs.265.92Cr) on accrual basis. Actual disbursement in the FY 2023-24 amounted to Rs.146.88 Cr (disbursement made against provision of Rs.137.43 Cr created @ 4.25% on SD for FY 2022-23) as per provisions in Supply Code, 2014. Hon'ble Commission may kindly approve the actual disbursement in 2023-24 amounting to Rs.146.88 Cr as per Regulation 29 (8) under SBU D.

5.23. Interest on PF:

Hon'ble Commission approved Rs.235.67 Cr towards interest on PF. But as per the audited accounts, the actual interest paid on PF was Rs.193.67 Cr, which was less than approval by Rs.42 Cr. Hon'ble Commission had considered average PF balance for 2023-24 at Rs. 3319.32 Cr but actual average PF balance was Rs.2926.28 Cr. The actual rate of interest remained at the approved rate of 7.10 % during the year. The actual interest as per the audited accounts may kindly be approved among SBUs as given below:

Table 5.16: Interest on GPF 2023-24 (Rs Cr)				
Item	**SBU G**	**SBU T**	**SBU D**	**Total**
Interest on GPF	9.94	18.85	164.88	193.67

5.24. Interest on Master Trust Bonds:

5.24.1. State Government, as per notifications dated 31.10.2013 and 28.01.2015, ordered creation of a Master Trust for meeting the unfunded liability of pension, gratuity and leave surrender as on 31.10.2013, in respect of the personnel transferred from erstwhile KSEB to KSEBL. Funding arrangements were put in place through issue of 2 series of Bonds for the total liability as on 31.10.2013 amounting to Rs.12418.72 Cr. Hon Commission recognized the funded pension liabilities as above and approved recovery of interest on KSEBL share of Bonds as per Tariff Regulations, 2021. Thus, Hon'ble Commission approved Rs.570.08 Cr towards interest on Master Trust bonds in the MYT order.

5.24.2. KSEBL, as per audited accounts, has not accounted pension and terminal liabilities under employee cost by virtue of operationalization of Master Trust. Instead, accounted 10% interest on Bonds issued to Master Trust amounting to Rs.570.08 Cr. Bifurcation of interest on bonds among SBUs are as detailed below:

Table 5.17: Interest on Master Trust Bonds 2023-24 (Rs Cr)

Item	SBU G	SBU T	SBU D	Total
Interest on Bonds	31.88	61.17	477.03	570.08

5.24.3. There is no deviation between interest approved on bonds for the year as per the MYT orders of the Hon'ble Commission and the actual interest as per audited accounts. However, there has been minor difference between approval and actual under SBU wise segregation of total interest. In the MYT order, Hon'ble Commission adopted employee cost ratio for 2017-18 as the basis for SBU wise segregation of allowable interest whereas the employee cost ratio for the year 2023-24 was considered for the purpose in the audited accounts.

5.25. Other Charges:

5.25.1. Hon'ble Commission has not approved any amount towards other interest. Actual amount booked as per accounts amounts to Rs.752.44 Cr and its ARU wise segregation is furnished below.

Table 5.18 Details of other charges under Interest and Finance Charges 2023-24 (Rs. Cr)

No	Particulars	SBU G	SBU T	SBU D	Total
1	Interest cost on power purchase			43.26	43.26
2	Generation based incentive	0.19			0.19
2	Other Bank charges	0.28	0.52	0.81	1.61
3	Decommissioning Cost	0.12	0.29	0.48	0.89
4	Finance cost of Fair valuation of the receivable from KWA arrear			706.89	706.89
	Total	**0.59**	**0.81**	**751.44**	**752.84**

(i) Interest cost on power purchase

It is humbly submitted that KSEBL has incurred Rs.43.26 Cr as interest on power purchase on account of various orders of Hon'ble CERC but accounted in separate heads as above. Details of interest are furnished below:

Table 5.19 Details of interest on power purchase expenses during the year 2023-24 (Rs Cr)			
Sl No	**Payee Name**	**Reason of Interest**	**Amount**
1	NLC	Debit Note dud 29.06.2023 as per CERC Order dated 4.5.2023 on IA No.33/2023 in petition No.223/MP/2022 accounted vide DB No.18-22 of 8/2023 (5,56,60,228+3,44,36,731+4,93,52,524+6,41,89,739+1,42,97,312)	21.79
2	NTPC Ltd	NTPC-RSTPS-1 Invoice No.603110318, dtd 3.8.2023 for the month of 7/2023	16.93
3	NTPC Ltd	NTPC-Simhadri, Invoice No.60313132849, dtd 5.9.2023 for the month of 8/2023	1.32
4	NTPC Ltd	NTPC-RSTPS 1, Invoice No.603211188, dtd 4.11.2023 for the month of 11/2023	0.68
5	Others	Adjustments	2.54
	Total		**43.26**

(ii) **Finance cost on Fair valuation of amount receivable from KWA arrear**

It is humbly submitted that KSEBL has been regularly following up the Government of Kerala for early realization of Government arrears including that of the Kerala Water Authority arrears. It is humbly submitted that an amount of Rs.2068.07 crore (PrincipalRs.1728.47 Cr + Interest Rs.339.61) is due from KWA towards electricity charge arrears including surcharge at the reduced rate of interest applicable for One Time Settlement. Government vide G.0.(Rt)No.45/2024/POWER dated 12-03-2024 have decided to take over the dues of Rs.2068 Crore. The amount of Rs.2068 Crore will be released to KSEBL in 10 equal annual installments of Rs.206.80 crore to KSEBL from the Financial Year 2024-25 onwards. Since the amount is a deferred payment over a period of ten years, the fair valuation of the arrears has to be done mandatorily as per Ind AS and the financial cost on this account to the tune of Rs.706.89 Cr is charged to the P&L Account. The matter is duly disclosed in note 8.1 of the audited accounts. Hon'ble Commission may kindly note that the fair valuation impact is not captured in the regulatory accounts. Hence KSEBL is not preferring this claim in true up. The Hon'ble Commission may kindly note that, since the arrears are to be received over the next ten years, KSEBL had to borrow money to make good the fund shortage arising due to delay in clearance of arrears. Therefore, it is humbly prayed that the Commission may kindly grant interest on such additional borrowings required, while truing up the accounts of the ensuing years.

5.25.2. Interest on decommissioning cost of Rs.0.89 Cr is the interest provided on the provision for commissioning liability build up at the discounted value to its future liability as per Ind AS -16. KSEBL is not claiming the interest on decommissioning liability in true up. Other bank charges comprise bank charges and demand drafts omission and other levies made by banks etc. of the following:

Table 5.20 Other bank charges (Rs Cr)	
Stamp duty on bill discounting facility	0.36
Other Bank Charges	1.25
TOTAL	**1.61**

5.25.3. Hon'ble Commission may kindly approve the expenses as per Table 5.22 above (as reduced by the interest on decommissioning cost of Rs.0.89 Cr) amounting to Rs.45.06 Cr as furnished below:

Table 5.21 Details of other charges under Interest and Finance Charges (Rs. Cr)					
No	**Particulars**	**SBU G**	**SBU T**	**SBU D**	**Total**
1	Interest cost of power purchase			43.26	43.26
2	Generation based incentive	0.19			0.19
3	Other Bank Charges	0.28	0.52	0.81	1.61
	Total	**0.47**	**0.52**	**44.07**	**45.06**

5.26. Based on the submission in preceding paragraphs, Hon'ble Commission may kindly approve interest and finance charges as summarized below:

Table 5.22: Interests and finance Charges (Rs Cr)								
	Approval				**True up requirement**			
Item	**SBU G**	**SBU T**	**SBU D**	**Total**	**SBU G**	**SBU T**	**SBU D**	**Total**
Interest on Term Loan	118.52	316.61	501.91	937.04	104.46	350.82	483.76	939.04
Security Deposit Interest			156.11	156.11			146.88	146.88
Master Trust Bond Interest	30.78	64.18	475.19	570.10	31.88	61.17	477.03	570.08
WC/OD interest	6.87	23.99		30.86	9.21	30.86		40.07
PF interest	12.73	26.50	196.44	235.67	9.94	18.85	164.88	193.67
Other charges					0.47	0.52	44.07	45.06
Carrying cost			211.91	211.91			321.24	321.24
GROSS TOTAL	**168.90**	**431.23**	**1541.56**	**2141.69**	**155.96**	**462.22**	**1637.86**	**2256.04**

5.27. Additional contribution to Master Trust

5.27.1. Hon'ble Commission provisionally approved Rs.400 Cr towards unfunded actuarial liability in the MYT order. Tariff Regulations provide that the annual pension contribution by KSEBL to the Master Trust based on the actuarial valuation is allowed to be recovered through tariff on an annual basis. Statutory transfer scheme as well as tripartite agreement provides for the same.

5.27.2. It is respectfully submitted that as per the Actuarial valuation as on 31.03.2024, the increase for the year 2023-24 has been Rs.1468.96 Cr. But on restating the amount, the impact on the accounts is to the tune of Rs.2258.50 Cr and the same was captured in

audited accounts for 2023-24 as follows:

(i) Liability pertaining to 2023-24 amounting to Rs.729.98 Cr has been booked under employee cost for the year and

(ii) Remaining portion, pertaining to earlier years Rs.1528.52 Cr under other comprehensive income in P&L account.

It is submitted that the restatement of the liability was done in accounts for the year 2023-24 for rectifying the observation made by C&AG on the supplementary audit of the accounts for the year 2022-23 that the total terminal liability of employees is understated due to short provisioning against the closing defined obligation as per actuarial valuation as on 31.03.2023. This was done for complying the provisions of the Ind AS 19.

5.27.3. The net liability as on 31.03.2024 amounted to Rs.30177.32 Cr. Details are disclosed under Note 51 of the audited accounts.

5.27.4. Kerala Service Rules as applicable in the Government are applicable to employees of KSEBL. The details of terminal benefits paid to retired employees in FY 2023-24 through Master Trust amounts to Rs.2375.56 Cr as detailed below:

Table 5.23: Amount paid to pensioners by Master Trust during 2023-24

Month	Rs. Cr	Month	Rs. Cr
Apr-23	169.13	Oct-23	192.02
May-23	153.91	Nov-23	206.18
Jun-23	165.08	Dec-23	209.21
Jul-23	172.75	Jan-24	255.92
Aug-23	234.05	Feb-24	236.04
Sep-23	196.05	Mar-24	185.15
		TOTAL	**2375.56**

5.27.5. It is humbly submitted that KSEBL has fully remitted the amount due on Master Trust bonds during 2023-24 along with interest for belated payment in earlier years. A sum of Rs.1588.87 Cr has been paid more than the liability on Master Trust bonds as on 31.03.2024 as detailed below and disclosed under note No 51.3of the audited accounts.

Table 5.24 Details of amount payable to Trust on Bonds issued as on 31.03.2024 (Rs. Cr)

1	Payable to Trust as per audited accounts 2022-23	-742.27
2	Principal repayment due on 20yr bond in 2023-24	407.20
3	Principal repayment due 10 yr bond in 2023-24	382.36
4	Interest on 20yr bond-2023-24	570.08
5	Interest on 10yr bond-2023-24	169.32
6	Received from Government	0.00
7	Interest on belated transfer of fund to Trust in earlier years	0.00
8	Total payable in 2023-24	786.69
9	Payment made during the year	2,375.10
10	Payable to Trust on 31.03.2024 as per audited accounts	-1588.87

5.27.6. Considering the above, it is humbly requested that the Hon'ble Commission may true up

Rs.400 Cr for 2023-24 towards additional contribution to Master Trust, since KSEBL has no other source to meet these expenses as detailed below.

Table 5.25: Interest on Master Trust Bonds (Rs Cr)				
Item	**SBU G**	**SBU T**	**SBU D**	**Total**
Additional contribution to Master Trust	21.60	44.98	333.42	400.00

5.27.7. Hon'ble Commission may kindly note that additional contribution to Master Trust was approved in the previous control period also. The additional actuarial liability for the year 2023-24 amounts to Rs.1468.96 Cr and the same has been duly accounted for in the accounts. The Hon'ble Commission may be pleased to consider the actuarial additional liability over Rs.400 Cr amounting to Rs.1068.96 Cr (Rs.1468.96 Cr – Rs. 400 Cr) with interest as and when the scheme for unfunded liability is finalized.

5.28. Repayment of Bonds issued to Master Trust

5.28.1. Hon'ble Commission, in line with Regulation 34 (iv) of Tariff Regulation, 2021, had approved Rs. 407.20 Cr for the year 2023-24 in the MYT order dated 25.06.2022. Hon'ble High Court of Kerala, as per judgment dated 07.09.2023 in WP (C) 19205/2023, set aside the enabling Tariff Regulation 2021 (Regulation 34 (iv)) owing to fault with the procedure. Hon'ble Court however observed that by following the procedure contemplated for framing regulations, even the aspects contained in Regulation 34(iv) can be included. In view of the order of the Hon'ble High Court on the above W.P.(C), Hon'ble Commission has published the draft Second Amendment notification on 1.12.2023. After conducting public hearing and considering the suggestions and objections of the stakeholders, the final notification viz. Kerala State Electricity Regulatory Commission (Terms and conditions for Determination of Tariff) (Second Amendment) Regulation, 2024 was published on 27th February 2024. The Hon'ble Commission has also issued the Truing up order for the year 2022-23 on 28.06.2024 duly allowing the additional contribution of the Master Trust by virtue of the second amendment dated 27th February 2024.

5.28.2. In view of the above Hon'ble Commission may kindly approve a claim of Rs.407.22 Cr for truing up for the year 2023-24 under bond repayment as furnished below:

Table 5.26: Repayment of Master Trust Bonds (Rs Cr)				
Item	**SBU G**	**SBU T**	**SBU D**	**Total**
Repayment of Master Trust Bonds	21.99	45.81	339.42	407.22

5.29. Depreciation:

5.29.1. Depreciation as per audited accounts amounts to Rs.1919.11 Cr has been worked out for the entire GFA without considering consumer contribution and grants. From this amount, claw back depreciation has been determined at Rs.398.31 Cr and credited to Other Income under Note 30 of Annual accounts. Since depreciation can be claimed strictly in line with Tariff Regulations, 2021 in regulatory accounts, it is humbly requested that the

income accounted under note 30 towards claw back as per accounts may not be considered in True up, as has been followed consistently.

5.29.2. Hon'ble Commission may kindly note that KSEBL has not been providing depreciation on increase in the value of Plant and Machinery notified by the government while revesting. This in turn lead to adverse audit qualification by the statutory auditors. The recurrence of the same audit qualification year after year was also seriously commented on by the CAG during the audit for the FY 2021-22. Hence to avoid the qualification, KSEBL has provided depreciation on the revalued asset as well. Accordingly, the depreciation as per account for 2023-24 is inclusive of Rs. 565.59 Cr towards enhancement in asset value at the time of re vesting. This is duly disclosed in Note No. 36 of audited accounts. Normative depreciation for 2023-24, in line with Tariff Regulation 2021, has been computed at Rs.778.19 Cr as detailed below:

Table 5.27: Normative Depreciation for 2023-24 (Rs Cr)

	Particulars	**SBU-G**	**SBU-T**	**SBU-D**	**KSEBL**
1	Approved GFA as on 31.03.2023 as per true up 2022-23	5695.73	9241.16	15961.16	30898.05
2	Less: Adjustments in 2022-23				
	Cost of damaged assets	0.00	0.00	805.39	805.39
3	Adjusted GFA as on 31.03.2023	5695.73	9241.16	15155.77	30092.66
4	Assets >13 years old (GFA as on 1-4-2010)	3401.75	3253.94	3529.34	10185.03
5	Fully depreciated Assets (assets up to 1-4-1993)	367.83	242.31	476.90	1087.04
6= (4-5)	Assets having life 13-30 yrs	3033.92	3011.63	3052.44	9097.99
7	Value of land on above*	80.61	208.34	8.33	298.29
8	Grants and contributions till 31.03.2010	0.00	0.00	2802.79	2802.79
9	Grants and contributions till 01.04.1993 - fully Depreciated	0.00	0.00	151.13	0.00
10= (8-9)	Grants and contributions on assets having life from 13 to 30 years	0.00	0.00	2651.66	2651.66
11= 6-(7+10)	Assets having life 13-30 years	2953.31	2803.29	391.45	6148.04
12= (11*1.42%)	Depreciation for Assets 13-30 years (@1.42%)	41.94	39.81	5.56	87.30
13= (3-4)	Assets < 13 years old (1-4-2010 to 31-3-	2293.98	5987.22	11626.43	19907.63
14	Value of land (from 01.04.2010 to 31.03.2023) *	113.86	294.28	13.19	421.32
15	Adjustment if any	0.00	0.00	0.00	0.00
16= (14-15)	Adjusted value of land	113.86	294.28	13.19	421.32
17	Grants and contributions (1-4-2010 to 31-3-2023)	92.61	759.61	6097.12	6949.34
18 = (13-16-17)	GFA < 13 years old as on 01.04.2023	2087.51	4933.33	5516.12	12536.97
19	Asset addition for 2023-24 as per this Submission	405.92	614.36	788.63	1808.91
20 = (18+19)	Closing balance of Assets <13 years Old	2493.43	5547.69	6304.75	14345.88

	Particulars	SBU-G	SBU-T	SBU-D	KSEBL
21= (18+20)/2	Average Value of Assets <13 Years old	2290.47	5240.51	5910.44	13441.43
22 = (21*5.14%)	Depreciation for assets <12 years (@5.14%)	117.73	269.36	303.80.	690.89
22= (12+21)	Total Depreciation for assets for 2023-24	159.67	309.17	309.36	778.19

*Details furnished in Table below.

5.29.3. Land value considered in the above computation is determined as detailed below:

Table 5.28 Computation of land value for the purpose of depreciation (Rs Cr)

Sl No	Particulars	SBU G	SBU T	SBU D	TOTAL
1	Land value 31.3.2023-Data form and audited accounts.	1519.17	502.62	22.52	2044.31
2	Revaluation 31.10.2013	1277.00	0.00	0.00	1277.00
3= (1-2)	Land value excluding revaluation	242.17	502.62	22.52	767.31
4	More than 30 years -As on 1.4.1992		0.00	0.00	47.70
5= (3-4)	For computation of normative depreciation (1 to 30 years)	194.47	502.62	22.52	719.61
6	Land value > 13 years ie 01.04.1992 to 31.03.2010 (proportionate)	80.61	208.34	9.33	298.29
7= (5-6)	Land value < 13 years	113.86	294.28	13.19	421.32

5.29.4. Consumer contribution, grants and subsidy has been considered in the above computation as detailed below:

Table 5.29 Contributions grants for the purpose of depreciation (Rs Cr)

Sl No	ACCOUNTS	SBU-G	SBU-T	SBU-D	KSEBL
1	Grant till 31.10.2013	00.0	00.0	4169.87	4169.87
2	01.10.2013 TO 31.03.2023	92.61	759.61	4881.17	5733.39
3 = (1+2)	Till 31.03.2023	92.61	759.61	9051.04	9903.26
4	Less: Grants applicable on fully depreciated assets – Receipt till 01.04.1993	00.0	0.00	151.13	151.13
5= (3-4)	Grant obtained -1 to 30 years old	92.61	759.61	8899.91	9752.13
6	Less: 13 to 30 years -1992 to 2010			2802.79	2802.79
7= (5-6)	Grant < 13 years -01.04.2010 to 31.03.2023	92.61	759.61	6097.12	6949.34

5.30. Consumer contribution, Grants and Subsidy

5.30.1. Details of Consumer Contribution and grants as on 31.03.2024 as per accounts amounted to Rs.4928.95 Cr (Note no. 23 & 28 of accounts) as detailed below:

Table-5.30: Details of consumer contribution and grants as on 31.03.2024 (Rs. Cr)			
No	**Particulars**	**Amount**	**Amount**
1	Consumer contribution	3,531.831	
2	Subsidies	175.24	
3	Grant	2586.49	
4	Local bodies etc	761.16	
5	De commissioning liability (IND AS)	11.28	
6	Adjustments	-295.09	
	Total	**6770.91**	
	Less Grant withdrawn	0.00	6770.91
6	Less: Clawed back		
7	2013-14	9.12	
8	2014-15	24.65	
9	2015-16	39.39	
10	2016-17	77.10	
11	2017-18	97.39	
12	2018-19	145.18	
13	2019-20	200.21	
14	2020-21	245.69	
15	2021-22	309.88	
16	2022-23	324.04	
17	2023-24	369.32	
	Total	**1841.97**	
	Less: Claw back reversed	0.00	1841.97
17	Contribution and grants as on 31.03.2024		4928.95

5.30.2. The Hon'ble Commission may kindly note that decommissioning liability considered under this head does not involve cash flow. Therefore, actual receipt of Consumer Contribution and grants till 31.03.2024 amounted to Rs.6759.63 Cr. SBU wise break up of consumer contribution and grants as on 31.03.2024 as per accounts are furnished in Appendix 5 hereto.

5.30.3. Hon'ble Commission may kindly note that the receipt of contribution and grants during the year amounts to Rs.1037.36 Cr.

5.31. Others Rs.22.82 Cr:

5.31.1. The details of other expenses as per the audited accounts and detailed explanation of various items are provided there under.

Table 5.31 DETAILS OF OTHER EXPENSES (Note No. 38)					
No	**Particulars**	**SBU G**	**SBU T**	**SBU D**	**Total**
1	Discount to Consumers	0.00	0.00	4.00	4.00
2	Loss on account of flood/ cyclone, compensation etc	0.07	0.56	4.68	5.31
3	Miscellaneous Losses and Write Offs	0.00	0.00	13.51	13.51
	TOTAL	**0.07**	**0.56**	**22.19**	**22.82**

5.31.2. Discount to consumers (Rs.4.00 Cr) denotes discount allowed for advance remittance. Loss on account of flood/ cyclone, compensation etc. (Rs.5.30 Cr) denote the following:

Table 5.32 Details of losses on account of flood, compensation etc.

Particulars	Rs Cr
Compensation For Injuries, Death & Damage-Staff	0.72
Compensation For Injuries, Death & Damage-Outsider	4.59
TOTAL	**5.31**

5.31.3. Miscellaneous losses and write off consist of the following:

Table 5.33 Break-up of Miscellaneous losses and write offs (Rs Cr)

Particulars	Rs Cr
Withdrawal of Prior Period Credit to Revenue	13.42
Amount round of in various ARUs	0.09
Total	**13.51**

Withdrawal of prior period credit to revenue represents write off revenue recognized in earlier years. A sum of Rs.13.51 Cr was written off during the year as detailed in Appendix 11.

5.31.4. In view of the above submission, Hon'ble Commission may kindly true up Rs 22.82 Cr as tabulated below:

Table 5.34 DETAILS OF OTHER EXPENSES sought for true up 2023-24					
No	**Particulars**	**SBU G**	**SBU T**	**SBU D**	**Total**
1	Discount to Consumers			4.00	4.00
3	Loss on account of flood/ cyclone, Compensation etc.	0.07	0.56	4.68	5.31
4	Miscellaneous Losses and Write Offs			13.51	13.51
	TOTAL	**0.07**	**0.56**	**22.19**	**22.82**

5.32. Exceptional items Rs (-) 752.52 Cr: Expenses incurred for the restoration of assets damaged due to natural calamities are accounted under this head. This head also includes compensation of loss taken over by the Government of Kerala, the details of which are furnished below. Item wise break up is furnished below:

Table 5.35 Details of exceptional items (Rs Cr)	
O&M work related to Natural Calamity-Plant& Machinery	0.21
Payment to contractors-O&M work related to natural calamity-Building	0.01
Payment to contractors- O&M work related to natural calamity-civil works	0.02
Payment to contractor-natural calamity-Hydraulic works	0.01
Payment to contractors for the O&M work related to natural Calamity	14.95
Compensation of Loss taken over by Government of Kerala	(-)767.72
TOTAL	**(-)752.52**

Compensation of Loss taken over by Government of Kerala

5.32.1. The Fifteenth Finance Commission has recommended performance - based additional borrowing space of 0.50 per cent of Gross State Domestic Product to States linked to performance in the Power Sector. The objectives of the additional borrowing space are to improve the operational and economic efficiency of the sector and to promote a sustained increase in paid electricity consumption. This special dispensation has been for each year for a four-year period from 2021-22 to 2024-25. For each financial year from 2021-22, this borrowing space will be in addition to the regular borrowing limit. Government of Kerala has availed additional borrowing for power sector reforms for 2021-22 and 2022-23 by agreeing to take over the future losses of all public sector DISCOMs in the State in graded manner as given below and submitted an undertaking to this effect.

2022-23	2023-24	2024-25	2025-26 & onwards
60% of the loss for the year 2021-22	75% of the loss for the year 2022-23	90% of the loss for the year 2023-24	100% of the loss for the year 2024-25 & onwards

The Government of Kerala had taken over 75% of the loss sustained by KSEBL in 2022-23 amounting to Rs 767.72 Cr to fulfill the entry level condition for additional borrowing space of 0.50% on GSDP linked to performance in the power sector. As per the audited accounts of KSEBL, a total loss of Rs.1023.62 Cr is reported for 2022-23 and 75% of the same comes to Rs.767.715 Cr. In order to fulfill the entry level conditions for availing 0.5% of GSDP as additional borrowing for the financial year 2023-24, Government have decided to take over the loss of Rs.767.715 Cr. ie. 75 % of the total loss of Rs.1023.62 Cr incurred by the KSEBL for the financial year 2022-23 vide G.O. (110 No. 8908/2023/Fin. dated 16/12/2023. The amount was released to KSEBL and the same was accounted during the financial year 2023-24 under the head Exceptional Items. It is humbly submitted that the Hon'ble Commission while truing up the accounts for the year 2022-23 has considered the amount of Rs.767.72 Cr while determining the trued-up revenue gap. Hence It is humbly prayed that the amount in the accounts may not be considered while truing up for the year 2023-24 as the same has

already been considered in 2022-23.

5.32.2. It may kindly be seen that KSEBL has been able to contain R&M expenses at optimum levels despite the ageing assets that are being maintained by KSEBL. Hon'ble Commission may kindly approve Rs.15.20 Cr over and above the normative claim. SBU wise break up of other expenses are furnished below, which may kindly be approved.

Table 5.36: Exceptional items (Rs Cr)				
Item	**SBU G**	**SBU T**	**SBU D**	**Total**
Exceptional items-R&M	0.02	0.18	15.00	15.20

5.33. O&M Expenses: Hon'ble Commission approved Rs.4456.29 Cr under this head on normative basis against which a sum of Rs.4925.69 Cr was actually incurred during the year 2023-24, exceeding approval by Rs.469.40 Cr. as summarized below:

Table 5.37: Details of O&M expenses 2023-24 (Rs Cr)									
		Approval				**KSEBL Accounts**			
No	**Particulars**	**SBU G**	**SBU T**	**SBU D**	**Total**	**SBU G**	**SBU T**	**SBU D**	**Total**
1	Employee cost					267.97	519.84	3638.25	4426.06
2	Less: Capitalized					47.85	97.40	344.09	489.34
3	Balance					220.11	422.44	3294.16	3936.71
4	R&M expenses					22.89	32.10	215.03	255.02
5	A&G expenses					53.13	135.09	541.98	730.20
6	Less: Capitalized					4.81	0.68	5.76	11.25
7	Balance					48.33	134.41	536.22	718.94
8	Total O&M Expenses	206.08	644.81	3605.40	4456.29	291.35	588.95	4045.39	4925.69

5.33.1. Employee cost: It is respectfully submitted that gross employee cost during the year amounts to Rs. 4426.06 Cr. Item-wise tabulation is furnished in Appendix 6.

Table 5.38: Gross Employee Cost for the year 2023-24 (Rs. Cr)				
Particulars	**SBU G**	**SBU T**	**SBU D**	**Total**
Gross employee cost	267.97	519.84	3638.25	4426.06

5.33.2. R&M expenses: SBU wise split up of R&M expenses as per accounts is furnished below:

Table 5.39: R&M expenses for 2023-24 (Rs Cr)				
Item	**SBU G**	**SBU T**	**SBU D**	**Total**
R&M expenses	22.89	32.10	215.03	255.02

5.33.3. A&G expenses: Hon'ble Commission may kindly note that net A&G expenses as per accounts amounts to Rs.718.94 Cr as detailed in Note 37 of the Annual Statement of Accounts. Expenses under this head contain predominantly, the following expenses:

Table 5.40: Administrative & General Expenses 2023-24 (Rs Cr)		
1	Conveyance & travel	75.38
2	Other operative expenses	295.51
3	Expenditure on Filament free Kerala	0.42
4	Electricity Duty u/s 3(I), KED Act	159.19
5	Power factor incentive to consumers	56.49
6	Freight & other expenses	43.21
7	Rent, rates & taxes	16.91
8	Printing, Stationery, Telephone, postage, internet charges etc.	33.58
9	Notary fee and other expenses relating to KSERC & CGRF	7.98
10	Expenditure NILAAVU Scheme	8.72
11	All Other A&G expenses	32.81
12	**Gross A&G Expenses**	**730.20**
13	Less: Expenses Capitalized	11.26
14	**Net A&G Expenses**	**718.94**

The Hon'ble Commission may kindly note that a sum of Rs.56.49 Cr has been extended to consumers towards Power factor incentive. In this regard, the following submission is made for the kind consideration of the Hon'ble Commission. KSEBL respectfully submits that the Hon'ble Commission may permit exclusion of power factor incentive from A&G expenses and the same deducted from revenue from sale of power in this petition.

5.33.4. It may kindly be seen that an amount of Rs.0.42 Cr has been incurred for the filament free Kerala campaign in which LED lamps were supplied to consumers. An income of Rs.0.56 Cr has been booked under Income from bulb, scrap, tender form etc. (Note 29 B of audited accounts). Break up details are furnished in Appendix 10. It is humbly requested that the Hon'ble Commission may consider the cost of bulbs booked under A&G expenses as a reduction from sale proceeds of LED Bulbs while truing up.

5.33.5. One of the major expense items booked under A&G expense is the section 3(1) duty payable by KSEB to the Government. Section 3(1) duty is a statutory levy. Hon'ble Commission has not been admitting section 3(1) duty as a revenue expenditure quoting the provisions in the "Kerala Electricity Duty Act- 1963" that "(3) The duty under this section on the sales of energy should be borne by the Licensee and shall not be passed on to the consumers". Accordingly, since the year 2003-04, Hon'ble Commission has not admitted duty as pass through in tariff. It is humbly submitted that KSEB had already taken up the matter with the Government to enable Section 3(1) duty as pass through in tariff and also filed a second appeal before the Hon'ble Supreme Court of India in the matter, which is pending disposal as of date.

5.33.6. KSEBL has incurred a sum of Rs.8.72 Cr during 2023-24 (Rs.5.67 Cr in 2021-23 and 2022-23) towards Nilaavu scheme on account of expenditure incurred by KSEBL in connection with dismantling old streetlights and installing led streetlights. Hon'ble Commission, as per order dated 03.12.2020 in OP No. 38/2020, has directed KSEBL to submit all the details including the direct and indirect cost and revenue earned as per Government order dt 10.08.2020 for taking the role of SPV for the implementation of the scheme in

the truing up petition for the years concerned. KSEBL earned revenue of Rs.1.92 Cr as revenue during the year 2023-24 (Rs.13.08 Cr in 2022-23) and accounted under Miscellaneous receipts under Note 29 B.

5.33.7. It was prayed before the Hon'ble Commission in the TU petition for 2022-23 that KSEBL may be permitted to furnish details of expenses (including the amount applicable for 2021-22 and 2022-23) and revenue pertaining to the scheme in the truing up petition for the year 2023-24.

5.33.8. It is humbly submitted that the expenses booked amounting to Rs. 14.39 Cr during 2021-22, 2022-23 and 2023-24 were spent in the respective years. Since the project is still going on and is expected to be completed by end of 2024-25, it is humbly submitted before the Hon'ble Commission that the expenses and income on Nilaavu scheme may be trued up in 2024-25.

5.33.9. Based on the Tariff Regulation, KSEBL is entitled to claim normative O&M expenses of Rs.4598.22 Cr for 2023-24 in line with Tariff Regulation, 2021, as explained in the respective chapters against Rs.4456.29 Cr approved by the Hon'ble Commission, as summarized below.

Table 5.42: Normative O&M expenses for 2023-24 (Rs. Cr)

Sl. No	Particulars	SBU G	SBU T	SBU D	Total
1	True up claim as per Tariff Regulation, 2021	189.46	625.20	3783.56	4598.22
2	Approved as per Order dated 25.06.2022	206.08	644.81	3605.40	4456.29
3	Variance over approval	(-)16.62	(-) 19.91	178.16	141.93

5.33.10. Hon'ble Commission may kindly true up O&M expenses for the year at Rs.4593.65 Cr as detailed below:

Table 5.43: Normative O&M expenses for 2023-24 (Rs. Cr)

Sl. No	Particulars	SBU G	SBU T	SBU D	Total
1	As per norms	189.46	625.20	3783.56	4598.22

5.34. Revision of O & M norms:

As per Tariff Regulation, 2021 Hon'ble Commission determined the norms for the 5-year control period from 2022-23 to 2026-27 based on the O&M cost approved in the ARR for 2018- 19 (as the trued-up figures were not available). In addition, a sum of Rs. 248.47 Cr was also considered for pay revision due from July/ August 2018. It was stated in Annexure 7 that "During February 2021, KSEB Ltd has implemented the Pay Revision with effect from July/August 2018 for its Officers and staff. As per Hon'ble APTEL decision in OP 1/2013 and 19/2013 in the Order dated 10.11.2014, the effect of pay revision have to be allowed as part of employee cost. The final norms of KSEB Ltd can be arrived at only after the truing up of the figures from the financial year 2018-19 is over and after the sub judice matter is resolved. The norms will be reworked

accordingly. KSEB Ltd may file the petition for the approval of ARR & ERC for the Control period based on the provisional figures as mentioned in Tables 2 to 7 above."

5.34.1. In compliance, KSEBL filed the MYT petition for the approval of ARR & ERC for the Control period based on the provisional figures as mentioned in Tables 2 to 7 of Annexure 7 and the Hon'ble Commission has decided the same as per order dated 25.06.2022.

5.34.2. It may kindly be seen that the final norms of KSEBL can be arrived at only after the truing up for the financial year 2018-19 is over and after the redetermination of allowable employee strength and implementation of pay revision. Hon'ble Commission has issued truing up order for 2018-19 (base year considered for the present control period) on 14.06.2022 and determined the allowable staff strength at 30321 as per order dated 14.07.2022. Pay revision since July / August 2018 has also been implemented and the financial commitment for 2018- 19 has been trued up as per order dated 03.10.2023 in the matter of truing up for 2021-22.

5.34.3. Therefore, it was submitted in True up petition for 2022-23 that the provisional norms contained in Tariff Regulation, 2021 can be finalized, which would enable KSEBL to claim its eligible O&M expenses in the truing up petition for 2022-23 onwards. To facilitate the process, KSEBL had computed and submitted the norms strictly in line with the methodology adopted by the Hon'ble Commission and directions contained in the Tariff Regulation, 2021 for the kind consideration of the Hon'ble Commission. Hon'ble Commission vide true up order dated 28.6.2024 did not consider the submission of KSEBL. It has been viewed that the Commission is yet to take a decision to revise norms for O&M expense by considering Tariff Regulation 2021 and may amend the same after completing the procedure. It is fervently appealed before the Commission to initiate the process to amend Tariff regulation without further delay,

5.34.4. In view of the above, the revised norms, its impact on the O & M cost etc., are furnished below for kind consideration of the Hon'ble Commission. The normative values as per Tariff Regulation, 2021 and revised values are enclosed as Appendix 11 and 12. Details of inflation are furnished in Appendix 13.

5.34.5. The computation methodology is briefly described below:

(i) The values trued-up for O&M expenses for 2018-19 has been Rs. 2721.49 Cr, which does not include pay revision expenses as well as employee cost for additional employee strength approved. is furnished below:

Table 5.44 Approved O&M expenses for 2018-19 as per TU order dated 14.06.2022						
Table 2.20, 2.23 & 2.24 of the order						
O&M expenses eligible for SBU-G as per the Tariff Regulations, 2018				**SBU T**	**SBU D**	**Total**
Particulars	**Existing stations**	**New stations**	**Amount (Rs. Cr)**			
Employee expense	95.34	0.575	95.91	264.93	1876.16	2237
R&M expenses	23.09	0.139	23.23	79.56	251.54	354.33
A&G expenses	5.34	0.032	5.38	18.73	106.05	130.16
Total	**123.77**	**0.746**	**124.52**	**363.23**	**2233.75**	**2721.49**

(ii) **Cost of additional staff strength:** - Based on the methodology adopted by Hon'ble Commission to disallow employee cost over 2009 level during 2009-10 to 2017-18, the additional employee cost (for 3146 nos) eligible for inclusion in base value has been computed at Rs. 137.49 Cr.

(iii) **Impact of Pay revision:** - Hon'ble Commission, as per order dated 03.10.2023, approved Rs. 181.18 Cr for 27175 employees. Thus, the allowable pay revision expenses for 2018- 19 computed as follows:

(a) Proportionate expenses for 30321 nos since it has been ordered to capture employee strength of 30321 nos from 2022-23 onwards. This amounts to Rs. 202.15 Cr (Rs. 181.18 Cr *30321 / 27175).

(b) Based on the methodology adopted by the Hon'ble Commission in the Tariff Regulation, the annualized pay revision commitment for 2018-19 amounted to Rs. 285.40 Cr (Rs. 202.15/8.5*12) against Rs. 248.47 Cr provisionally considered in the Tariff Regulation, 2021.

(iv) Thus, it may kindly be seen that the base level O&M cost to be considered for norm determination for the control period from 2022-23 onwards amounts to Rs. 3144.38 Cr against Rs. 2961.93 Cr considered in the Tariff Regulation, 2021. The increase of Rs.182.45 Cr, works out to 6.16 % of the provisionally considered base value of Rs. 2961.93 Cr. Item wise details are furnished below:

Table 5.45: Comparison of base value for norm determination (Rs. Cr)

O&M break up	Base value considered in TR, 2021(Rs.in Cr)				Revised base value (Rs Cr)				Difference (Rs Cr)			
	SBU-G	SBU-T	SBU-D	Total	SBU-G	SBU-T	SBU-D	Total	SBU-G	SBU-T	SBU-D	Total
Employee cost-2018-19	95.63	264.93	1873.91	2234.47	95.91	264.93	1876.16	2237	0.28	0	2.25	2.53
R&M expenses-2018-19	23.16	79.56	246.25	348.97	23.23	79.56	251.54	354.33	0.07	0	5.29	5.36
A&G expenses-2018-19	5.36	18.73	105.92	130.02	5.38	18.73	106.05	130.16	0.02	0	0.13	0.14

O&M break up	Base value considered in TR, 2021(Rs.in Cr)				Revised base value (Rs Cr)				Difference (Rs Cr)			
	SBU-G	SBU-T	SBU-D	Total	SBU-G	SBU-T	SBU-D	Total	SBU-G	SBU-T	SBU-D	Total
O&M expenses-2018-19	124.15	363.23	2226.08	2713.46	124.52	363.22	2233.75	2721.49	0.37	-0.01	7.67	8.03
PR expenses	11.36	33.15	203.96	248.47	12.24	33.8	239.37	285.40	0.88	0.65	35.41	36.93
Additional employee cost for 2018-19					5.89	16.27	115.31	137.49	5.89	16.27	115.31	137.49
Total	135.51	396.38	2430.04	2961.93	142.65	413.29	2588.43	3144.38	7.14	16.91	158.39	182.45
Proportion					**4.29**	**11.84**	**83.87**	**100**				

(v) The revised O&M norms for 2022-23 to 2026-27 are worked out based on the revised base values. For arriving at the O& M norms for 2022-23 to 2026-27, the base year 2018- 19 value is escalated at the rate of 4.84% till 2021-22 and 4.454% thereafter as has been adopted in the Regulation.

5.34.6. The O&M expenses for 2023-24 based on the normative values thus determined amounts to Rs.3948.30 Cr as summarized below. It is humbly submitted that the actual escalation rate based on inflation in 2023-24 (3.41%) has been considered on the norms determined for 2022-23. Hon'ble Commission may kindly note that the increase in O&M cost on account of norm revision computed by KSEBL amounts to Rs.254.17 Cr, which works out to 6.44 % on the normative O&M cost as per prescribed norms in Tariff Regulation, 2021.

Table 5.45 A Comparison of O&M expenses (Rs Cr)					
Sl. No	**Particulars**	**SBU G**	**SBU T**	**SBU D**	**Total**
1	O&M expenses as per norms contained in Tariff Regulation, 2021 (excluding R&M for SBU (D)	170.38	625.64	3152.28	3948.30
2	O&M expenses based on revised norms (excl R&M for SBU D)	175.66	652.13	3374.22	4202.01
3= (2-1)	Difference	5.28	26.49	221.94	253.71
4= (3/1) X 100	Variation in %	3.09	4.23	7.04	6.42

5.35. Pay revision arrears

5.35.1. The Hon'ble Commission, while deliberating the provision created for pay revision during 2018-19 to 2020-21, had observed that the expenses towards pay revision can be approved only on payment basis. It was also ordered to claim the expenses on payment basis along with supporting documents. In view of the direction KSEBL had preferred claim in the true up petition for the arrears disbursed during 2021-22 to the extent of 2 instalments. The background of pay revision was also appraised in detail before the Hon'ble Commission. The pay revision arrears of work men and officers from 01.08.2018 to 31.12.2020 and 01.07.2018 to 31.12.2020 respectively were ordered to be credited to the GPF account of the respective workmen / officer in 4 instalments on 01.06.2021,

01.01.2022, 01.06.2022 and 01.01.2023. Two instalments were disbursed in 2021-22 and the remaining 2 instalments in 2022-23. During the year 2023-24, an amount of Rs.7.93 Cr was disbursed as pay revision arrears hence this amount is proposed to be trued up during this year under SBU-D. Hon'ble Commission, while issuing orders on true up for the year 2022-23 has ordered to furnish government approval for pay revision implemented in KSEBL. KSEBL take up the matter with the Government of Kerala for obtaining orders on the above. KSEBL is yet to receive the same and since pay revision arrears is a genuine claim of KSEBL. Hon'ble Commission may kindly approve an amount of Rs.7.93 Cr.

5.36. Return on Equity: Return on equity is claimed @14% on the Government equity in KSEBL amounting (Rs.3499.05 Cr) to Rs.489.86 Cr. The function wise break up of equity is made based on equity allocation among SBUs in the Trifurcated Balance sheet as detailed below:

Table5.48: Allocation of Equity and RoE (Rs. Cr)				
Particulars	**SBU G**	**SBU T**	**SBU D**	**Total**
Equity Capital	831.27	857.05	1810.73	3499.05
RoE @14% on above	116.38	119.99	253.50	489.87

5.37. Amortization of intangible assets (Software development cost):

KSEBL have developed in-house software on major functional areas viz Human Resource Information System, Accounting Software, billing etc. The employees cost attributable to the software developments were capitalized as intangible asset and was amortized during 2023-24 as detailed below:

Table 5.48(A): Amortization of Intangible Assets 2023-24 (Rs Cr)				
Item	**SBU G**	**SBU T**	**SBU D**	**Total**
Amortization of Intangible Assets – Software Development Cost	1.20	1.20	9.64	12.04

The amount is duly disclosed under note 36 of the audited accounts. Hon'ble Commission may kindly approve the same.

5.38. Amortization of intangible assets (Edamon-Kochi line compensation and Pugalur-Trichur line compensation):

An amount of Rs.9.30 Cr is claimed towards the line compensation charges in respect of Edamon-Kochi line compensation and Pugalur–Trichur line compensation as detailed in para 3.22 to 3.34 of Chapter 4 in respect of SBU-T.

Table 5.48(B): Amortization of Intangible Assets 2023-24 (Rs Cr)		
Sl No	**Transmission line**	**Rs. Cr**
1	Edamon-Kochi	8.06
2	Pugalur-Trichur	1.24
	Total	9.30

5.39. Gains on achievement of T&D loss reduction target: As per Tariff Regulations, 2021, KSEBL

is entitled to retain part of the gains on account of achieving loss reduction in excess of approved target. Share of KSEBL amounts to Rs.131.59 Cr as detailed in Chapter 4. Hon'ble Commission may kindly approve the same.

5.40. Incentive on transmission availability: KSEBL is eligible to get the incentive on achieving target transmission availability as per the Tariff Regulation, 2021. Hence an amount of Rs10.49 Cr is claimed in true up as detailed in Chapter 3 .This may please be approved.

5.41. Non-Tariff Income and Other income: Hon'ble Commission has approved an amount of Rs.801.27 Cr under Non-Tariff income & Other income for the year 2023-24. Actual non-tariff income during the year as per accounts amounted to Rs.1984.04 Cr. It may kindly note that the disclosure in accounts and truing up petition regarding non-tariff income differ as the revenue from operations are depicted in accounts against revenue from tariff in truing up petition. Therefore, certain components of income disclosed under revenue from operations (Note 29 A) in accounts amounting to Rs.246.80 Cr are considered under non-tariff income in this petition. Out of Rs.1984.04 Cr, the remaining Rs.376.51 Cr is disclosed under Other operating income (Note 29 B) and Rs.1360.73 Cr under Other income (Note 30) of audited accounts as detailed below.

Table 5.49: Non-tariff income for 2023-24 (Rs. Cr)

Sl. No	Particulars	SBU G	SBU T	SBU D	Total
A	Revenue booked under Note No. 29 A				
1	Miscellaneous receipts / charges from interstate trading of energy			17.76	17.76
2	Meter rent / service line rental			108.38	108.38
3	Recovery of theft / malpractices			2.01	2.01
4	Wheeling charges recoveries			2.57	2.57
5	Miscellaneous charges from consumers			116.08	116.08
	Revenue booked under Note No. 29 (Revenue from Operations) considered as Non-Tariff Income (A)			246.80	246.80
B	Revenue booked under Note No 29 B				
6	Rebate received			114.68	114.68
7	Interest-Advance to suppliers / contractors	1.80	3.35	63.84	68.98
8	Income from sale of bulb, scrap, tender forms etc	2.63	20.72	38.78	62.14
9	Miscellaneous receipts	6.69	17.39	106.62	130.71
	Sub Total (B)	**11.12**	**41.46**	**323.92**	**376.51**
C	**Other income as per Note 30 of Annual accounts**				
1	Interest on belated payment charges				612.78
2	Interest-Staff loans and advances				0
3	Interest from loans & others			0.08	0.08
4	Interest from banks	3.64	4.22	7.59	15.45
5	Interest on tax deducted at source			1.07	1.07
6	Claw back of grants	4.12	52.12	342.07	398.31
7	Reversal of provision for doubtful debts			241.54	241.54

8	Other receipts	25.45	30.28	35.77	91.50
	Sub Total (C)	33.21	86.62	1240.90	1360.73
D= (A+B+C)	**Income as per Accounts considered as non-tariff income**	**44.33**	**128.08**	**1811.62**	**1984.04**

5.42. Miscellaneous charges from consumers (Note 29 A- Rs.116.08 Cr) are detailed in Appendix 7. Miscellaneous receipts / charges from inter-state trading of energy (Note No 29. A- Rs.1 7.76 Cr) are detailed in Appendix 8. Miscellaneous receipts (Note 29 B- Rs.130.71 Cr) are detailed in Appendix 9. Income from sale of bulb, scrap/tender form (Note 29 B) amounting to Rs.62.14 Cr as detailed in Appendix 10.

5.43. KSEBL had submitted in truing up petition for 2018-19 that KWA dues till 30.09.2018 were taken over by the Government of Kerala, (to be settled with interest in 4 annual installments commencing from 2019-20 onwards) and interest income on this count may be recognized on actual receipt basis. It was prayed that the revenue recognized in books during 2018-19 may not be trued up. It is respectfully submitted that GoK has released 2 installments due for 2019-20 and 2020-21 during the year 2020-21 and interest earned on collection has duly been offered under non-tariff income for true up in 2020-21. Hon'ble Commission was pleased to true up the same as prayed by KSEBL.

5.44. During the year 2022-23, Rs. 100 Cr was received as part of fourth instalment of KWA arrear taken over by the Government and interest pertains to the receipt amounts to Rs.19.88 Cr, which was trued up in 2022-23.

5.45. During the year 2023-24, an amount of Rs.171.67 Cr was received as part of the fourth installment of KWA arrears taken over by the Government and interest pertains to the receipts amount to Rs.34.12 Cr, which may kindly be trued up.

5.46. Government of Kerala had taken over the arrears of Kerala Water Authority to the tune of Rs.2068.07 Cr vide GO. (Rt) No.45/2024/POWER, dated 12.03.2024 and ordered to release the amount in 10 equal annual installments of Rs.206.80 Cr to KSEBL from the financial year 2024-25 onwards. KSEBL has booked an amount of Rs.339.61 Cr being the interest on KWA arrears on accrual basis and is included under 'Interest on belated payment of charges' head under Non-tariff Income. It is prayed that the revenue recognized in books during 2023-24 may not be trued up and may be considered in future on actual receipt basis.

5.47. The Government of Kerala vide order GO(Ms)No.30/2024/FIN, dated 12.03.2024 was pleased to net off the dues to KSEBL to the tune of Rs.534.84 Cr as on 31.10.2023 in respect of State Government Departments. Local bodies, State PSUs and Public institutions (except dues of Kerala Water Authority) against the electricity duty. Accordingly, the netting off was effected and duly accounted for in the year 2023-24. An amount of Rs.118.36 Cr was credited under the head Interest on belated payment of charges from consumers. This amount is proposed for truing up during the year 2023-24 as the amount was adjusted against the electricity duty.

5.48. Income from sale of scrap/tender form amounting to Rs.62.14 Cr includes Rs.0.56 Cr towards sale proceeds of LED bulbs. Further, the cost of LED bulbs amounting to Rs.0.42 Cr has been accounted under A&G expenses. Hon'ble Commission may kindly note that while approving

Non-Tariff Income for earlier years, Hon Commission was pleased to exclude the LED purchase value from NTI while truing up since the cost of bulb is not trued up under A&G expenses.

5.49. Details of income to be trued up on account of LED transaction are furnished below:

Table- 5.50: Details of income to be trued up in 2023-24 (Rs Cr)		
Particulars		
Sale proceeds of LED bulbs (Note 29 B of Annual Accounts)	0.56	
Cost of LED bulbs (Note 37 of Annual Accounts)	0.42	0.14

5.50. Out of Rebate received is Rs.114.68 Cr, mostly being rebate allowed by CGS. As already submitted in para 5.33, KSEBL earned revenue of Rs.1.92 Cr as revenue from Nilavu scheme during the year 2023-24 and accounted under Miscellaneous receipts under Note 29 B. Since the expenses connected with the project are not fully captured in books, Hon'ble Commission may true up the same later.

5.51. In view of the above submission, Hon'ble Commission may kindly approve Rs.1341.85 Cr, against Rs.801.27 Cr approved as per order dated 25.06.2022 as detailed below:

Table 5.51: Non-tariff income for 2023-24 (Rs. Cr)					
Sl. No	**Particulars**	**SBU G**	**SBU T**	**SBU D**	**Total**
1	Income as per Accounts considered as non-tariff income as per Table 5.49	44.33	128.08	1811.62	1984.04
2	Less: Items sought to be excluded in TU				
i	Claw back of grant-Considered separately	4.12	52.12	342.07	398.31
ii	Cost of LED Bulbs booked under A&G expenses			0.42	0.42
iii	Income under Nilaavu Scheme			1.92	1.92
Iv	Reversal of Provision for doubtful debts			241.54	241.54
v	Unrealized interest in respect of KWA arrear taken over by the Government of Kerala			339.61	339.61
Vi	Total exclusions (I to iv)	4.12	52.12	925.56	981.80
3	Add: Items sought to be considered in True up				
	Interest on KWA arrears released by GOK			34.12	34.12
4	**NT Income sought for TU 2023-24 (1-2+3)**	**40.21**	**75.96**	**920.18**	**1036.36**

5.52. Revenue from Tariff: Annual Statement of Accounts discloses Revenue from operation, which is inclusive of certain items that are considered under non-tariff income in the Regulatory accounts. Reconciliation of total income as per audited accounts and income considered in Truing up petition is furnished below:

Table 5.52: Revenue from tariff for 2023-24 (Rs. Cr)

Sl No	Particulars	SBU G	SBU T	SBU D	Total
1	Revenue from Operation as per Accounts (Note No. 29)				20441.75
2	Less: Items considered under non-tariff income (Table 5.49 Subtotal A+B)	11.12	41.46	570.72	623.31
3= (1-2)	Revenue from Tariff and external sale as per Accounts				19818.44
4	Less: Power factor incentive accounted under A&G Expenses			56.49	56.49
5= (3-4)	Revenue from tariff & external sale for truing up purpose				19761.95
	Break up				
8	Revenue from sale within the State including fuel surcharge			19757.87	19757.87
9	Less: Power factor incentive			56.49	56.49
10= (8-9)	Revenue domestic sale net of PFI			19701.38	19701.38
11	Add: Revenue from external sale			60.57	60.57
12= (10+11)	**Total**			**19761.95**	**19761.95**

5.53. Revenue from sale within the State including fuel surcharge amounts to Rs.19,757.87 Cr and external sales Rs.60.57 Cr as detailed in Chapter 4.

5.54. Amortization of past approved revenue gap:

Hon'ble Commission had ordered to amortize past revenue gap to the tune of Rs.850 Cr during the year. It is humbly submitted that in the absence of regulatory surplus during the year, KSEBL is not able to amortize any portion of past gap during the year as ordered by the Hon'ble Commission.

5.55. Variation between audited accounts and truing up petition: Net profit as per audited accounts for 2023-24 has been Rs.218.51 Cr (excluding RoE). As per the audited accounts KSEBL has incurred operational loss of Rs.534.01 Crs. But on accounting the receipt of compensation from the Government of Kerala to the tune of Rs.767.72 Crs for taking over the 75% loss of KSEBL for the year 2022-23, the company reported a net profit of Rs.218.51 Crs. But on accounting the past actuarial liability to the tune of Rs.1528.52 Crs in the "Other Comprehensive Income" part of the profit and loss statement the company is reporting a carryover loss of Rs.1310.01 Cr (1528.52-218.51) to the Balance sheet. The following table summarizes the variation between loss as per audited accounts and TU petition. Detailed information is furnished in **Appendix 14.**

Table 5.53: Comparison of Accounts and TU Petition (Rs Cr)

Sl. No	Particulars	As per Accounts	TU Petition	Difference
1	Interest & Finance charges	2409.63	2256.04	-153.59

Table 5.53: Comparison of Accounts and TU Petition (Rs Cr)				
Sl. No	**Particulars**	**As per Accounts**	**TU Petition**	**Difference**
2	Additional contribution to Master Trust	0.00	400.00	400.00
3	Depreciation	1998.40	778.20	-1220.20
4	O&M Expenses	4910.67	4598.22	-141.93
5	Pay revision expenses	0.00	7.93	7.93
6	Power purchase & PGCIL charges	12982.59	12982.59	0.00
7	RoE	0	489.87	489.87
8	Others	22.82	22.82	0.00
9	Generation of power	0.34	0.34	0.00
10	Exceptional items	-752.52	15.40	767.92
11	Gains out of T&D loss reduction and Incentive	0.00	131.59	131.59
12	Amortization of past gap	0.00	0.00	0.00
13	Amortization of compensation-Edamon-Kochi	0.00	8.06	8.06
14	Amortization of compensation-Trichur-Pugalur	0.00	1.24	1.24
15	Amortization of Intangible assets	12.04	12.04	0.00
16	Incentive for achievement of target transmission availability	0.00	10.49	10.49
17	Repayment of Bond	0.00	407.22	407.22
18	ARR (1 to 17)	21583.97	22122.05	-538.88
19	Revenue from tariff	20381.18	19701.38	-679.80
20	External sale	60.57	60.57	0.00
21	Non-tariff income	1360.73	1036.35	-324.38
22	ERC	21802.48	20798.30	-1004.18
23	Revenue (Gap)/Surplus	218.51	(1323.75)	-1542.26

5.56. Comparison of approval and truing up sought: The following Table summarizes the variation between approval granted by the Hon'ble Commission as per order dated 25.06.2022 and truing up sought for the year 2023-24. Detailed information is furnished in Appendix 15.

Table 5.54: Comparison of Approval and TU Petition (Rs Cr)				
Sl. No	**Particulars**	**As per MYT order**	**TU Petition**	**Difference**
1	Fuel charges for generation	0.00	0.34	0.34
2	POP & Interstate transmission	10564.23	12982.59	2418.36
3	Interest & Finance charges	2141.68	2256.04	114.36
4	Depreciation	737.76	778.20	40.44
5	O&M Expenses	4456.29	4598.22	141.93
6	Other expenses	0.00	22.82	22.82
7	Gains on T&D loss reduction & incentive	0.00	131.59	131.59
8	Master trust additional contribution	400.00	400.00	0.00
9	RoE	489.87	489.87	0.00
10	Amortization of past gap	850.00	0.00	-850.00
11	Amortization of compensation-Edamon-Kochi	14.94	8.06	-6.88
12	Amortization of compensation-Tirchur-Pugalur	0.00	1.24	1.24

13	Amortization of Intangible assets	0.00	12.04	12.04
14	Incentive for achievement of target transmission availability	0.00	10.49	10.49
15	Exceptional items	0.00	15.40	15.40
16	Pay revision	0.00	7.93	7.93
17	Repayment of Bond	407.20	407.20	0.00
18	ARR (1 to 14)	20061.99	22122.05	-2060.06
19	Revenue from tariff	15873.80	19701.38	3827.58
20	External sale	382.16	60.57	-321.59
21	Non-tariff income	866.94	1036.35	169.41
22	ERC	17122.90	20798.30	3675.40
23	Revenue (Gap)/Surplus	(2939.09)	(1323.75)	(1615.34)

CHAPTER- 6 : SUMMARY OF TRUE-UP AND PRAYER

6.1 An item wise comparison of the ARR, ERC and Revenue gap approved by the Hon'ble Commission, actual as per audited accounts and the true up Sought by KSEBL are given in the Table below.

Table 6.1 SUMMARY OF TRUE-UP FOR FY 2023-24

Particulars	Approved	Actual	Sought for TU	Deviation from approval
Generation Of Power	0.00	0.34	0.34	0.34
Purchase of power	10564.23	12982.59	12982.59	2418.36
Interest & Finance Charges	2141.68	2409.63	2256.04	114.36
Depreciation	737.76	1998.40	778.20	40.44
O&M cost	4456.29	4910.67	4598.22	141.93
Other Expenses	0.00	22.82	22.82	22.82
Additional contribution to Master Trust	400.00	0.00	400.00	0
Gains on T&D loss reduction	0.00	0.00	131.59	131.59
Edamon Kochi Compensation			8.06	8.06
Thrissur Pugalur Compensation			1.24	1.24
Incentive for achievement of Transmission target availability	0	0.00	10.49	10.49
Amortization of intangible assets-software	14.94	12.04	12.04	-2.9
Amortization of past gap	850.00	0.00	0.00	-850
RoE	489.87	0	489.87	0
Exceptional item	0.00	-752.52	15.40	15.40
Pay revision expenses	0.00	0.00	7.93	7.93
Repayment of bond	407.22	0	407.22	0
Net Expenditure (A)	**20061.99**	**21583.97**	**22122.05**	**2060.06**
Non-Tariff Income	866.94	1360.73	1036.35	169.41
Revenue from tariff & external sale	16255.96	20441.75	19761.95	3505.99
Total Income (B)	**17122.9**	**21802.48**	**20798.3**	**3675.40**
Revenue surplus /(Gap) (B-A)	**(2939.09)**	**218.51**	**(1323.75)**	**(1615.34)**

6.2 The Statutory Auditors have audited and certified the accounts for the year 2023-24. This truing up petition is prepared on the basis of the audited accounts. The petitioner is furnishing the details of accounts as per formats prescribed in the Tariff regulation, 2021. However, should the Hon'ble Commission require any further details or information; the petitioner will be furnishing them as and when required.

Prayer

KSEBL humbly requests before the Hon'ble Commission that:

(1) Truing up of Expenses and Revenue as per the Audited Accounts of KSEBL for the year 2023-24 as explained in this petition may kindly be approved, in view of the care and caution taken by the KSEBL for carrying out the functions of the Board as a public utility as per the statutory provisions under the Electricity Act, 2003 and also as per the directions, orders and regulations issued by the Hon'ble Commission, policies and directions issued by the State and Central Government and other statutory bodies within the provisions of the Electricity Act-2003.

(2) O&M norms may kindly be revised based on trued up values for 2018-19, approved pay revision expenses for 2018-19 and additional employee cost for the approved employee strength, as envisaged in Tariff Regulation, 2021.

(3) Final true up orders may kindly be issued after considering eligible O&M cost based on revised O&M norms, as envisaged in Tariff Regulation, 2021.

(4) The revenue gap as per the petition may be accounted as regulatory asset or any other appropriate means deemed fit by the Hon'ble Commission according to the provisions of law.

On Behalf of KSEB Limited

Deputy Chief Engineer (TRAC)
in charge of Chief Engineer
(Commercial & Tariff)
